VIGNAN'S

FOUNDATION FOR SCIENCE, TECHNOLOGY & RESEARCH

(Deemed to be University) - Estd. u/s 3 of UGC Act 1956

C24 Curriculum & Course Contents

B.Tech. Computer Science & Engineering

w.e.f. batch 2024-28

FOREWORD

The vital aspect of Computer Science & Engineering program is problem solving through programming. It is an essential skill required to carry out profession of Software Engineer. CSE graduates practice the activities (analysis, design, development, testing and maintenance) of software development life cycle to solve complex and challenging real world problems.

R22 Curriculum enables our students to start with the basic science, basic engineering and introductory programming courses. Further, they learn the mathematical foundations of computing; get hands-on experience in building software solutions using various technologies for various real-world problems and pursue advanced courses such as Artificial Intelligence, Machine Learning, Cloud Computing, Big Data & Analytics, Internet of Things and much more.

Salient Features of R 22 Curriculum:

- *Multidisciplinary holistic education with continuous learning and continuous assessment.*
- *Lateral entry and lateral exit options.*
- *Credit Earning by credit transfer.*
- *Honors/ Research Honors/ Minor/ Add-on Diploma/ Add-on Certification and Dual B.Tech.+ M.Tech./ MBA Degree of 5 Years.*
- *Semester drop option to pursue innovation, incubation, entrepreneurial and advanced exploratory activities and subsequent re-entry.*

Emphasis on continuous formative assessment with a creative summative assessment will facilitate the student to "Move away from high stake examinations – towards more continuous and comprehensive evaluation".

The Board of Studies of B.Tech. CSE Programme consists of a right mix of eminent personalities from Academic, Research and Industry Organizations, besides experienced faculty members of the University.

External BoS Members:

1. Prof. C. R. Rao, Professor SCIS, University of Hyderabad.
2. Prof. R. B. V. Subramanyam, Professor, Department of CSE, NIT Warangal.
3. Dr. B. Venkata Ramana, Assoc. Professor & HoD, Department of CSE, IIT Tirupathi.
4. Dr. Nagesh Bhattu Sristy, Asst. Professor, Department of CSE, NIT AP.

I thank all the BOS members, Academic Council Members and University authorities for their continuous support and encouragement towards design of this innovative curriculum for CSE.

Dr. S. V. Phani Kumar
Head, Department of CSE
VFSTR Deemed to be University

VIGNAN'S
Foundation for Science, Technology & Research
(Deemed to be UNIVERSITY)
-Estd. u/s 3 of UGC Act 1956

VISION

To evolve into a Centre of Excellence in Science & Technology through creative and innovative practices in teaching – learning, towards promoting academic achievement and research excellence to produce internationally accepted, competitive and world class professionals who are psychologically strong & emotionally balanced, imbued with social consciousness & ethical values.

MISSION

To provide high quality academic programmes, training activities, research facilities and opportunities supported by continuous industry - institute interaction aimed at promoting employability, entrepreneurship, leadership and research aptitude among students and contribute to the economic and technological development of the region, state and nation.

Department of COMPUTER SCIENCE & ENGINEERING

VISION of the department

To evolve as a centre of high repute in Computer Science & Engineering and create computer software professionals trained on problem solving skills imbued with ethics to serve the ever evolving and emerging requirements of IT Industry and society at large.

MISSION of the department

M_1: Imparting quality education through well designed curriculum, innovative teaching and learning methodologies integrated with professional skill development activities to meet the challenges in the career.

M_2: Nurture research and consultancy activities amongst students and faculty by providing State-of-art facilities and Industry-Institute Interaction.

M_3: Developing capacity to learn new technologies and apply to solve social and industrial problems to become an entrepreneur.

B.Tech in Computer Science and Engineering

Program Educational Objectives (PEOs)

PEO1: Graduates acquire extensive technical knowledge and related skills required to demonstrate themselves as professionals or pursue higher education.

PEO2: Graduates adapt to upskilling and excel in their careers despite future technological changes, and demonstrate research aptitude to generate innovative engineering solutions.

PEO3: Graduates acquire the potential to contribute for the field of computing as well as for the societal development by demonstrating professional, social and ethical practices.

Program Specific Outcomes (PSOs)

The students will be able to

PSO1: **Application Development Skills:** Design and development of web applications using various technologies such as HTML, JSP, PHP, ASP and ASP.NET to cater the needs of the society.

PSO2: **Enrich Research Skills:** Offer solutions which impact geo-socio-economic and environmental scenarios by using Machine Learning, Artificial Intelligence and IoT.

Program Outcomes (POs)

PO1: Engineering knowledge

PO2: Problem analysis

PO3: Design/development of solutions

PO4: Conduct investigations of complex Problems

PO5: Modern tool usage

PO6: The engineer and society

PO7: Environment and sustainability

PO8: Ethics

PO9: Individual and team-work

PO10: Communication

PO11: Project management and finance

PO12: Life-long learning

Student Outcomes (SOs) - Engineering Accreditation Commission (EAC)

SO1: An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

SO2: An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as a global, cultural, social, environmental, and economic factors.

SO3: An ability to communicate effectively with a range of audiences.

SO4: An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.

SO5: An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.

SO6: An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.

SO7: An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

Student Outcomes (SOs) - Computing Accreditation Commission (CAC)

SO1: Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions.

SO2: Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.

SO3: Communicate effectively in a variety of professional contexts.

SO4: Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.

SO5: Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.

SO6: Apply computer science theory and software development fundamentals to produce computing-based solutions.

COURSE STRUCTURE – C24 (M.P.C stream)

Induction Programme

Course Title	L	T	P	C	Course category
Orientation Session	0	2	0	1	Binary grade

I Year I Semester

Course Title	L	T	P	C	Course category
Linear algebra & Ordinary differential equations / Calculus	3	2	0	4	Basic Sciences
Engineering Physics / Engineering Chemistry	3	0	2	4	Basic Sciences
Basic of Electrical & Electronics Engineering/ Engineering Graphics	2	0	2	3	Basic Engineering
Programming in C	2	0	4	4	Basic Engineering
English Proficiency & Communication Skills (PET)	0	0	2	1	Humanities
IT Tools & Cyber security	0	2	2	2	Basic Engineering
Environmental Studies / Management Science	2	2	0	3	Basic Sciences / Humanities
	12	**6**	**12**	**21**	
Physical Fitness, Sports & Games / Self-empowerment & Gender Sensitization	0	0	2	1	Binary grade
Constitution of India / Indian Knowledge Systems	0	2	2	1	Binary grade
	0	**2**	**4**	**2**	
	36			**23**	

I Year II Semester

Course Title	L	T	P	C	Course category
Linear algebra & Ordinary differential equations / Calculus	3	2	0	4	Basic Sciences
Engineering Physics / Engineering Chemistry	3	0	2	4	Basic Sciences
Basic of Electrical & Electronics Engineering/ Engineering Graphics	2	0	2	3	Basic Engineering
Problem Solving through Python	2	0	2	3	Basic Engineering
Technical English Communication	2	0	2	3	Humanities
Environmental Studies / Management Science	2	2	0	3	Basic Sciences / Humanities
	14	**4**	**8**	**20**	
Physical Fitness, Sports & Games / Self-empowerment & Gender Sensitization	0	0	2	1	Binary grade
Constitution of India / Indian Knowledge Systems	0	2	0	1	Binary grade
	0	**2**	**2**	**2**	
	30			**22**	

B. TECH – CSE
COURSE STRUCTURE – C24

II Year I Semester

Course Code	Course Title	L	T	P	C	Course Category
24MT203	Discrete Mathematical Structures	3	2	0	4	Basic Sciences
22TP201	Data Structures	2	2	2	4	Basic Engineering
22CS201	Database Management Systems	2	2	2	4	Professional core
24CS204	Object-Oriented Programming through JAVA	2	2	2	4	Professional core
24CS203	Computer Architecture	3	2	0	4	Professional core
24CS205	Front end application Development	0	2	2	2	Professional core
24CS299	Design Thinking and Engineering Orientation	0	0	2	1	Basic Engineering
24CS801	Industry interface course (Modular course)	0	0	2	1	Department Elective
	Sub Total	**12**	**12**	**12**	**24**	
	Life Skills	0	0	2	01	Binary Grade
	Sub Total	**12**	**12**	**14**	**25**	
	Total	**38**			**25**	

II Year II Semester

Course Code	Course Title	L	T	P	C	Course Category
22ST202	Probability and Statistics	3	2	0	4	Basic Sciences
22CS206	Design and Analysis of Algorithms	2	2	2	4	Professional core
22CS207	Operating Systems	2	0	2	3	Professional core
24CS206	Theory of Computation	2	2	0	3	Professional core
24CS207	Full Stack Development	0	2	2	2	Professional core
	Open Elective – 1 (NPTEL)	2	2	0	3	Open Elective
24CS201	Field projects	0	0	2	1	Project
	Sub Total	**11**	**10**	**08**	**20**	
	UHV	1	0	0	1	Binary Grade
	Minors/Honours - 1	3	0	2	4	Minors/Honours
	Sub Total	**15**	**10**	**10**	**25**	
	Total	**35**			**25**	

III Year I Semester

Course Code	Course Title	L	T	P	C	Course Category
22TP301	Soft Skills Laboratory	0	0	2	1	Humanities
24CS301	Optimization Techniques	2	2	0	3	Basic Sciences
24CS302	Artificial Intelligence	2	0	2	3	Professional core
24CS303	Computer Networks	2	2	2	4	Professional core
24CS304	Mobile Application Development	0	2	2	2	Professional core
	Department Elective-1	3	0	2	4	Department Elective
	Open Elective-2	2	0	2	3	Open Elective
24EN301	Professional Communication Skills	0	0	2	1	Humanities
	Sub Total	**11**	**06**	**14**	**21**	
	Minors/Honours - 2	3	0	2	4	Minors/Honours
	Sub Total	**14**	**06**	**16**	**25**	
	Total	**36**			**25**	

III Year II Semester

Course Code	Course Title	L	T	P	C	Course Category
24CS305	Computing Ethics	1	2	0	2	Humanities
24TP301	Quantitative aptitude & Logical reasoning	1	2	0	2	Humanities
24CS306	Machine Learning	2	0	2	3	Professional core
22CS407	Cryptography and Network Security	2	0	2	3	Professional core
	Department Elective-2	2	0	2	3	Department Elective
	Department Elective-3	3	0	2	4	Department Elective
	Open Elective-3	2	0	2	3	Open Elective
24CS307	Inter Departmental Project	0	0	2	1	Project
	Sub Total	**13**	**04**	**12**	**21**	
	Minors/Honours - 3	3	0	2	4	Minors/Honours
	Sub Total	**16**	**04**	**14**	**25**	
	Total	**34**			**25**	

IV Year I Semester

Course Code	Course Title	L	T	P	C	Course Category
24CS401	Software Project Management	2	2	0	3	Professional core
24CS402	Parallel & Distributed Computing	2	2	0	3	Professional core
24CS403	Privacy and Intrusion Detection	2	2	0	3	Professional core
	Department Elective-4	3	0	2	4	Department Elective
	Department Elective-5	3	0	2	4	Department Elective
	Department Elective-6	3	0	2	4	Department Elective
	Sub Total	**15**	**06**	**06**	**21**	
	Minors/Honours - 4	3	0	2	4	Minors/Honours
	Sub Total	**18**	**06**	**08**	**25**	
	Total	**32**			**25**	

IV Year II Semester

Course Code	Course Title	L	T	P	C	Course Category
22CS404 / 22CS405	Internship / Project Work	0	2	22	12	Project
	Sub Total	**0**	**02**	**22**	**12**	
	Minors/Honours - 5	0	2	6	4	Minors/Honours
	Sub Total	**0**	**04**	**28**	**16**	
	Total	**32**			**16**	

Department Electives

Sl. No.	Course Code	Course Title	L	T	P	C
1	22CS801	Advanced Data Structures	2	2	2	4
2	22CS802	Advanced JAVA Programming	2	2	2	4
3	22CS803	Computer Graphics	2	2	2	4
4	22CS804	Deep Learning	3	0	2	4
5	22CS805	Digital Image Processing	2	2	2	4
6	22CS807	Mobile Ad-hoc Networks	3	0	2	4
7	22CS809	Text Mining	3	0	2	4
8	22CS810	Numerical Algorithms	3	2	0	4
9	22CS811	Operating System Design	3	0	2	4
10	22CS813	Simulation and Modeling	3	0	2	4

Open Electives offered to Non-Computing Branches

Sl. No.	Course Code	Course Title	L	T	P	C
1	22CS851	Database Systems	2	0	2	3
2	22CS852	Mobile Application Design and Development	2	0	2	3
3	22CS853	Java Programming	2	0	2	3
4	22CS854	Python Programming	2	0	2	3
5	22CS855	Design and Development of Internet Applications	2	0	2	3

Honours

Sl. No.	Course Code	Course Title	L	T	P	C
1	22CS951	Advanced Graph Algorithms	3	0	2	4
2	22CS952	Biometrics	3	0	2	4
3	22CS953	Parallel and Distributed Computing	3	2	0	4
4	22CS954	Internet of Things	3	0	2	4
5	22CS955	Wireless Sensor Networks	3	0	2	4
6	22CS956	Capstone Project	0	2	6	4

Minor Courses

Sl. No.	Course Code	Course Title	L	T	P	C
1	22CS901	Introduction to Python Programming	2	2	2	4
2	22CS902	OOPS through JAVA	2	0	4	4
3	22CS903	Database Management Systems	2	2	2	4
4	22CS904	Web Technologies	2	0	4	4
5	22CS905	Mobile Application Development	2	0	4	4
6	22CS906	Design and Analysis of Algorithms	2	2	2	4
7	22CS907	Operating Systems and Shell Programming	2	2	2	4
8	22CS908	Computer Networks	3	0	2	4
9	22CS909	Capstone Project	0	2	6	4

24MT101 LINEAR ALGEBRA AND ORDINARY DIFFERENTIAL EQUATIONS

Hours per week:

L	T	P	C
3	2	0	4

PREREQUISITE KNOWLEDGE: Basics of matrices, Differentiation and Integration.

COURSE DESCRIPTION AND OBJECTIVES:

The goal of this course is to build a grasp of the principles of mathematics through matrices, differential equations and applications that serves as an essential tool in several engineering applications.

MODULE-1

24L+16T+0P=40 Hours

UNIT-I: MATRICES

Definition of matrix, Types of matrices, Algebra of matrices, adjoint of a matrix, inverse of a matrix through adjoint and elementary row operations, Rank of a matrix, Echelon form, Normal form, Eigen values and Eigen vectors (up to 3 x 3 matrices only) and properties (without proofs).

UNIT-II: APPLICATIONS OF MATRICES

Consistency of system of linear equations, Solution of system of linear equations having unique solution and involving not more than three variables by Gauss elimination method and Gauss Jordan method; Cayley-Hamilton theorem (without proof), Power of a matrix, Inverse of a matrix; Strength of materials and strength of beams using Eigen values and Eigen vectors.

PRACTICES:

- Compute inverse of a matrix if exists.
- Explain with suitable examples how rank of matrix is independent of the elementary operations.
- Explain with suitable examples how rank of matrix is unique.
- Discuss with suitable examples when Eigen values and Eigen vectors are possible for a matrix.
- Discuss the possibility of solution of a system of equations.
- Discuss when inverse and power of a matrix exist using Cayley-Hamilton theorem.

MODULE-2

24L+16T+0P=40 Hours

UNIT-1: ORDINARY DIFFERENTIAL EQUATIONS (ODE)

First Order Differential Equations: Introduction to ODE, variable separable method, homogenous and non-homogenous differential equations, linear differential equations, Bernoulli's equations.

Second Order Differential Equations: Linear differential equations with constant coefficients with **RHS** of the form e^{ax}, x^n, sin(ax) or cos(ax).

UNIT-2: APPLICATIONS OF ODE

Applications of ODE: Newton's law of cooling, Law of natural growth and decay; LR circuit.

PRACTICES:

- Check the order and degree of an ODE.
- Find solution for any four ordinary differential equations by applying suitable method.
- Find numerical solution for any four ordinary differential equations by applying suitable method.
- Discuss some applications of ODE.

SKILLS:

- Find rank of a matrix using different methods.
- Compute the Eigen values and Eigen vectors of a matrix.
- Find analytical solution of a differential equation using appropriate method.
- Demonstrate any one numerical method to solve differential equation.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Apply the concepts of rank, Eigen values and Eigenvectors of a matrix and finding inverse of a matrix and powers of a matrix.	Apply	1	1, 2, 3, 9, 10, 12
2	Apply differential equations in real life problems.	Apply	2	1, 2, 3, 9, 10, 12
3	Analyse the solution of a system of linear equations and find it.	Analyze	1	1, 2, 3, 9, 10, 12
4	Inspect the analytical method for solving differential equations and applications.	Analyze	2	1, 2, 3, 9, 10, 12

TEXT BOOKS:

1. B. S. Grewal," Higher Engineering Mathematics", 44 Edition, Khanna Publishers, 2018.
2. N. P. Bali, K. L. Sai Prasad, "A Textbook of Engineering Mathematics I, II, III", 2nd Edition, Universal Science Press, New Delhi, 2018.

REFERENCE BOOKS:

1. H. K. Dass and Er. RajanishVerma, "Higher Engineering Mathematics", 3rd revised edition, S. Chand & Co., 2015.
2. B. V. Ramana, "Advanced Engineering Mathematics", TMH Publishers,11th edition 2011.
3. T. K.V. Iyengar et. al, "Engineering Mathematics, I, II, III", S. Chand & Co., New Delhi,2018.

24MT102 CALCULUS

Hours per week:

L	T	P	C
3	2	0	4

PREREQUISITE KNOWLEDGE: Differentiation, Integration, Vectors.

COURSE DESCRIPTION AND OBJECTIVES:

The goal of this course is, students should be able to adapt their knowledge and techniques to solve other similar application problems. Make aware students about the importance and symbiosis between Mathematics and Engineering. To develop the mathematical skill so that students are able to apply mathematical methods & principals in solving problem from Engineering fields. Students will learn applications of differentiation, integration and vectors to use in solving problem from Engineering fields.

MODULE-1

24L+16T+0P=20 Hours

UNIT-1: CALCULUS

Differential Calculus

Functions, Limits, Continuity, Differentiability, Tangent and normal, Mean value theorems; Analyzing functions for Extreme value and critical points, Intervals on which a function is increasing or decreasing, Relative (local) extrema, Absolute (global) extrema, Concavity and inflection, graphical transformation.

UNIT-2: Integral calculus

Double integral, Triple integral, Change of order, integration in polar coordinates, Finding Areas and Volumes.

PRACTICES:

- Write basic formulas of differentiation and integration
- Finding the differentiation and integration of functions of several variables.
- What is the tangent and normal?
- Find the velocity and acceleration of given function.
- Tell about maxima and minima.
- Discuss the given function is increasing or decreasing.

MODULE-2

24L+16T+0P=20 Hours

UNIT-1: VECTOR CALCULUS

Introduction to vectors, Vector algebra, Scalar and vector point functions, Gradient, Divergence and Curl.

UNIT-2: APPLICATIONS OF VECTOR CALCULUS

Normal vector, Directional Derivate, Solenoidal and Irrotational vectors, Scalar potential; Line integral, surface integral, volume integral, Green's theorem for plane, Gauss divergence theorem, Stokes' theorem (without proofs).

PRACTICES:

- Tell difference between a vector and scalar.
- Algebra of a vectors.
- Find the area and volume bounded by the curves.
- Differentiate between line, surface and volume integrals.
- Applications of vector differential operators.
- Discuss the tangent and normal vector.

SKILLS:

- Needs to evaluate integration and differentiation of all functions
- Understand how to choose the limits from the given curves
- Need to understand the geometrical representation of each given curve

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Apply differentiation and integration for solving engineering problems	Apply	1	1, 2, 3, 4, 9, 10, 12
2	Apply vector differential operators in real life problems	Apply	2	1, 2, 3, 4, 9, 10, 12
3	Analyse a function for its extreme values	Analyze	1	1, 2, 3, 4, 9, 10, 12
4	Evaluate line, surface and volume integrals	Evaluate	2	1, 2, 3, 4, 9, 10, 12

TEXT BOOKS:

1. N. P. Bali, K. L. Sai Prasad, "A Textbook of Engineering Mathematics I, II, III", 2nd Edition, Universal Science Press, New Delhi, 2018.
2. B. S. Grewal, "Higher Engineering Mathematics", 44 Edition, Khanna Publishers, 2018.

REFERENCE BOOKS:

1. H. K. Dass and Er. Rajanish Verma, "Higher Engineering Mathematics", Third edition S. Chand and Co., 2015.
2. B. V. Ramana, "Advanced Engineering Mathematics", TMH Publishers, 2017.
3. T. K.V. Iyengar et al, "Engineering Mathematics, I, II, III", S. Chand and Co., New Delhi, 2018.
4. Erwin Kreyszig, "Advanced Engineering Mathematics", 10th Edition, John Wiley & Sons, Inc, 2010.

24PY102 - ENGINEERING PHYSICS

Hours per week:

L	T	P	C
3	0	2	4

PREREQUISITE KNOWLEDGE: Basics of Quantum Mechanics and Semiconductors

COURSE DESCRIPTION AND OBJECTIVES:

This course ensures commensurable understanding of the principles of Quantum Mechanics and applying them to find the various physical properties of conductors and semiconductors. This embark perspective outlook on optoelectronic devices and optical fibres in view of semiconductor physics.

MODULE- 1

24L+0T+16P=40 Hours

UNIT 1: QUANTUM MECHANICS

Quantum Mechanics: Introduction to Quantum Mechanics, Concept of wave and particle duality of radiation, de Broglie's concept of matter waves, Schrödinger's time-independent wave equation, Eigen values and Eigen functions, Particle confined in a one-dimensional infinite potential well and finite potential well, Quantum Tunnelling, Applications of Quantum laws in SEM and Quantum dots.

UNIT 2: FREE ELECTRON THEROY

Free electron theory of solids: Classical free electron theory, expression for electrical conductivity, Quantum frees electron theory, Fermi-Dirac distribution, electronic specific heat of solids, Density of states, Bloch's Theorem, Classification of solids based on energy bands.

PRACTICES

- Photoelectric Effect-Determination of Planck's constant.
- I-V characteristics of a *p-n* junction diode.
- Determination of energy bandgap of a *p-n* junction diode.
- Thermistor-Measurement of resistivity of a thermistor.

MODULE-2

24L+0T+16P=40 Hours

UNIT 1: SEMICONDUCTOR PHYSICS

Introduction, Classification of semiconductors; Direct and indirect bandgap semiconductors, Intrinsic and extrinsic semiconductors.

Intrinsic semiconductors; Variation of intrinsic carrier concentration, Fermi level, and electrical conductivity with temperature.

Extrinsic semiconductors; Effect of temperature on carrier concentration, energy band diagrams Hall Effect.

UNIT-2: OPTOELECTRONICS

Classification of optoelectronic devices; Photovoltaic cell, LED.
LASERs; Introduction, characteristics, population inversion, pumping, Semiconductor diode Laser and applications of Lasers.
Optical fibers; Acceptance angle, Numerical aperture, applications.

PRACTICES:

- Hall Effect - Determination of Hall coefficient.
- Laser - Determination of wavelength.
- Optical fibre - Determination of Numerical aperture and acceptance angle.
- Solar cell – Determination of Fill factor & Efficiency.
- Study of I-V Characteristics of Light Emitting Diode

SKILLS:

- Apply the quantum laws to understand the electron dynamics of solids.
- Assess the importance of semiconductors in terms of its properties,
- Comprehend the significance of optoelectronic devices.

COURSE OUTCOMES:

Upon successful completion of the course, students will have the ability to:

CO No.	Course outcomes	Blooms level	Module No.	Mapping with POs
1	Understand and analyze Quantum Mechanics principles to solve physical problems	Analyze	1	1, 2, 3, 4, 9, 10
2	Determine of energy levels and classification of solids based on energy bands	Evaluate	1	1, 2, 4, 5, 9, 10
3	Apply the knowledge of semiconductors to illustrate and to assess their properties	Apply	2	1,3,4,5,6, 9, 10
4	Interpret the process of optoelectronic transition and their functionality	Evaluate	2	1,2,3,5, 9, 10

TEXT BOOKS:

1. S.M. Sze, "Semiconductor Devices Physics and Technology", John Wiley & Sons, 3rd Edition, 2015.
2. C. Kittel, "Introduction to Solid State Physics", Wiley, 9th Edition, 2019.

REFERENCE BOOKS:

1. N. Zettili, "Quantum Mechanics: Concepts and Applications", Wiley, 3rd Edition, 2022.
2. P. K. Palanisamy, "Engineering Physics", Scitech Publications, 3rd Edition, 2013.
3. S. O. Pillai, "Solid State Physics", New age International publishers, 8th Edition, 2018.
4. M.A. Wahab, "Solid State Physics: Concepts and Applications", Narosa Publishing House, 3rd Edition, 2015.
5. M. N. Avadhanulu and P.G. Kshirsagar, "A Textbook of Engineering Physics", S. Chand Publishing, 11th Edition, 2010.

24CT101 - ENGINEERING CHEMISTRY

Hours per week:

L	T	P	C
3	0	2	4

PREREQUISITE KNOWLEDGE: Concept of bonding, chemical reactions and electrochemical cell

COURSE DESCRIPTION AND OBJECTIVES:

The course aims to cover the importance of chemistry and its applications in engineering disciplines particularly focusing on developing new engineering materials (such as polymers, nanomaterials etc.) and understanding their property for scientific and engineering applications. In addition, the students are also expected to acquire knowledge on electrochemistry and construction of batteries and fuel cells.

MODULE-1

24L+0T+16P=40 Hours

UNIT-1

POLYMER TECHNOLOGY

Introduction and Classification of Polymers, Polymerization techniques, Preparation, properties and applications of PE, Nylon-6,6; Elastomers-Natural rubber and properties, vulcanization, synthetic rubbers-Buna N and S; Introduction to conducting polymers (Polythiophene).

UNIT-2

ELECTROCHEMICAL DEVICES

Introduction to Electrochemistry, Electrochemical series, Galvanic and electrolytic cells, EMF of an electrochemical cell, Reference Electrode-Standard Hydrogen Electrode; Classification of batteries, Battery characteristics, Construction, working and applications of: Lead-acid storage cell, Lithium-ion battery; Fuel Cells- Classification, Construction, working and applications of H_2-O_2 fuel cell and solid oxide fuel cell; Concept of Redox-flow batteries.

PRACTICES:

- Synthesis of Bakelite
- Preparation of Nylon-6,6
- Determination of Fe (II) by dichrometry method
- Determination of strength of Weak acid by pH-method
- Determination of strength of acid by conductometry

MODULE-2

24L+0T+16P=40 Hours

UNIT-1
MATERIALS CHEMISTRY

Engineering Materials: Refractories – classification, refractoriness RUL, Chemical & Thermal stability; Lubricants – classification, viscosity, viscosity index, flash and fire points,

cloud and pour points and mechanical stability; Abrasives-types and properties; Ionic liquids as lubricants

Nanomaterials: Introduction, classification, properties, Top-down (Ball Milling) and Bottom-up (Sol-Gel) synthetic methods; Synthesis, properties and applications of Carbon Nanotubes and Graphene; Introduction to nanocomposite and membrane technology.

UNIT-2

CHEMISTRY FOR ELECTRONICS AND COMPUTING

Introduction to electronic memory devices, Classification of electronic memory devices; Organic- Inorganic hybrid materials; Display Systems- Electromagnetic spectrum, Interaction of light with matter, Photo and electro active materials properties and applications (OLED's, QLED's); Introduction and applications of Computational Chemistry.

PRACTICES:

- Determination of viscosity of oil (Biodiesel, castor oil and coconut oil)
- Purity of lubricant- Saponification and neutralization number
- Synthesis of magnetic Iron oxide nanoparticles
- Estimation using Colorimetry
- Chemsketch and Avogadro

SKILLS:

- Synthesize various polymers.
- Synthesize nanomaterials.
- Identify the properties of different industrially relevant engineering materials.
- Understand the different components of an electrochemical cell.
- Design electrochemical cell such as battery.
- Identify the types of energy conversion/storage systems.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Apply various synthetic methodologies for preparing different types of polymers for various engineering applications.	Apply	1	1, 2, 4, 9, 10, 11, 12
2	Evaluate the output of different electrochemical cells and ability to design various batteries and fuel cells for different applications.	Evaluate	1	1, 2, 3, 5, 9, 10, 11, 12
3	Analyze and design different engineering materials and nano materials by evaluating their properties for enhancing the productivity.	Analyze	2	1, 2, 3, 5, 6, 7, 9, 10, 11, 12
4	Analyze possible devices and display systems and their relevance to electromagnetic spectrum.	Analyze	2	1, 2, 3, 5, 6, 7, 9, 10, 11, 12

TEXT BOOKS:

1. S. Chawala, "A Textbook of Engineering Chemistry Engineering Materials and Applications", Dhanpat Rai Publications, 3rd Edition, 2015.
2. K. S. Maheswaramma and M. Chugh, "Engineering Chemistry", Pearson, 1st Edition, 2015

REFERENCE BOOKS:

1. VR Gowariker, NVViswanathan, Jayadev, Sreedhar, "Polymer Science", New age Int.Publishers, 4th Edition, 2021
2. Kirby W.Beard, "Linden's Handbook of Batteries", Fifth Edition, McGrawHill, 2019.
3. MaliniS, KSAnantha Raju, "Chemistry of Engineering materials", CBSpublishers PvtLtd., 2022
4. T. Pradeep, "Nano: The Essentials; Understanding of Nano Science and Technology" Tata McGraw-Hill, New Delhi, 2012.
5. Takatoshi Tsujimura, "OLED Display Fundamentals and Applications", Wiley–Blackwell, 2012
6. Frank Jensen, "Introduction to Computational Chemistry", Second Edition, John Wiley & Sons Ltd, 2007.
7. Avogadro User Manual - https://avogadro.cc/docs/

24CT103 – ENVIRONMENTAL STUDIES

Hours per week:

L	T	P	C
2	2	0	3

PREREQUISITE KNOWLEDGE: General awareness regarding environmental problems and importance of environmental protection.

COURSE DESCRIPTION AND OBJECTIVES:

It is a multidisciplinary subject where we deal with different aspects using a holistic approach. It is evolving to be the education for sustainable and ethical development both at a local and global level. It helps to prepare the next generation and to plan appropriate strategies for addressing environmental issues. It identifies and create solutions that conserve to manage ecosystem and biodiversity and also helps to eliminate pollutants, toxicants, preserve air, water and soil quality. Environmental education recognize impacts of global issues, enhances the public awareness and helps to take decisions towards environmentally responsible actions.

MODULE-1

16L+16T+0P=32 Hours

UNIT-1: Sustainable Development, Natural Resources and Ecosystems

Sustainable Developments - Concepts and SDGs; Natural Resources (forest, water, land), Energy Resources (Renewable and non-renewable resources); Global energy scenario; Concept of Ecology and Ecosystem, Ecological Succession, Population Ecology, Biotic Interactions.

UNIT-2: Biodiversity and Ecosystem conservation

Biodiversity, Threats to Biodiversity, Man environment interactions; Biodiversity conservation (*in-situ* and *ex-situ* conservation strategies); Protected areas, Traditional Knowledge in biodiversity conservation, Ecosystem protection.

PRACTICES AND FIELD PROJECTS:

- Visit to biogas plant, solar power plant, waste to energy power plant (document preparation)
- Detailed study of any environmental problem/situation, Identification of factors pertaining to that problem with conclusion (Students can choose any environmental problem in the surrounding. (document preparation)
- Field project**:** river/pond/lake/forest/grassland/hill/mountain ecosystem study and report writing.
- Herbarium Sheet preparation
- Preparation of Compost
- Activity on Renewable Energy Resource Research

MODULE-2

16L+16T+0P=32 Hours

UNIT-1 Pollution Control and Environmental Sanitation

Emerging Local and Global Environmental Issues; Environmental Pollution Control Monitoring, Strategies and devices in industries; Clean technologies; Waste Management-Municipal, Industrial, Biomedical, Hazardous, Radioactive Waste, and E-waste management; Waste-water Treatment (Water Purification Technologies, Adsorption Studies).

UNIT-2 Environmental Health and Safety

Occupational Health and Hygiene; Toxicity Studies (Heavy Metal Toxicity, Xenobiotic and Carcinogenicity, Effects on Human health by Agents like Asbestos, Tobacco etc.); Bioremediation Technologies.

PRACTICES AND FIELD PROJECTS

- Field Project: Water and Waste-water analysis and Remediation techniques, Visit to Sewage Treatment Plant.
- Field Project: Assessment of Plastic Pollution on Campus and nearby Places with Possible Remediation Strategies
- Disposal of laboratory waste- Individual Practice and Report Writing, Lab Safety Awareness
- Report Writing: 12 principles of Green Chemistry for Environmental Sustainability
- Report Writing: Environmental Management (ISO 14001), Eco-mark, Eco labeling
- Lab Safety and Local Disaster Management Plan

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Ability to apply knowledge, research skills and field techniques to protect environment.	Apply	1	6,7,12
2	Analyze the biodiversity of different ecosystems and formulate various conservation approaches	Analyze	1	7,8,10,12
3	Collect the data using research skills and analyze the presence of various environmental pollutants for problem solving	Analyze	2	6,7,12
4	Design and apply remediation technologies for environmental sustainability	Apply	2	3,6,7,9,12
5	Survey of various waste management approaches and recommend their implementation strategies	Evaluate	2	7,8,9,10,12

TEXT BOOKS:

1. A. Kaushik and C. P. Kaushik, “Perspectives in Environmental Studies”, 7th edition, New Age International Publishers, 2021.
2. B. Joseph, “Environmental Studies”, 3rd edition, McGraw Hill Education, 2017.

REFERENCE BOOKS:

1. Mahua Basu & S.Xavier, "Fundamentals of Environmental Studies", Cambridge University Press, 2017.
2. Sanjay K. Sharma, Ackmez Mudhoo, "Green Chemistry for Environmental Sustainability", CRC Press, 2019.
3. Ram Chandra, "Environmental Waste Management,"CRC Press, 2020.
4. Srujana Kathi, "Environmental Toxicoloy", Notion Press, 2021.
5. James E. Girard, "Principles of Environmental chemistry", 2nd edition, Jones and Bartlett Publishers, 2009.

24EE101- BASICS OF ELECTRICAL AND ELECTRONICS ENGINEERING

Hours Per Week:

L	T	P	C
2	0	2	3

PREREQUISITE KNOWLEDGE: Electrostatics and Electromagnetism.

COURSE DESCRIPTION AND OBJECTIVES:
This course provides an insight into the functioning of basic electrical components like resistor, inductor and capacitor. It deals with the constructional and operational details of AC machines. It also deals with the basic electronic components like P-N junction diode, Zener diode, Transistor and their characteristics.

MODULE-1

16L+0T+16P=32 Hours

UNIT-1

FUNDAMENTALS OF ELECTRIC CIRCUITS:
DC Circuits: Concept of network, Active and passive elements, Voltage and current sources, Concept of linearity and linear network, Unilateral and bilateral elements, R, L and C as linear elements, Ohm's Law, Kirchhoff's Laws, Application to simple series, Parallel circuits, Mesh and nodal analysis of resistive circuits with DC source.

AC circuits: Generation of AC voltage, Frequency, Average value, R.M.S. value, Form factor, Peak factor for sinusoidal only.

UNIT-2

SEMICONDUCTOR DEVICES
Classification of semiconductors, P-N junction diode -operation and its characteristics, Half wave rectifier - operation, efficiency; Full wave rectifiers -types, operation, efficiency; Zener diode and its characteristics, Zener diode as Voltage regulator.

Bi polar junction transistor- operation, types (NPN & PNP)

PRACTICES:

- Verification of Ohm's law.
- Verification of Kirchhoff's current law.
- Verification of Kirchhoff's voltage law.
- Determination of R.M.S. Values of sinusoidal waveform.
- Verification of PN junction diode characteristics under both forward and reverse bias.
- Verification of Zener diode characteristics under reverse bias.

MODULE-2

16L+0T+16P=32 Hours

UNIT-1

ANALYSIS OF AC CIRCUITS

Analysis of single - phase AC circuits consisting of R, L, C, RL, RC (series and parallel)

(simple numerical problems). Introduction to three phase system, Relation between phase and line quantities of voltages and currents in star and delta connected systems (Elementary treatment only).

UNIT-2

AC MACHINES:

Electromagnetism: Concepts of Magneto motive force, Reluctance, Flux and flux density, Concept of self-inductance and mutual inductance, Coefficient of coupling.

Static AC Machine: Principle of operation of single phase transformer, Constructional features, EMF equation (simple numerical problems).

Rotating AC Machine Principle of operation of three phase induction motor, Slip ring and squirrel cage motors, Torque equation; Constructional details of synchronous machine.

PRACTICES:

- Transformation ratio of a single phase transformer at different loads.
- Measurement of Energy in single phase resistive load circuit.
- Measurement of Power in single phase resistive load circuit.
- Determination of impedance in complex AC circuits.
- Verification of line and phase quantities in a balanced three phase system.

SKILLS:

- Distinguish between linear and nonlinear elements by looking at VI characteristics
- Develop a simple loop generator.
- Design a voltage regulator using Zener diode.
- Design a half wave rectifier using PN junction diode.
- Design a full wave rectifier using PN junction diodes.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Analyze the resistive circuits with independent sources and find its solution..	Analyze	1, 2	1, 2, 6, 9
2	Solve the AC (single and three phase) and DC circuits using different methods.	Apply	1, 2	1, 2, 9 ,12
3	Apply the concepts off electromagnetism for its applications.	Apply	2	1, 2, 3, 9, 12
4	Examine the different electrical equipment.	Evaluate	2	1, 2, 9, 12
5	Acquire the knowledge of semiconductor devices to create circuits.	Create	1	1, 2, 3, 9, 12

TEXT BOOKS:

1. V. K. Mehta, "Principles of Electrical Engineering and Electronics", 1st edition, S.Chand & Co., Publications, New Delhi, 2019.
2. D.P. Kothari, "Basic Electrical and Electronics Engineering", 2nd edition, TMH, New Delhi, 2017.

REFERENCE BOOKS:

1. Millman and Halkias, "Electronic Devices and Circuits", Mc Graw Hill, 2006.
2. A.K. Thereja and B.L.Thereja, "Electrical Technology", Vol.–II, S. Chand & Co., Publications, 2020.
3. U. Bakshi and A. Bakshi, "Basic Electrical Engineering", 1st edition, Technical Publications, Pune, Nov 2020

24ME101 - ENGINEERING GRAPHICS

Hours per week:

L	T	P	C
2	0	2	3

PREREQUISITE KNOWLEDGE: Basics of Geometry.

COURSE DESCRIPTION AND OBJECTIVES:
Engineering graphics is the language of engineers and is the most effective way of communicating and sharing technical ideas in the form of pictures/drawings. The objective of this course is to familiarize the students with the conventional concepts of engineering drawing and computer aided drawing.

MODULE-1

16L+0T+16P=32 Hours

UNIT-1

ENGINEERING CURVES
Types of lines; Lettering, Dimensioning, Geometric constructions - lines, polygons (Angle, ARC, General and Inscribe in circle method), Conical curves (General method), Ellipse by Oblong method.

UNIT-2

ORTHOGRAPHIC PROJECTIONS OF POINTS, LINES & PLANES
Principles of projection; Projections of points; Projection of straight lines - Inclined to one plane, inclined to both planes; Projection of planes - Inclined to one plane.

PRACTICES:

- Construction of polygons using different methods (i.e. ARC, Angle, General).
- Inscribe a regular hexagon & pentagon in a circle of the given diameter.
- Tracing of conical curves (Ellipse, Parabola, Hyperbola) by using General Method.
- Draw the projections of the points situated in all the 4 quadrants.
- Draw the projections of a line when it is inclined to one plane (HP or VP).
- Draw the projections of a line when it is inclined to both the planes (HP &VP).
- Draw the projections of a plane when it is inclined to one plane (HP or VP).

MODULE-2

16L+0T+16P=32 Hours

UNIT-1

PROJECTIONS OF SOLIDS
Projection of solids axis inclined to one reference plane - Prisms, pyramids, Cylinder and cone.

DEVELOPMENT OF SURFACES
Development of lateral surfaces of simple solids - Prisms, Pyramids, Cylinder and cone.

UNIT-2

ORTHOGRAPHIC VIEWS

Conversion of pictorial views into orthographic views.

Drafting Using Computer Package: Introduction to 2D modelling software - AutoCAD; Conversion of Isometric view into Orthographic views of simple castings; Conversion of Orthographic views into Isometric view of simple solids - Prisms, Pyramids, Cylinders and cones.

PRACTICES:

- Draw the projections of Prisms, when they are inclined to one reference plane (HP or VP).
- Draw the projections of Pyramids, when they are inclined to one reference plane (HP or VP).
- Draw the projections of cylinder & cone, when they are inclined to one reference plane (HP or VP).
- Draw the complete surface development of prisms & pyramids with the given dimensions.
- Draw the complete surface development of cylinder & cone with the given dimensions.
- Draw the orthographic view's (i. e. front view, top view, and side view) of the given pictorial view of the sketches by using AutoCAD.
- Draw the Isometric view of simple solids (Prisms & Pyramids) by using AutoCAD.
- Draw the Isometric view of simple solids (Cylinder & Cone) by using AutoCAD.

SKILLS:

- Convert isometric views of objects into orthographic views and vice versa.
- Visualize the shape of the 3D components.
- Create pictorial views by using AutoCAD
- Know projections by visualization.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Communicate the technical ideas in the form of drawings.	Apply	1	1,2,3,5
2	Apply the drawing skills in representing various geometrical features.	Apply	1	1,2,3,5
3	Develop orthographic projections and isometric views of various objects.	Apply	1	1,2,3,5
4	Estimate the lateral surface area of regular geometrical solids.	Analyze		1,2,3,5
5	Sketch simple objects and their pictorial views using AutoCAD.	Analyze	2	1,2,3,5

TEXT BOOKS:

1. J Hole, “Engineering Drawing”, Tata McGraw-Hill, 2nd Edition, 2019.
2. N D Bhatt, “Engineering Drawing”, Charotar Publication, 53rd Edition, 2014.

REFERENCE BOOKS:

1. Basant Agrawal and C.M. Agrawal “Engineering Drawing”, Tata Mc Graw- Hill, 2nd Edition 2018.
2. K L Narayana, “Engineering drawing”, SciTech Publications, 3rd Edition, 2011.
3. Colin H. Simmons, Dennis E. Maguire, Manual of Engineering Drawing, 2nd Edition, 2003.

24CY101 - IT TOOLS AND CYBER SECURITY

Hours per week:

L	T	P	C
0	2	2	2

PREREQUISITE KNOWLEDGE: Nil

(OBJECTIVE:) COURSE DESCRIPTION AND OBJECTIVES.

The coursework will introduce cyber security concepts and provide awareness about cybercrimes and preventive measures and safe practices to be followed. quantum physics concepts and their applications in quantum computing. It also familiarizes students with the to learn and practice the usage of software tools such as word, excel, ppt and LaTex, text and image editors.

MODULE-1

0L+16T+16P=16 hours

UNIT-I

Practicing exercises using software tools:

- Write a passage in a Word document. Give the passage a title, then format it according to the guidelines below:
  - Add the time and date, and make sure the title is bold, italicized, and underlined.
  - Increase the font size, style, and line spacing by two-fold.
  - Set the margins to 1.75 on the right and 1.5 on the left.
  - Set a border for the passage.
- Create a timetable using Table format of Word for your programme
- Preparation of resume
- Preparation of calendar for the current month
- Create an invitation for your department function or college day using the text box and other options
- Create a call letter for an interview using mail merge.
- Create value-added webpages about course curriculum using Hyperlink, styles, layout etc.
- In Excel, create a student mark statement and determine the total, average, and percentage of marks
- Create a payroll system in Excel for 5 employees given Basic salary, 40% for D.A;12% for HRA; 10% for TA; 10% for CCA; 12% for PF; IT 10%. Calculate the gross and net salary
- Create a pivot table to analyse your worksheet data.

UNIT-II

Practicing exercises using software tools:

- Prepare a presentation on your university using PowerPoint.
- Prepare a presentation on launch of a product.
- Prepare a brochure of a value added course
- Create a hierarchy chart of your organization
- Design a Magazine cover.
- Design a poster inviting all students of your department Festival.
- Installation and demonstration of LaTeX.
- Prepare professional pdf documents using LaTeX.
- Prepare LaTex document containing mathematical equations, tables and images.

MODULE-2

0L+16T+16P=16 hours

UNIT-I

Introduction to Cyber security and Cybercrimes: Defining Cyberspace, Issues and challenges of cyber security, Cyber Hygiene, Classification of cybercrimes, Common cybercrimes targeting computers and mobiles, cyber-crime against women and children, social engineering attacks, reporting of cyber-crimes, Remedial and mitigation measures, Legal perspective of cyber-crime, IT Act 2000.

Social Media Overview and Security: Types of Social media platforms, Social media privacy, Challenges, opportunities and pitfalls in online social network, Security issues related to social media, Best practices for the use of Social media.

UNIT-II

E-Commerce and Digital Payments: Main components of E-Commerce, Elements of E-Commerce security, E-Commerce threats, E-Commerce security best practices. Modes of digital payments, Digital payments related common frauds and preventive measures.

Device Security: End Point device and Mobile phone security, Password policy, Security patch management, Importance of regular data backups, Downloading and management of third party software, Device security policy, Significance of host firewall and Ant-virus, Management of host firewall and Anti-virus, Wi-Fi security, Configuration of basic security policy and permissions.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Create word documents, presentations and spread sheets by applying various tools.	Apply	1	2, 5
2	Usage of latex tool for report writing	Apply	2	5, 6
3	Apply the concept of Cyber security, issues and challenges associated with it, cyber-crimes, their nature, legal remedies and the process of reporting crimes and best practices to be taken up in the usage of social media.	Apply	1	6, 8
4	Apply the basic concepts related to E-Commerce, digital payments, RBI guidelines and preventive measures against digital payment frauds.	Apply	2	6, 12
5	Use basic tools and technologies to protect their devices.	Apply	2	6

SKILLS:

- Integration of various components of a computer system.
- Trouble shooting of components of a computer system.
- Installation of OS and its various tools/ applications.
- Usage of IT tools such as MSWord, LaTex etc.
- Creating the documents using MS-Word and LaTex.
- Analysing and visualizing data with excel.
- Developing vari- ous power point presentations.

TEXT BOOKS:

1. Fundamentals of Computers by Reema Thareja, Oxford University Press 2nd edition 2019, India.
2. Lamport, Leslie (1994). LaTeX: A Document Preparation System, User's Guide and Reference Manual (2nd ed.). Pearson
2. Cyber Security Understanding Cyber Crimes, Computer Forensics and Legal Perspectives by Sumit Belapure and Nina Godbole, Wiley India Pvt. Ltd.

REFERENCE BOOKS:

1. Information Warfare and Security by Dorothy F. Denning, Addison Wesley.
2. John Walkenbach, Herb Tyson, Michael R.Groh and FaitheWempen, "Microsoft Office 2010 Bible", Wiley.

24CS101-PROGRAMMING IN C

Hours per week:

L	T	P	C
2	0	4	4

PREREQUISITE KNOWLEDGE: Fundamentals of Problem Solving

COURSE DESCRIPTION AND OBJECTIVES:

This course is aimed to impart knowledge on basic concepts of C programming language and problem solving through programming. It covers basic structure of C program, data types, operators, decision making statements, loops, functions, strings, pointers, structures and unions. At the end of this course, students will be able to design, implement, test and debug complex problems using features of C.

MODULE- 1

16L+0T+32P=48 Hours

UNIT-1

INTRODUCTION TO ALGORITHMS AND PROGRAMMING LANGUAGES

Introduction to Algorithms: Introduction to computational thinking – Problem solving with algorithms and Flowcharts-Introduction to C: Structure of a C program - pre-processor statement, inline comments, variable declaration statements, executable statements; C Tokens - C character set, identifiers and keywords, type qualifiers, type modifiers, variables, constants, punctuations and operators.

Data Types and Operators: Basic data types; Formatted I/O; Reading and writing characters; Operators - assignment, arithmetic, relational, logical, bitwise, ternary, address, indirection, size of, dot, arrow, parentheses operators; Expressions - operator precedence, associative rules.

Control Statements: Introduction to category of control statements; Conditional branching statements - if, if– else, nested-if, if – else ladder, switch case;

Iterative statements - for, while, do - while, nested loops; **Jump statements** - break, jump, go to and continue.

UNIT-2

FUNCTIONS& POINTERS

User-defined functions: Function declaration - definition, header of a function, body of a function, function invocation; Call by value, Call by address. Recursion; Library Functions; Storage classes; Scope of a variable.

Pointers: Declaration, Initialization, Multiple indirection, Pointer arithmetic.

PRACTICES:

Questions on Data Handling

- Write a program to accept a number as input from the user which denotes the temperature in Celsius, convert it to Fahrenheit reading and print it
- Write a C program to find maximum between two numbers using conditional operator.
- Write a C program to find maximum between three numbers using conditional operator.

Questions on Decision Making

- Write a program to accept two numbers as input and print the greater of the 2 numbers
- Write a program to accept 3 numbers as input and print the maximum of the 3
- Write a program to accept a number as input and check if it is positive, negative or zero.
- Write a C program to check whether a year is leap year or not.
- Write a C program to find all roots of a quadratic equation.
- Write a C program to input marks of five subjects Physics, Chemistry, Biology, Mathematics and Computer. Calculate percentage and grade according to following:

  Percentage >= 90% : Grade A

  Percentage >= 80% : Grade B

  Percentage >= 70% : Grade C

  Percentage >= 60% : Grade D

  Percentage >= 40% : Grade E

  Percentage < 40% : Grade F

Questions on Loops

- Write a program to accept a number as input print the sum of its digits
- Write a program to accept a number as input, reverse the number and print it
- Write a program to accept a number as input and check if it is an Armstrong number
- Write a program to accept a number as input and check if the prime number or not.
- Write a program to accept 2 numbers as input and find their LCM
- Write a program to accept a number as input and check if it is a palindrome
- Write a program to accept 2 numbers as input and print their HCF
- Write a program to accept a number as input and check if a number is strong number or not.
- Write a program to generate prime numbers between two intervals given as input
- Write a program to accept a number N as input from the user and print the following pattern. Sample N = 5.

  1

  12

  123

  1234

  12345

- Write a program to accept a number N as input from the user and print the following pattern. Sample N = 5.

  A

  AB

  ABC

  ABCD

  ABCDE

Questions on switch statement

- Write a C program print total number of days in a month using switch case.
- Write a C program to create Simple Calculator using switch case.

Questions on functions

- Write a C program to check whether a number is prime number or not using a function.
- Write a C program to check whether a number is an Armstrong number using a function.
- Write a C program to check whether a number is a perfect number using a function.
- Write a C program to swap two numbers using pointers.
- Write a program that finds and returns the sum, largest and average of three integer numbers using a function and pointers.
- Write a C program to find factorial of any number using recursion.
- Write a C program to generate nth Fibonacci term using recursion.
- Write a C program to find GCD (HCF) of two numbers using recursion.

MODULE-2

16L+0T+32P=48 Hours

UNIT-1

ARRAYS & STRINGS

Arrays: Introduction; Types of arrays; Single dimensional array - declaration, initialization, usage, reading, writing, accessing, memory representation, operations; Multidimensional arrays. Passing arrays to functions; Relationship between arrays and pointers, Scaling up - array of arrays, array of pointers, pointer to a pointer and pointer to an array; Dynamic memory allocation functions – Creating arrays dynamically and performing operations on them.

Strings: Character array, Reading strings from the standard input device, displaying strings on the standard output device, Importance of terminating a string, Standard string library functions.

UNIT-2

STRUCTURES, UNIONS &FILES

Structures: Defining a structure, declaring structure variable, Operations on structures, Pointers to structure - declaring pointer to a structure, accessing structure members using pointer; Array of structures, Nested structures, passing structures to functions - passing each member of a structure as a separate argument, passing structure variable by value, passing structure variable by reference/ address; Typedef and structures.

Unions: Defining a union - declaring union variable, operations on union;

Files: Introduction to files, Streams, I/O using streams – opening a stream, closing stream; Character input, Character output, File position indicator, End of file and errors, Line input and line output, Formatted I/O, Block input and output.

PRACTICES:

Questions on 1-D Arrays

- Print the contents of an array from the left to the right
- Print the contents of an array from the right to the left
- Find the maximum and minimum elements in an unsorted array

- Sort a given list of numbers in ascending order.
- Replace every element in an array with the sum of its every other element
- Reverse the elements of an array (in place replacement)
- Write a program to find the second largest and smallest elements in an unsorted array

Questions on Strings

- Implement the string length function
- Implement the string copy function
- Implement the string concatenate function
- Implement the string compare function
- Implement the count words function
- Implement the string reverse function
- Write a program to accept a string as input and check if it is a palindrome
- Implement the strstr function.
- Write a program to print the character in the string with the count where count is the occurrence of the character
- Write a program to remove the duplicate characters in the given string

Questions on 2-D Arrays

- Print the contents of a 2D array in a zig-zag order
- Print the contents of a 2D array diagonal-wise
- Print the contents of a 2D array in the lower triangular order
- Find and print the maximum and minimum elements of each row of a matrix
- Write a program to find the sum of two matrices.
- Write a program that finds the product of two matrices.
- Write a program to find the transpose of a given matrix.

Questions on Files, Structures & Unions:

- Ramesh wants to do addition on complex numbers. He did it with regular practice but Charan asked him to do with the help of structures by following below Criteria.
  Write a C program that defines a structure named 'Complex' consisting of two floating point members called "real and imaginary". Let c1 and c2 are two Complex variables; compute the sum of two variables.

  Example:
  c1= 2 8
  c2= 6 4
  Sum= 8.000000+12.000000i
- Write a C program to accept details of 'n' employee (eno, ename, salary) and display the details of employee having highest salary. Use array of structure.
- Write a C program that creates a student file containing {Roll No, Student Name, Address, Stream}, where the data will be inserted and display the list of students who are in CSE (Stream=CSE).

  Input: A file name
  Output: The attributes such as Roll_No, Student_Name, Stream, Address.
  Sample Input
  201fa4200 Raja CSE Guntur
  201fa4201 Bala IT Tenali
  Sample Output

201fa4200 Raja CSE Guntur

- Write a C program that reads content from an existing text file and write the same in a new file by changing all lowercase alphabetic character to upper case. (Existing file may contain digit and special characters).

  Example:
  Input: Enter the file name.
  Output: New file with updated content

- Write a C program to count the occurrences of the given string in a file.

  Example:
  Input: Enter the File name to read the string to be counted.
  Output: Display the count of occurrences of the string.

SKILLS:

- Analysis of the problem to be solved.
- Select static or dynamic data structures for a given problem and manipulation of data items.
- Application of various file operations effectively in solving real world problems.
- Develop C programs that are understandable, debug gable, maintainable and more likely to work correctly in the first attempt

COURSE OUTCOMES:

Upon completion of the course, the student will be able to achieve the following outcomes:

No.	Course Outcome	Blooms Level	Module No.	Mapping with POs
1	Identify suitable data type for operands and design of expressions having right precedence	Apply	1,2	1
2	Apply decision making and iterative features of C Programming language effectively	Apply	1,2	1
3	Select problem specific data structures and suitable accessing methods	Analyze	1,2	1,2
4	Design and develop non- recursive and recursive functions and their usage to build large modular programs and also able to design string manipulation functions	Create	1,2	3
5	Develop C programs that are understandable, debuggable, maintainable and more likely to work correctly in the first attempt	Evaluate	1,2	3,4

TEXT BOOKS:

1. Behrouz A. Forouzan, Richard F.Gilberg, "Programming for Problem Solving", 1st edition, Cengage publications, 2019.
2. Ajay Mittal, "Programming in C - A Practical Approach", 1st edition, Pearson Education, India, 2010.

REFERENCE BOOKS:

1. Reema Thareja, "Computer Fundamentals and Programming in C", 1st edition, Oxford University Press, India, 2013.
2. Herbert Schildt, "C: The Complete Reference", 4th edition, Tata McGraw-Hill, 2017.
3. Byron S Gottfried, "Programming with C", 4th edition, Tata McGraw-Hill, 2018.

24CS102– PROBLEM SOLVING THROUGH PYTHON

Hours per week

L	T	P	C
2	0	2	3

PREREQUISITE KNOWLEDGE: Prior knowledge of any programming language and object-orientedconcepts is helpful but not mandatory.

COURSE DESCRIPTION AND OBJECTIVES:

This course offers sufficient knowledge required to understand the fundamental concepts of Python programming language. This course enables students to choose appropriate data structures (lists, dictionaries, tuples, sets, strings) for the given problem. In addition, the students will be able to createreliable, modular and reusable applications using Object- Oriented Programming approaches. At the end they will get an idea of how to access database using python programming, develop web applications, and using web Services using python Programming.

MODULE-1

16L+0T+16P=32 Hours

UNIT-1

INTRODUCTION

Introduction to python, Variables, Assignment, Keywords, Built-in functions, Indentation, Comments, Basic data types - integers, float, complex, Boolean, strings; Python program development, running python using REPL shell, Python scripts.

Operators and Expressions: Operators- arithmetic operators, comparison (relational) operators, assignment operators, logical operators, bitwise operators, membership operators, identity operators;Expressions and order of evaluations.

Control Structures: Conditional control structures - if, else if, else; Loop control structures - for, while, for... else, while…else, nested loops, break, continue, pass.

UNIT-2

PYTHON DATA STRUCTURES AND FUNCTIONS

Data Structures: Lists, Tuples, Sets, Strings, Dictionary - creation, accessing, basic operators and methods; List comprehensions.

Functions: Defining functions, calling functions, Passing arguments - keyword arguments, default arguments, positional arguments, variable-length arguments; Types of functions- anonymous functions,fruitful function, recursive functions; Scope of the variables- global and local variables, Development ofrecursive and non-recursive functions.

PRACTICES:

- Given two integers N1 and N2, write a program to find their product without using multiplication (‘*’) operator and loops.
- Given two integers N1 and N2 having same value, write a program to check whether N1 and N2 points to the same object or not.
- Given an Integer N, write a program to check whether given number is even or odd

without using modulus operator.

- Given coordinates of centre of a circle, radius and a point coordinate, write a program to check whether the given point lies inside or on the circle, or outside the circle.
- Print the Nth prime number. you need to find the Nth prime number, where n is given by user
  prime numbers: 2, 3, 5, 7, 11, 13 ...
  if n = 3 then prime = 5
  Sample Input: 3
  Sample Output: 5
- Write a program to find the sum of digits in a given integer.
- Given an integer N as an input, decides the geometrical figure for which the area has to be calculated, for example N=1 for circle, N=2 for rectangle, and N=3 for triangle. Write a program to display the area of the respective figure.
- Given an integer amount X, write a program to find the minimum number of currency notes $ (500, 100, 50, 20, 10, 5, 2, 1) required for the given amount.
  Input:
  575
  Where input is the amount for which we have to calculate the number of currency notes.
  Output: 4
  Explanation: Total amount = 1(500-dollar note) + 1(50-dollar note) + 1(20-dollar note) + 1(5-dollar note) = 575, hence the minimum number of notes required is 4.
- Given a list L of N numbers (integers), Write a program to find the sum of the elements of given list L with the corresponding elements of the reverse of list L. If list L has elements [1,2,3], then reverse of the list L will be [3,2,1] and the resultant list should be [4,4,4].
- Given a positive integer number n. Write a program to generates a dictionary d which contains (i, i*i*i) such that i is the key and i*i*i is its value, where i is from 1 to n (both included). Print the content of the dictionary d.
- Write a program to create a data structure to store student information such as regd no, name, percentage of marks, phone number and display the student details based on the order of percentage of marks.
- For a Given a string, design and implement functions to perform the following:
  a) remove vowels in the given string.
  b) count number of uppercase and lowercase letters.
  c) remove all special characters.
  d) check whether it is a palindrome or not.
  e) swap case of each letter.
- Create a function that receives 3 numbers and returns the median, i.e. the number that is not the min and not the max, but the one in between.
- Given two lists of integer numbers, write a function to perform the following operations.
  a. print elements that are common in both the lists. (Print without duplicates).
  b. print elements that are present in the first list and not present in second list.
  c. print elements that contain the first element of the first list and last element of the second list.
  d. print largest number of both the lists.
  e. print smallest number of both the lists.

- Create a function to find Common Elements in Two Lists Using Sets
- Implement a program to calculate GCD using recursive functions.
- Given two non-negative integers num1 and num2 represented as strings, return the product of num1 and num2, also represented as a string.
  Input: num1 = "2", num2 = "3"
  Output: "6"
- Create a Python function that rotates a list by a specified number of positions to the right. For instance, if the list is [1, 2, 3, 4, 5] and the rotation count is 2, the resulting list should be [4, 5, 1, 2, 3].
  Example:

  Input: [1, 2, 3, 4, 5], 2
  Output: [4, 5, 1, 2, 3]

  Input: [1, 2, 3, 4, 5], -1
  Output: [2, 3, 4, 5, 1]

  Input: ['a', 'b', 'c', 'd'], 3
  Output: ['b', 'c', 'd', 'a']

MODULE-2

16L+0T+16P=32 Hours

UNIT-1

MODULES, PACKAGES AND OOPS

Creating modules, Import Statement, From...Import Statement, Name Spacing, Creating user definedmodules.

Standard Modules: sys, math, date, time, os, random and itertools modules.

Packages: Package Initialization, Importing * From a Package, Sub packages.
Object Oriented Programming: Classes and Objects, Attributes and Methods, Constructors and Destructors- (__init__and__del__methods), Encapsulation - bundling of data and methods within a class, promotes data integrity and prevents unauthorized access, use of access modifiers to control access to attributes and methods; Inheritance, Method Overriding, Polymorphism, Abstraction, Class and Instance Variables - Differentiate between class variables (shared among all instances) and instance variables (unique to each instance), Access Modifiers-access modifiers and their role in controlling access to class members. **Operator Overloading:** customization of operators for user- defined classes- define methods such as add,_sub,_mul_, etc.

UNIT-2

ERRORS AND EXCEPTIONS AND FILES

Introduction to Exceptions, Handling Exception, Try Except Else and Finally Block, Raising Exceptions.

File Processing: Reading and writing files -creating a new file, writing to a file, reading text files, openingand closing files, reading, writing, tell (), seek (), rename ().

PRACTICES:

- Program that defines a function large in a module which will be used to find larger of two values and called from code in another module
- Write a program to find the number of lines, words and characters in the file.
- Write a program to append the data to an already existing file.
- Write a program that reads a file line by line. Each line read from the file is copied to another file with line numbers specified at the beginning of the line.
- Define the following functions that are more robust to erroneous input data
  a) To divide two numbers (To handle Zero Division Error).
  b) To read two integer numbers and display them (To handle Value Error).
  c) To display elements of a list (To handle Index Error).
  d) To open a file and display file contents (To handle File Not Found Error).
- Write a python program to handle multiple exceptions using raise keyword.
- Write a program that uses class to store the name and marks of students. Use list to store the marks in 3 subjects
- Write a program with class Employee that keeps a track of the number of employees in an organization and also stores and display their name, designation and salary details.
- Write a program that has classes such as Student, Course and Department. Enroll a student in a course of a particular department
- Write a program that has an abstract class Polygon. Derive two classes Rectangle and Triangle from Polygon and write methods to get the details of their dimensions and hence calculate the area.
- ● Write a Program that overloads the + operator on a class Student that has attributes name and marks
- Write a Program to overload the - operator to subtract two Distance objects.
- Write a program to overload the * operator to multiply two complex numbers

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Analyze various features of programming language and their application in problem solving in computer programming to write, compile, and debug programs in python language.	Analyze	1	1, 2
2	Make use of different data types to design programs involving decisions, loops, and functions.	Apply	1	1, 2, 5
3	Analyze the usage of different data structures for practical and contemporary applications for a given problem.	Analyze	1	1, 2, 3, 5
4	Choose appropriate programming paradigms, interrupt and handle exceptions using files to propose solution through reusable modules	Apply	2	1, 2, 3,5
5	Develop applications for a range of problems using object-orientedprogramming techniques.	Apply	2	1, 2,3,4,5

TEXT BOOKS:

1. Kenneth A. Lambert, "The Fundamentals of Python: First Programs", Cengage Learning, 2011.
2. Mark Lutz, "Learning Python", 5th edition, Orielly Publishers, 2013.
3. Reema Thareja, 'Python Programming using problem solving approach", oxford university press,2017

REFERENCE BOOKS:

1. Introduction to Computation and Programming Using Python. John V. Guttag, The MIT Press.
2. James Payne, "Beginning Python using Python 2.6 and Python 3", Wrox publishing.
3. Paul Gries, "Practical Programming: An Introduction to Computer Science using Python 3",The Pragmatic Bookshelf, 2nd edition, 4 Oct. 2013.
4. Allen B. Downey, "Think Python", 1st edition, Orielly publishing.

24EN101 – ENGLISH PROFICIENCY AND COMMUNICATION SKILLS

Hours per week

L	T	P	C
0	0	2	1

PREREQUISITE KNOWLEDGE: Basics of grammar, Read and understand for global context, Cultural sensitivity and Basic writing skills.

COURSE DESCRIPTION AND OBJECTIVES:

English Proficiency and Communication Skills seeks to develop the students' abilities in grammar, speaking, reading, writing and overall comprehension skills. The course will provide students an exposure on a wide range of language use in everyday situations. It will make the students to equip with functional English and make them use it confidently in their professional and social contexts. Finally,
students will strengthen their reading, writing, listening and speaking skills in English

MODULE-1

0L+0T+16P=16 Hours

UNIT-1

MY LIFE AND HOME – MAKING CHOICES – HAVING FUN

Reading: Understanding main message, factual information global meaning, specific information and paraphrasing.
Writing: Developing hints based mail, Writing short messages/paragraphs.
Listening: Understanding short monologues or dialogues and choose the correct visual.
Speaking: Express simple opinions /cultural matters in a limited way.
Vocabulary: Discerning use of right word suiting the context, B1 Preliminary word list.
Grammar: Frequency Adverbs, State Verbs, AFV and Prepositions.

UNIT-2

ON HOLIDAY - DIFFERENT FEELINGS – THAT'S ENTERTAINMENT!

Reading: Longer text for detailed comprehension, gist and inference.
Writing: Developing notes and responding to penfriends or 'e-pals'.
Listening: Understand straightforward instructions or public announcements.
Speaking: Describing people, things and places in a photograph.
Vocabulary/Grammar: Comparatives and Superlatives, Gradable and non-gradable adjectives, Cloze tests.

PRACTICES:

- Developing hints based mail.
- Writing short message.
- Writing paragraphs.
- Expressing opinions and cultural matters.
- Understanding short monologues.
- Understanding straightforward instructions and public announcements.
- Describing people, things and places in a photograph.

MODULE-2

0L+0T+16P=16 Hours

UNIT-1

GETTING AROUND – INFLUENCES - STAY FIT AND HEALTHY

Reading: Reading for understanding coherence of the text and drawing inferences.
Writing: Reading an announcement from a magazine or website for preparing an article.
Listening: Discussion activities and listening to understand the gist of each short dialogue.
Speaking: Snap Talks, Make and respond to suggestions, discuss alternatives and negotiate agreement.
Vocabulary / Grammar: Punctuation, Prepositions, Phrasal Verbs, B1 Preliminary word list.

UNIT-2

LOOKS AMAZING! – THE NATURAL WORLD – EXPRESS YOURSELF!

Reading: Content, Communicative Achievement, Organisation and Language.
Writing: Developing a story with clear links to the given opening sentence.
Listening: An interview for a detailed understanding of meaning and to identify attitudes
and opinions.
Speaking: Discuss likes, dislikes, experiences, opinions, habits, etc.
Vocabulary / Grammar: Modals, Conditionals, Verb forms (Time and Tense).

PRACTICES:

- Listening to understand the gist of each short dialogue.
- Listening to an interview for a detailed understanding of meaning and to identify attitudes and opinions.
- Preparing an article.
- Discuss for alternatives and negotiate agreement.
- Discussion on likes, dislikes, experiences, opinions, habits, etc.

SKILLS:

- Use of appropriate grammar and vocabulary with syntactic patterns in short texts.
- Read and extract the main message, global meaning, specific information, detailed comprehension, understanding of attitude, opinion and writer purpose and inference.
- Listen to understand key information, specific information, gist and detailed meaning and to interpret meaning.
- Understand questions and make appropriate responses and talk freely on everyday topics.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Apply to read and grasp content on a range of topics/texts related to their everyday life like notifications, advertisements, travel brochures, news reports, articles.	Apply	1	7, 8, 9, 10, 12
2	Apply suitable strategies to achieve comprehension, like listening for main points and checking comprehension using contextual clues etc.	Apply	1	7, 8, 9, 10, 12
3	Demonstrate vocabulary beyond that of the familiar subjects.	Analyze	1, 2	7, 8, 9, 10, 12
4	Show sufficient control of English grammar and sentence variety to coherently organise information at sentence and discourse levels.	Evaluate	2	7, 8, 9, 10, 12
5	Use functional English to communicate and interact effectively in everyday situations.	Create	2	7, 8, 9, 10, 12

TEXT BOOKS:

1. Emma Heyderman and Peter May, "Complete Preliminary", Student's Book with Answers, 2nd edition, Cambridge University Press, 2019.

REFERENCE BOOKS:

1. Annette Capel and Rosemary Nixon, "Introduction to PET", Oxford University Press, 2009.
2. Adrian Doff and Craig Thaine, "Empower Pre intermediate", Cambridge University Press, 2015.
3. Louise Hashemi and Barbara Thomas, "Objective PET", Cambridge University Press, 2010.

24EN102 – TECHNICAL ENGLISH COMMUNICATION

Hours per week

L	T	P	C
2	0	2	3

PREREQUISITE KNOWLEDGE: Basic sentence formation, understanding contextual meanings, basic writing skills and moderate fluency in English.

COURSE DESCRIPTION AND OBJECTIVES:

In this course students will read, analyze, and interpret material from technical and general fields, and practice reading, writing, listening and speaking skills to gain exposure and functional English on a variety of contemporary topics. The overall course objective is to provide English for Specific Purposes(ESP) instruction to enhance students' reading, writing, listening and speaking skills through a practice in the language. It will aim to build students' confidence and motivation through exposure to academic skills like Note making/taking, Paraphrasing, Summarizing, Report Writing, Making Presentations etc., so as to generate interest in the language from an ESP perspective. Finally, students are expected through the course to gain key strategies and expression for communicating with professionals and non-specialists.

MODULE-1

16L+0T+16P=32 Hours

UNIT - 1

GENETICS

Reading: Reading for **Note Making** Sub skills: Reading for global understanding (skimming), specific information (scanning), understanding main ideas and supporting ideas, guessing contextual meanings from the text. -Vocabulary building: commonly used roots, prefixes, and suffixes.

Writing: Note making, organising main points and sub points, numbering and sequencing, suggesting titles, paraphrasing and summarising. **Functional grammar:** Common Errors in Articles and Prepositions (Handout).

Listening: Listening for **Note Taking**: top down and bottom up approach, listening for main ideas and supporting points.

Speaking: Presentation in teams - ideas on the topic summarised, making a PPT, effective introductions and conclusions, logical organisation of content, using appropriate structure and cohesive devices.

UNIT - 2

NEW-AGE ENTREPRENEURS

Reading: Reading: predicting, skimming, scanning, reading for inference, extrapolative reading **Vocabulary building:** Academic vocabulary from the text: synonyms, antonyms, Words often confused.

Writing: Paragraph writing; writing a topic sentence, supporting sentences, effective introductions and conclusions, use of cohesive devices. Types of Paragraphs: Descriptive, narrative, argumentative and expository. **Functional grammar**: Common Errors in Verb forms and Conditional sentences (Handout) Model verbs.

Listening: Listening for identifying parts from a description, listening to and sorting information, listening for specific information.

Speaking: Narrating/Retelling an incident, using suitable cohesive devices/discourse markers Speaking of past and present habits/ activities/events - Speaking of future plans.

PRACTICES:

- Note making.
- Summarizing.
- Paragraph Writing.
- Reading Comprehension
- Error Correction and Restructuring.
- Vocabulary building.
- Process Description.
- Tenses.

MODULE-2

16L+0T+16P=32 Hours

UNIT - 1

SOCIAL MEDIA

Reading: Reading for factual information researching for supporting evidence - skimming, scanning, **Vocabulary building:** One-word substitutes.

Writing: Letter Writing - E-mail writing – New age communication – Format, protocol, and style- WhatsApp, Facebook and Twitter **Functional grammar:** Common Errors in Sub-Verb Agreement and Modals.

Listening: Listening to a **Business Presentation**: Listening for deducing information, for abstract details and specific details, listening for taking a message.

Speaking: Making a presentation with a PPT on a topic assigned- organising the presentation using appropriate discourse markers - presenting a point of view - Extempore.

UNIT - 2

HEALTH AND NUTRITION

Reading: Reading for data interpretation and information transfer from graphical aids to text reports (pictograms. tables, graphs, pie charts, flow charts), deducing specific information and general information -**Vocabulary building:** business vocabulary, collocations**,** idioms and phrasal verbs.

Writing: Writing a **Report**: Drafting general and factual reports - **w**riting an overview - an effective introduction - organising information into paragraphs (Stages of writing: planning /organising /writing /editing /rewriting) - **Functional grammar:** transformations and miscellaneous common errors.

Listening: Listening to a Ted talk and sorting information – taking notes from a discussion.

Speaking: Group Discussion – prerequisites -generating content - initiating a discussion - expressing one's opinion ~ leading a discussion - agreeing/ disagreeing to someone's view - cutting into a speech - body language and voice modulation.

PRACTICES:

- E-mail writing.
- Letter writing.
- Report writing.
- Process Description
- Data Interpretation
- Messaging in Social media.
- Writing Reviews.
- Group Discussion

SKILLS:

- Apply different subskills like skimming, scanning, reading for information, reading for inference etc. to understand different kinds of text.
- Apply different sub skills like top down, bottom up approaches to listening.
- Use functional vocabulary relevant to engineering and technology to express ideas lucidly.
- Use appropriate sentence structure, cohesive devices to construct simple text in regular correspondence like e-mails and letters.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Apply a variety of strategies to interpret and comprehend spoken texts/ discourse using contextual clues.	Apply	1	6, 7, 8, 9, 10, 12
2	Apply appropriate reading strategies to interpret content / material related to engineering and technology domain.	Apply	1	6, 7, 8, 9, 10, 12
3	Participate in discussions and make short presentations on general and technical topics.	Analyze	1, 2	6, 7, 8, 9, 10, 12
4	Possess an ability to write clearly on topics related to technology and workplace communication.	Evaluate	2	6, 7, 8, 9, 10, 12
5	Choose functional language, grammar structures, cohesive devices and skills of organisation to express clearly in speaking.	Create	2	6, 7, 8, 9, 10, 12

TEXT BOOKS:

1. N P Sudharshana & C Savitha, "English for Technical Communication", Cambridge University Press, 2016.

REFERENCE BOOKS:

1. Balasubramanian T, "A Text book of Phonetics for Indian Students", Orient Longman, New Delhi, 1989.
2. Krishnaswamy, N and Sriraman, T, "Current English for Colleges", Trinity publications, 2016.
3. Mohan Krishna and Meera Banerjee, "Developing Communication Skills", Macmillan India Ltd. New Delhi, 1990.
4. Ashraf Rizvi M, "Effective Technical Communication", 2nd Edition, McGraw Hill Education, 2017.
5. Narayanaswamy V R, "Strengthen your Writing", Third Edition Orient Black Swan, New Delhi, 2005.
6. Naterop, Jean, B. and Rod Revell, "Telephoning in English", 3rd Edition Cambridge University Press, Cambridge, 2004.

24MS101 MANAGEMENT SCIENCE

L	T	P	C
2	2	0	3

PREREQUISITE KNOWLEDGE: Basic knowledge on the concepts of Management.

COURSE DESCRIPTION & OBJECTIVES: The objective of this course is to examine the ramifications of managerial practices within the field of engineering and Students will understand the significance of operation management and develop the ability to analyse the markets, customers, competitors, and accordingly plan HR function effectively.

MODULE- 1

16L+16T+0P=32Hours

UNIT-1

INTRODUCTION TO MANAGEMENT

Concepts of Management and organization - nature, importance and Functions of Management, Evolution of Management Thought - Taylor's Scientific Management Theory, Fayol's Principles of Management, Administrative Management Theory, Mayo's Hawthorne Experiments, Maslow's Theory of Human Needs, Herzberg's Theory, Leadership Styles, Corporate Social Responsibility in Indian perspectives.

UNIT-2

OPERATIONS MANAGEMENT

Principles of plant location, Principles and types of plant layout, Operations and productivity, Project Management - Project Planning, Project Scheduling and Project Controlling, Need for Inventory control, EOQ, ABC Analysis, Purchase Procedure and Stores Management. Statistical process Control: Control Charts for variables – Setting Mean Charts Limits (X Charts) and Setting Range Chart Limits (R Charts). Control Charts for attributes- P-charts and C-charts.

PRACTICES:

- Design a comprehensive management strategy for a startup company, incorporating elements of planning, organizing, leading, and controlling.
- Develop a quality management plan for a software development project, outlining procedures for testing and validation with a view of project management procedure.
- Critically evaluate the effectiveness of different styles of leadership in crisis situations versus during times of stability with suitable examples from different organisations.
- Determining the various statistical controlling methods such as, X and R Charts for a specific company with the data.

MODULE- 2

16L+16T+0P=32Hours

UNIT- 1

HUMAN RESOURCES MANAGEMENT

Concepts of Human Resource Management, Basic functions of HR Manager - Manpower planning, Recruitment, Selection, Training and Development, Placement, Wage and Salary Administration, Promotion, Transfer, Separation, Performance Appraisal, Grievance Handling and Welfare Administration, Job Evaluation and Merit Rating.

UNIT- 2

MARKETING MANAGEMENT

Evolution of Marketing, Functions of Marketing Selling Vs Marketing, 4 P's of Marketing – Product Mix - Product Life Cycle, Place Mix – Channels of Distribution, Price Mix – Pricing Methods, Promotion Mix – Tools of Promotions. 7p's of marketing - Product, Price, Promotion, Place, People, Packaging, and Process.

PRACTICES:

- Select any Designation in an organization and describe its job description and job specifications.
- Develop a procedure on how to deal with grievances at your work with suitable example.
- Analyze marketing mix competencies for product as well as services.
- Analyze the Product Mix strategies of a select company with appropriate examples.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to

Co No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Analyze the nature and consequence of management.	Analyze	1	1,2,4,6,8,9, 11,12
2	Explore significance of operations management.	Analyze	1	1, 2, 3, 4,5,7,10
3	Plan and control the HR function effectively.	Apply	2	1,2,3,4,5,6, 8,11,12
4	Evaluate the markets, customers, and competition.	Evaluate	2	1,2,4,5,6,7, 9,10

TEXT BOOKS:

1. Stoner, Freeman, Gilbert, "Management", Pearson, 6th edition, 2018.
2. Aryasri, "Management Science", McGraw Hill, 4th edition, 2014.

REFERENCE BOOKS:

1. Kotler Philip & Keller Kevin Lane, "Marketing Management", Pearson, 15th Edition,2020.
2. Koontz & Weihrich, "Essentials of Management", Mc Graw Hill,11th Edition, 2020.
3. PC Tripathi, PN Reddy & Ashish Bajpai "Principles of Management", Mc Graw Hill, 7th Edition, 2021.

24SA101 - ORIENTATION SESSION

Hours per week:

L	T	P	C
0	2	0	1

PREREQUISITE KNOWLEDGE: Nil

COURSE DESCRIPTION AND OBJECTIVES:

This course is intended to help the students in getting adjusted with the new environment, to develop bondage with other students and faculty along with learning the institutional policies. The purpose of the orientation course is to create awareness on various issues that are essential for any individual.

It engages with the new students as soon as they come into the institution, before regular classes start. At the start, the incumbents learn about the institutional policies, processes, practices, culture and values, and their mentor groups are formed under SAGE (Socializing, Associating, Governing,Experiencing)

Objectives:

- To familiarize the fresh entrants with the new environment and system in engineering education.
- To make the students a responsible citizens.
- To promote smooth transition from +2 level education to graduate program among students.

MODULE-1

0L+8T+0P=8Hours

UNIT-1

Socializing: Interaction with faculty members, Deans, peers, student bodies and alumni.

UNIT-2

Associating: Eco-friendly practice and sustainable living, teaching sessions to school children of adopted villages, Interaction with farmers and identifying the problems, Field Visits.

PRACTICES

- Stress assessment and management
- Sustainable living
- Iconic visit

MODULE-2

0L+8T+0P=8Hours

UNIT-1

Governing: Brief about regulations and assessment pattern, Interaction with the functionaries i.e. Chancellor, Vice Chancellor and Registrar

UNIT-2

Experiencing: Physical activity, Creative arts, Literary activities, Customizing Daily routines and Habits, Practicing Meditation

PRACTICES

- Yoga and meditation
- Sports and Games
- Any of the one creative art

COURSE OUTCOMES:

Upon successful completion of the course, students will have the ability to:-

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Become familiar with the institute ethics and culture	Analyze	1	7,8,9,10,12
2	Cope up with stress management through meditation	Analyze	1	7,8,9,10,12
3	Socially conscious towards the society and environment	Apply	2	7,8,9,10,12
4	Build bonds with peers and faculty members	Apply	2	7,8,9,10,12

24SA103 - PHYSICAL FITNESS, SPORTS AND GAMES

Hours per week:

L	T	P	C
0	0	2	1

PREREQUISITE KNOWLEDGE: Nil

COURSE DESCRIPTION AND OBJECTIVES:

Physical fitness is a state of health and well-being and, more specifically, the ability to perform aspects of sports, occupations, and daily activities. Being physically active can improve brain health, help manage weight, reduce the risk of disease, strengthen bones and muscles, and improve ability to do everyday activities.

OBJECTIVES:

- To make an individual physically fit, mentally alert, emotionally balanced, socially well adjusted, morally true and spiritually uplifted.
- To make them agile and efficient for hard work and effective instruments for the achievement of their life's objectives.

MODULE – 1 & 2

The activities proposed under physical fitness are -

- Yoga and meditation
- 100 m
- 400 m
- Push Ups
- Individual game warm up
- Volleyball, Basketball, Kabaddi, Kho Kho, Badminton and Football- Basic skills of the game will be taught on rotation basis each week

PRACTICES:

- Yoga and meditation
- 100 m
- 400 m
- Push Ups

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Understand the knowledge about the practice of Yoga and its nature.	Analyze	1,2	1,2,6,9
2	Make familiar with the different system of the body.	Apply	1,2	1,2,9,12
3	Skills in explanation and demonstration of meditation.	Apply	2	1,2,3,9,12
4	Understood the information about the importance of diet and nutrition.	Evaluate	2	1,2,9,12
5	Ability to set up Yoga therapy center and skill to maintain case studies.	Create	1	1,2,3,9,12

TEXT BOOKS:

1. Sachitra Yoga pradipika: B. K. S. Ayyanger, ISBN 9788125016779
2. Yoga Chaitanya Pradipika: Yogacharya Dr. Raparthi Ramarao, Telugu, Year Published: 2003

REFERENCE BOOKS:

1. Asana pranayama Mudras Bandhas: Swami Satyananda Saraswati, Saraswati, Swami Satyananda, and Janez Kristijan Hiti. Asana pranayama mudra bandha. Bihar, India: Yoga Publications Trust, 1996.

24SA102 - SELF EMPOWERMENT & GENDER SENSITIZATION

Hours per week:

L	T	P	C
0	0	2	1

PREREQUISITE KNOWLEDGE: Gender Sensitisation and Personality Development

COURSE DESCRIPTION AND OBJECTIVES: To comprehend the dynamics of gender roles and relations within social and cultural contexts, and to evaluate legal and behavioral frameworks that promote gender equality and personal growth.

MODULE-I

0L+0T+16P=16Hours

UNIT-1

UNDERSTANDING GENDER:
Understanding of Gender, Gender Roles & relations, gender & culture, gender based violence,

UNIT-2

GENDER LAWS:

Gender & labour, Gender constitutional & Legal Perspectives.

PRACTICES:

- Conduct a survey on gender roles and relations in the banking and IT sectors to assess employee attitudes and resistance to policy changes.
- Undertake a study to identify the impact of cultural perceptions on gender-based violence and its effects on workplace productivity.
- Analyze the characteristics and components of gender laws to evaluate their effectiveness in promoting labor rights and reducing gender discrimination.
- Perform a study on the determinants of gender roles and culture by reviewing media content and social norms to understand their influence on attitudes.
- Conduct a comparative analysis of constitutional and legal perspectives on gender equality to assess their impact on labor practices and gender sensitivity in various industries.

MODULE-2

0L+0T+16P=16Hours

UNIT-1
SELF EMPOWERMENT:
Types of personality, Good Manners & Etiquettes, Ways to develop positive attitude

UNIT-2

PERSONALITY DEVELOPMENT: Decision Making Skills, Interpersonal Skills, Time Management, Stress Management, Conflict Management, Leadership Skills.

PRACTICES:

- Survey on different personality types and their impact on workplace performance and team dynamics.
- Undertake a study on the effectiveness of etiquette training programs in improving professional interactions and workplace relationships.
- Analyze strategies for developing a positive attitude and their impact on employee engagement and job satisfaction.
- Conduct a study on time management practices and their influence on productivity and work-life balance.
- Perform a study on conflict management techniques and their effectiveness in resolving workplace disputes and improving team cohesion.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Analyze the influence of gender roles and cultural norms on individual behavior and social structures.	Analyze	1	3
2	Apply legal perspectives and gender laws to assess their impact on gender equality and labor rights.	Apply	1	2
3	Analyze various personality types and develop strategies to apply good manners and etiquette in diverse settings.	Analyze	2	4
4	Apply decision-making and time management skills to enhance personal and professional effectiveness.	Apply	2	6
5	Analyze conflict management and leadership skills to effectively handle interpersonal and organizational challenges.	Analyze	2	1

REFERENCES:

1. A World of Equals: A Textbook on Gender, Susie Tharu, Uma Maheswari Bhrugubanda (2022), Orient Blackswan Pvt Limited
2. Chitra Chellam (2022), Personality Development, Sara Book Publication.

24SS101-CONSTITUTION OF INDIA

Hours per week:

L	T	P	C
0	2	0	1

PREREQUISITE KNOWLEDGE: High School-level Civics and Social Studies.

COURSE DESCRIPTION AND OBJECTIVES:

To provide students with a basic understanding of Indian Polity and Constitution and make students understand the functioning of government at the center and state level besides local self-government. This course also equips students with knowledge pertaining to fundamental rights and fundamental duties of a citizen in a democracy such as India.

MODULE-1

0L+16T+0P=16 Hours

UNIT-1

HISTORICAL BACKGROUND TO THE INDIAN CONSTITUTION

Meaning of the constitution law and constitutionalism; Historical perspective of the Constitution of India; Salient features and characteristics of the Constitution of India.

UNIT-2

FUNDAMENTAL RIGHTS, DUTIES, DIRECTIVE PRINCIPLES, AND AMENDMENT

Scheme of the fundamental rights - scheme of the Fundamental Right to Equality Article 14, scheme of the Fundamental Right to certain Freedom under Article 19, scope of the Right to Life and Personal Liberty under Article 21; Scheme of the Fundamental Duties and its legal status; Directive Principles of State Policy – its importance and implementation; Amendment of the Constitution - Powers and Procedure.

PRACTICES:

- Enactment of Constituent Assembly debates to further understand the rationale for the provisions of the constitution.
- Fundamental Rights in our popular culture - discussion in the movie Jai Bhim.
- Understanding the application of DPSPs in the law making procedure of the country
- Critically evaluating the challenges in the amendment of the constitution- reference to supreme court judgments.

MODULE-2

UNIT-1 **0L+16T+0P=16 Hours**

STRUCTURE AND FORM OF GOVERNMENT

Federal structure and distribution of legislative and financial powers between the Union and the States; Parliamentary Form of Government in India – The constitution powers and status of the President of India; Emergency Provisions-National Emergency, President Rule, Financial Emergency.

UNIT-2

LOCAL SELF GOVERNMENT

Local Self Government – Constitutional Scheme in India – 73rd and 74th Amendments (PRIs)

PRACTICES:

- Debate on the challenges to federalism in India.
- Critical presentation on the functioning of local self-government- references to the principle of decentralization
- Analyzing the implementation of reservation for women/marginalized communities in the local body elections.
- The financial challenges in the functioning of local self-governments- references to the devolution of the funds by the Finance commission of India (Article-280)

SKILLS:

- Understanding the basics of the Indian constitution.
- Know the fundamental rights, fundamental duties, and Directive Principles of State Policy.
- Fair knowledge about the functioning of various institutions in a democracy.

COURSE OUTCOMES:

Upon successful completion of the course, students will have the ability to:

CO No.	CO's	Blooms Level	Module No.	Mapping with PO's
1	Analyze major articles and provisions of the Indian constitution.	Analyze	1	6,8,9,10,12
2	Appreciation for the constitution and safeguarding individual rights.	Apply	1	6,8,9,10,12
3	Evaluating functions of various organs of the State in a democracy.	Evaluate	2	6,8,9,10,12
4	Evaluating distribution of powers at the various levels of the Government	Evaluate	2	6,8,9,10,12

TEXTBOOKS:

1. PM Bhakshi, "Constitution of India", Universal Law Publishing, 15th edition, 2018.
2. DD Basu "Introduction to Constitution of India", LexisNexis Publishing, 23rd edition, 2018.

REFERENCE BOOKS:

1. B. R. Ambedkar, "The Constitution of India" Educreation Publishing, India, 2020.
2. Subhash Kashyap, "Our Constitution", National Book Trust, India, 2nd edition, 2011.
3. Arun K. Thiruvengadam, "The Constitution of India: A Contextual Analysis", Hart Publishing India, 2017.

24SS102- INDIAN KNOWLEDGE SYSTEMS

Hours per week:

L	T	P	C
0	2	0	1

PREREQUISITE KNOWLEDGE: High School – level knowledge on Indian history and society.

COURSE DESCRIPTION AND OBJECTIVES:

The course introduces students to India's rich and diverse knowledge traditions and heritage. The profound insights, timeless wisdom, and diverse perspectives, originating from the ancient past have played a major role in shaping the country's cultural identity. The richness of India's intellectual heritage is vindicated by the existence of innumerable texts and thinkers in the country. By exploring India's knowledge traditions, the course aims to help students learn how the country's culture developed across the ages and to enable them look beyond conventional methods in finding solutions to scientific and technological issues.

MODULE-1

0L+16T+0P=16 Hours

UNIT-1

Overview of Indian Knowledge Systems (IKS)

Definition, Scope, and Significance of Indian Knowledge Systems; Various fields of study in IKS; Texts and sources of information for IKS; Vedic, Buddhist, Jain and other philosophies

UNIT-2

Ancient Contributions to Indian Knowledge Systems (IKS)

Contribution of ancient centers of learning; Contribution of prominent ancient philosophers

PRACTICES:

- Collecting information on Indian Knowledge Systems mentioned in the daily newspapers that report on unearthing new archeological discoveries, technologies, artforms, etc.
- Engaging in practical application of traditional indigenous techniques such as Yoga, Meditation, body art, martial arts viz. Karra saamu, etc.
- Group presentations on various indigenous sites of significance
- Poster display of profiles of prominent ancient philosophers

MODULE-2

0L+16T+0P=16 Hours

UNIT-1

Discipline-Specific Indian Knowledge Systems

Role of IKS within the discipline; Identifying and defining key concepts of IKS within the discipline; Demonstration of IKS via real-life examples (of technology, machinery, practices, artforms, architecture, etc.)

UNIT-2

IKS in the Modern World

Relevance of IKS in Contemporary Society; Challenges and Opportunities in Preserving and Utilizing IKS; Contributions of IKS to Global Knowledge Systems; Future Directions: Integrating IKS with Modern Science and Technology

PRACTICES:

- Exploring native technology, strategies, and methodologies specific to the discipline.
- Documenting various indigenous knowledge traditions.
- Using internet to gain awareness on various indigenous intellectual properties that are at risk of international appropriation.
- Group discussion on challenges to IKS from within the discipline

SKILLS:

- Analyzing and evaluating IKS practices in the context of modern challenges.
- Inculcates problem-solving skills by applying IKS principles to address contemporary issues.
- Sharpens research skills by gathering and analyzing information related to IKS.
- Inspires to work with others in preserving and promoting IKS.
- Motivates to learn and practice traditional skills like weaving, pottery, herbal medicine, etc.

COURSE OUTCOMES:

Upon successful completion of the course, students will have the ability to:

CO No.	CO's	Blooms Level	Module No.	Mapping with PO's
1	To understand the history, ancient knowledge and cultural practices of the people of India and apply its relevance to contemporary world	Apply	1, 2	6, 9, 10, 12
2	To analyze various case-studies from ancient India in the fields of science, technology, psychology, and society	Analyze	1, 2	6, 9, 10, 12
3	To analyze global implications of protecting indigenous knowledge systems	Analyze	1, 2	6, 9, 10, 12
4	To revive and revitalize ancient knowledge and evaluate the applicability of the concepts to contemporary scientific/technical methods, individual well-being and society at large	Evaluate	1, 2	6, 9, 10, 12

TEXTBOOKS:

1. B. Mahadevan, Vinayak Rajat Bhat, Nagendra Pavana R. N., "Introduction to Indian Knowledge System: Concepts and Applications", PHI Learning Private Limited, Delhi, 2023.
2. Mukul Chandra Bora, "Foundations of Bharatiya Knowledge System", Khanna Book Publishing, 2023.

REFERENCE BOOKS:

1. Dhirendranath Banerjee and Sanjit Kumar Sadhukan (ed.), "Ancient Indian Scientific Thought and Modern Theories: An Overview", National Mission for Manuscripts and D. K. Printworld, 2019.
2. Shaik Taufiq Khalil, "Indian Knowledge System: Arthshastra by Kautilya", Notion Press, 2023.
3. Chauhan Bhag Chand, "IKS: The Knowledge System of Bharata", Garuda Prakashan, 2023.

II

YEAR

COMPUTER SCIENCE AND ENGINEERING

B.Tech.

I SEMESTER

24MT203	-	Discrete Mathematical Structures
22TP201	-	Data Structures
22CS201	-	Database Management Systems
24CS204	-	Object Oriented Programming Through Java
XXXXXXX	-	Digital Logic Design
24CS205	-	Front end application Development
22SA201	-	Life Skills-I
24CS299	-	Design Thinking & Engineering Orientation

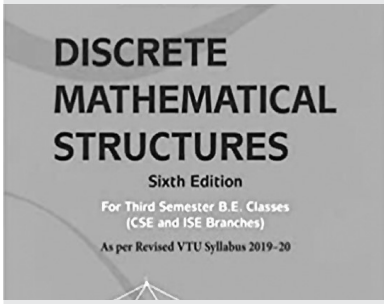
II SEMESTER

22ST202	-	Probabity and Statistics
XXXXXXX	-	Design and Analysis of Algorithms
22CS207	-	Operating Systems
XXXXXXX	-	Computer Organizatioin
24CS207	-	Full Stack Development
	-	Open Elective - 1 (NPTEL)
24CS201	-	Field Projects
22CS305	-	Industry interface course (Modular Course)*
	-	Minor / Honours - 1

COURSE CONTENTS

I SEM & II SEM

24MT203 DISCRETE MATHEMATICAL STRUCTURES



Source: https://www.amazon.in/Discrete-Mathematical-Structures-Dr-D-S-C/dp/9388478398

Hours Per Week :

L	T	P	C
3	2	0	4

PREREQUISITE KNOWLEDGE: Basic of set theory, Algebra.

COURSE DESCRIPTION AND OBJECTIVES:

The course objective is to provide students with an overview of discrete mathematics. Students will learn about of group and expected to demonstrate analytical and combinatorial methods such as propositional logic , Mathematical Induction, Boolean functions, combinatorial ,recurrence relation, generating function and graph theory

MODULE-1

UNIT-1 **8L+8T+0P=16 Hours**

ALGEBRAIC STRUCTURES AND BOOLEAN ALGEBRA

Group: Group, Subgroup, Lagrange's theorem, Introduction to rings and fields.

Boolean algebra: Boolean algebra, Truth table, Basic logic gates, Postulates of Boolean algebra, Principle of duality, Propositions, Connectives, Equivalence of Boolean functions.

UNIT-2 **8L+8T+0P=16 Hours**

APPLICATIONS OF LOGIC & BOOLEAN ALGEBRA

Normal form: CNF, DNF, CDNF, CCNF, Conversion of CNF to DNF and vice versa. Minimization of Boolean function (Karnaugh Maps), Generalized Pigeonhole Principle.

PRACTICES:

- Determine order of an element of group.
- Determine the order of a subgroup of a finite group.
- Determine whether algebraic structure is a Ring or a Field.
- Construct Truth table of propositions.
- Check whether propositions are equivalence.
- Obtain CNF, DNF of expression.
- Draw 2 – variables, 3-variables K-map.
- Minimize the Boolean function by K-map.
- Simplify the Boolean expression using Boolean algebra laws.
- Determine the homogeneous solution and particular solution for recurrence relation.
- Switching Circuit in Boolean algebra, Combination of two switches in a Circuit

MODULE-2

UNIT-1 **8L+8T+0P=16 Hours**

COMBINATORICS AND GRAPH THEORY

Combinatorics: The basics of counting, Permutations and combinations, Discrete Numeric Functions. Recurrence relations and Generating functions.

Graph theory: Graph terminology, Special types of graphs, Connected graph, Weighed graph, Graph Isomorphism, Euler and Hamiltonian paths and circuits, Planar graphs, Bipartite graph, Tree.

SKILLS:

- ✓ *Familiarity of concepts of statements, logic and truth tables.*
- ✓ *Analyze closed form of discrete numeric function.*
- ✓ *Know some basic properties of graphs, trees and related discrete structures.*

UNIT-2 **8L+8T+0P=16 Hours**

APPLICATIONS OF COMBINATORICS AND GRAPHS

Graph Coloring, Chromatic number, Matrix representation of graph, DFS, BFS algorithms, Minimum spanning tree, Kruskal's and Prim's algorithms.

PRACTICES:

- Represent sequence by Generating function.
- Solve Generation function by recurrence relation.
- Find degree of digraph and undirected graph.
- Identify Isomorphic graph, Euler circuit, Hamiltonian circuit.
- Determine matrix representation of graph.
- Determine the regions of planer connected graph.
- Draw Kn, Km,n.
- Determine the chromatic number of graphs.
- Find the weight of minimum spanning tree.
- Find number of arrangements that are possible.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Apply the concepts of mathematical logic and Boolean algebra.	Apply	1	1, 2, 9, 10, 12
2	Apply Karnaugh map to minimize.	Apply	1	1, 2, 9, 10, 12
3	Solve generating function by recurrence relations.	Apply	2	1, 2, 9, 10, 12
4	Model and solve real world problems using graphs and trees.	Apply	2	1, 2, 9, 10, 12

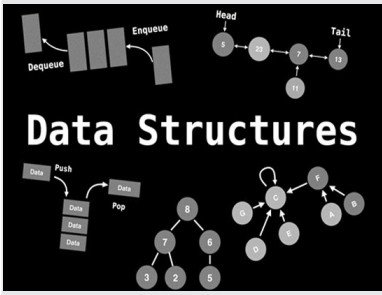
TEXT BOOKS:

1. Tremblay, J.P. and Manohar. R, "Discrete Mathematical Structures with Applications to Computer Science", 30th Reprint, Tata McGraw Hill Pub. Co. Ltd, New Delhi, 2017.
2. Rosen, K.H., "Discrete Mathematics and its Applications", 7th Edition, Tata McGraw Hill Pub. Co. Ltd., New Delhi, 2018.

REFERENCE BOOKS:

1. R.P. Grimaldi, "Discrete and Combinatorial Mathematics: An Applied Introduction", 4th Edition, Pearson Education Asia, Delhi, 2017.
2. S. Lipschutz and Mark Lipson., "Discrete Mathematics", Schaum's Outlines, Tata McGraw Hill Pub. Co. Ltd., New Delhi, 2021.
3. T. Koshy, "Discrete Mathematics with Applications", Elsevier Publications, 2015.
4. S B Singh. "Discrete Structures", Khanna Book Publishers Co-Pvt. Ltd. 2019.

22TP201 DATA STRUCTURES



Source: https://www.youtube.com/watch?v=Qmt0QwzEmh0

Hours Per Week :

L	T	P	C
2	2	2	4

PREREQUISITE KNOWLEDGE: Programming in C.

COURSE DESCRIPTION AND OBJECTIVES:

This course is aimed at offering fundamentals concepts of data structures and explains how to implement them. It begins with the basic concepts of data, data structures and then introduces the primitive and non-primitive data structures in detail. It forms the basis for understanding various ways of representing data and its usage in different computing applications.

MODULE-1

UNIT-1 **6L+6T+6P=18 Hours**

DATA STRUCTURES BASICS

Basic Terminology – data, information, datatype; Data Structures – Introduction, storage structures-sequential and linked storage representations; classification of data structures; Applications of data structures.

Sorting: Selection Sort, Bubble Sort, Insertion Sort, Quick Sort and Merge Sort.

Searching: Linear Search and Binary Search.

UNIT-2 **10L+10T+10P=30 Hours**

LINKED LISTS AND STACKS, QUEUES

Linked List: Introduction, Types of linked list – Singly linked list, doubly linked list and circular linked list, representation of linked list, Operations of linked list: Traverse forward/ reverse order, searching, insertion and deletion; Applications of linked lists.

Stack – Introduction, array and linked representations, implementation and their applications; Queue – Introduction, array and linked representations, implementation; Types – Linear, circular and doubly ended queues – operations; Applications of Queues.

PRACTICES:

PROBLEMS ON RECURSION – LEVEL 1

- Find the product of 2 numbers using recursion.
- Find the sum of natural numbers using recursion.
- Find the factorial of a number using recursion.
- Find the Nth term of Fibonacci series using recursion.
- Calculate the power using recursion.
- Write a recursive program for checking if a given number is a prime number.
- Given two integers write a function to sum the numbers without using any arithmetic operators.
- Convert a decimal to binary using recursion.
- Print all factors using recursion.
- Find the maximum product of digits among numbers less than or equal to N.

SKILLS:

- ✓ *Experienced to Store data and various types of data to handle.*
- ✓ *Ordering and sorting of data.*
- ✓ *Indexing and Searching of required data from large data sequences.*
- ✓ *Exposed to various characteristics such as Linear or non-linear, Homogeneous or heterogeneous and Static and Dynamic.*

PROBLEMS RECURSION – LEVEL 2

- Implement insertion sort recursively.
- Write a program to find the numbers less than N that are product of exactly 2 distinct prime numbers - using recursion.
- Implement selection sort recursively.
- Find the middle of a singly linked list using recursion.
- Find the sum of even numbers of an array using recursion.
- Check if a given array is in sorted order using recursion.
- Print alternate nodes of a linked list using recursion.
- Reverse a doubly linked list using recursion.
- Write a recursive function that returns all permutations of a given list.
- Implement bubble sort recursively.

PROBLEMS ON SORTING AND SEARCHING – LEVEL 1

- Implement the insertion sort function.
- Implement the bubble sort function.
- Implement the quick sort function.
- Implement the merge sort function.
- Implement the selection sort function.
- Implement linear search function.
- Implement binary search function.

PROBLEMS ON SLL – LEVEL 1

- Implement the insert function to insert nodes into a singly linked list (ascending order).
- Implement the insert function to insert nodes into a singly linked list (descending order).
- Implement the search node function.
- Implement the delete node function.
- Display forwards function.
- Display backwards function.
- Count the number of nodes in a singly linked list.
- Swap alternate nodes of a singly linked list.
- Move last node to the front of the linked list.
- Move first node to the last of the linked list.

PROBLEMS ON STACKS – LEVEL 1

- Implement two stacks using a single array.
- Given an array replace every element with nearest greater element on the right.
- Given a stack reverse the elements using only push and pop functions.
- Postfix evaluation using stack.
- Balance symbols.
- Find middle element in a stack.
- Remove middle element from a stack.
- Implement push and pop using linked list.
- Given an array of characters with the middle marked by X, check if the string is a palindrome.
- Maximum sum in sliding window.

PROBLEMS ON QUEUES – LEVEL 1

- Write a program to accept two numbers as input check if they are equal.
- Write a program to accept two characters as input and check if they are equal.
- Write a program to accept two numbers as input and print the greater of the 2 numbers.
- Write a program to accept two numbers as input and print the lesser of the 2 numbers.
- Write a program to accept 3 numbers as input and print the maximum of the 3.

- Write a program to accept 3 numbers as input and print the minimum of the 3.
- Write a program to accept a number as input and print EVEN if it is an even number and ODD if it is an odd number.
- Write a program to accept a number as input and check if it is divisible by 3. If it is divisible by 3 print YES else print NO.
- Write a program to accept a number as input and check if it is divisible by both 3 & 5. If it is divisible print YES else print NO.
- Write a program to accept a number as input and check if it is positive, negative or zero.

PROBLEMS ON DLL – LEVEL 1

- Implement insert function.
- Implement display forward function.
- Implement display backward function.
- Implement search function.
- Implement delete function.
- Reverse a doubly linked list from M to N.
- Find the sum of the odd and even nodes.
- Count odd keys of the linked list.
- Merge two sorted lists.
- Delete adjacent duplicate nodes.

PROBLEMS ON CLL – LEVEL 1

- Insert function (circular doubly linked list).
- Search function.
- Display forward.
- Display backward.
- Delete node (circular doubly linked list).
- Print the middle N nodes of a circular singly linked list.
- Move the last node of a circular singly linked list to the beginning.
- Delete adjacent duplicate nodes of a circular singly linked list.
- Delete nodes greater than a value from a circular doubly linked list.
- Find the sum of the nodes of a circular linked list.

PROBLEMS ON LINKED LIST – LEVEL 2

- Given 2 sorted linked lists, print the common elements.
- Reverse a list (using Stack).
- Given a pointer to a node (not the last node), delete the node.
- Reverse a list (Recursive).
- Reverse a list (Iterative).
- Reverse a singly linked list in pairs (recursive).
- Reverse a singly linked list in pairs (iterative).
- Check if a singly linked list is a palindrome or not.
- Remove the loop if exists.
- Given 2 linked lists with data in the ascending order, merge them into a single list.

MODULE-2

UNIT-1 **8L+8T+8P=24 Hours**

TREES

Trees: Basic Terminology, Types of Trees, Binary Tree – Introduction, properties, array and linked representations; Tree traversals and their implementation; Expression trees; BST – definition and operations, AVL trees – definition and construction; Applications of binary trees.

UNIT-2 **8L+8T+8P=24 Hours**

GRAPHS & HASHING

Graphs: Basic Terminology, Types of Graphs, Graphs representations – adjacency matric, adjacency list; Traversals - breath first search and depth first search; Applications of graphs.

Hashing: Introduction, Different hash functions, collision: avoidance and handling methods.

PRACTICES:

PROBLEMS ON BST – LEVEL 1

- Insert function.
- Insert function (recursive).
- Search function.
- Pre order traversal.
- Post order traversal.
- In order traversal.
- Level order traversal.
- Delete child node.
- Delete parent node.
- Delete nodes greater than a value from a circular doubly linked list.

PROBLEMS ON PRIORITY QUEUES – LEVEL 1

- Meeting rooms problem.
- Ugly number.
- Find median from data stream.
- Find the top K frequent elements.
- Find K Pairs with smallest sums.
- Find the Kth smallest element in a sorted matrix.
- Trapping Rain Water.
- Rearrange String k distance apart.
- Sort characters by frequency.
- Solve the maze problem.

PROBLEMS ON GRAPHS – LEVEL 1

- Implement Graph data structure.
- Implement BFS - iterative solution.
- Implement BFS - recursive solution.
- Implement DFS - iterative solution.
- Implement DFS - recursive solution.
- Check if given graph is strongly connected or not.
- Check if given graph is strongly connected or not - using DFS.
- Given a graph find the arrival and departure time of its vertices in DFS. Arrival time is the time when the vertex was explored for the first time, and departure time is the time at which all the neighbours are explored and are ready to backtrack.
- Given a directed acyclic graph and a source vertex, find the cost of the shortest path from source vertex to all other vertices present in the graph. If a vertex cannot be reached from given source vertex that distance may be printed as infinite.
- Given an undirected graph, check if the graph is 2 edge connected or not.

PROBLEMS ON HASHING – LEVEL 1

- Print a binary tree in vertical order.
- Find whether an array is subset of another array.
- Given an array A [ ] and a number x, check for pair in A [ ] with sum as x.

- Minimum operation to make all elements equal in array.
- Maximum distance between two occurrences of same element in array.
- Check if a given array contains duplicate elements within k distance from each other.
- Find duplicates in a given array when elements are not limited to a range.
- Most frequent element in an array.
- Smallest subarray with all occurrences of a most frequent element.
- First element occurring k times in an array.

PROBLEMS ON GRAPHS – LEVEL 2

- Find the shortest graph distances between every pair vertex in a given path. Assume that the graph does not have any negative edges.
- Find the shortest graph distances between every pair of vertices in a given path. The graph can have negative edges.
- Detect cycle in DFS.
- Count the number of connected components of a graph represented in the adjacent matrix.
- Count the number of connected components of a graph represented in the adjacent matrix - using DFS.
- Find a spanning tree - not necessarily a minimum spanning tree.
- Detect cycle in an undirected graph.
- Given an undirected graph, find its depth.
- Determine if a directed graph has a unique topological ordering.
- Given a directed acyclic graph and two vertices v and w, find the lowest common ancestor.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

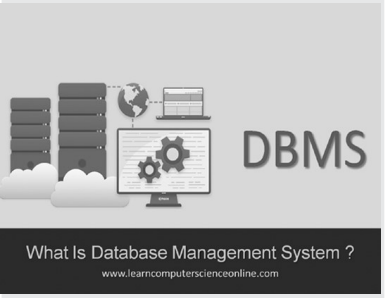
CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Explore the organization of several ADTs and the manipulation (searching, insertion, deletion, traversing) of data stored in various data structures.	Apply	1,2	1
2	Apply different data structures to solve a given problem.	Apply	1,2	1
3	Analyze the efficiency of using different data structures and choose the efficient data structure for solving a given problem.	Analyze	1,2	2
4	Develop new algorithms to solve various problems.	Create	1,2	3,4

TEXT BOOKS:

1. Reema Thareja, “Data Structures Using C”, 2nd Edition, Oxford University Press, 2014.
2. Seymour Lipschutz, “Data Structures with C”, 1st Edition, McGraw Hill Education, 2017.

REFERENCE BOOKS:

1. Ellis Horowitz and Sartaj Sahni, “Fundamentals of Data Structures”, illustrated edition, Computer Science Press, 2006.
2. Richard F. Gilberg and Behrouz A. Forouzan, “Data Structures: A Pseudocode Approach with C”, 2nd Edition, CENAGE Learning, 2005.
3. R G Dromey and Pearson, “How to solve it by Computer”, 2nd edition, Impression edition, 1998.



Source: https://www.youtube.com/watch?v=lDpB9zF8LBw

22CS201 DATABASE MANAGEMENT SYSTEMS

Hours Per Week :

L	T	P	C
2	2	2	4

PREREQUISITE KNOWLEDGE: Discrete Mathematical Structures.

COURSE DESCRIPTION AND OBJECTIVES:

This course presents an introduction to database management systems with an emphasis on how to organize, maintain and retrieve data efficiently from a relational database. It also focuses on requirements gathering and conceptual, logical, physical database design. The objective of the course is to enable the student to understand database design, expressing queries using SQL, query optimization and transaction processing.

MODULE-1

UNIT–1 **10L+6T+4P=20 Hours**

DATABASE SYSTEM CONCEPTS

Databases And Database Users: Introduction; Characteristics of the database approach; Actors on the scene; Advantages of using DBMS approach.

Database System Concepts and Architecture: Data models, Schemas and instances; Three-Schema architecture and data Independence; Database languages and interfaces; The database system environment; Centralized and Client-Server architectures for DBMS.

Conceptual Data Modeling and Database Design: Entity types, Entity sets, Attributes and keys; Relationship types, Relationship sets, Roles and structural constraints; Weak entity types; Relationship types.

UNIT–2 **6L+10T+12P=28 Hours**

RELATIONAL DATABASE DESIGN

Relational Database Design by ER–To-Relational Mapping: Relational Database design using ER-to-Relational mapping.

The Relational Data Model and Relational Database Constraints: Relational model concepts; Relational model constraints and Relational database schemas.

Relational Algebra: Unary relational operations - SELECT and PROJECT; Relational algebra operations from set theory; Binary relational operations - JOIN and DIVISION.

SQL: SQL data definition and data types; specifying constraints in SQL, Basic retrieval queries in SQL; INSERT, DELETE, and UPDATE statements in SQL.

PRACTICES:

- Design ER Model for various real time database applications.
- Development of Relational Database schemas for Company/Student/Sailors/ using DDL constructs of SQL.
- Apply various DML Commands such as select, insert, update etc. of SQL on Relational Database.
- Design of Relational Database schemas by specifying different types of Constraints.
- Apply various Relational Database operators (Arithmetic, Logical &comparison) and string-matching constructs of SQL.
- Expressing queries using Aggregate Functions of SQL on Relational Database.
- Queries on Relational Database using GROUP BY, HAVING and ORDER BY clauses of SQL.

MODULE-2

UNIT–1 **8L+8T+12P=28 Hours**

NORMALIZATION

Complex Queries, Triggers, Views: More complex SQL retrieval queries; Specifying constraints as assertions and actions as triggers; Views (virtual tables) in Pl/SQL.

Basics of Functional Dependencies and Normalization for Relational Databases: Informal design guidelines for relation schemas; Functional dependencies-inference rules, equivalence and minimal cover; Normal forms based on primary keys; Boyce-Codd normal form; Properties of relational decompositions, multi valued dependency, join dependencies.

UNIT–2 **8L+8T+4P=20 Hours**

TRANSACTION PROCESSING

Introduction To Transaction Processing Concepts and Theory: Introduction to transaction processing; Transaction and system concepts; Desirable properties of transactions; Characterizing schedules based on serializability.

Concurrency Control Techniques: Two-phase locking techniques for concurrency control, concurrency control based on timestamp ordering.

Database Recovery Techniques: Recovery concepts; Shadow paging; The ARIES recovery algorithm.

Indexing Structures for Files and Physical Database Design: Single level and multi-Level indexing; Dynamic multi-level indexing using B-trees and B+ trees.

PRACTICES:

- Design and Development of company database and expressing Nested queries using SQL.
- Design and Development of student database and specifying queries using set operations.
- Design and Development of sailor's database and specifying queries using different types of JOINs.
- Implementation of PL/SQL programs with Control Structures.
- Implementation of PL/SQL programs with Procedures.
- Implementation of PL/SQL programs with Function.
- Implementation of PL/SQL programs with Triggers.
- Creation and dropping of VIEWS.
- Relation R has eight attributes ABCDEFGH. Fields of R contain only atomic values. F = {CH -> G, A -> BC, B -> CFH, E -> A, F -> EG} is a set of functional dependencies (FDs) so that F+ is exactly the set of FDs that hold for R. How many candidate keys does the relation R have?
- Apply various DCL and TCL constructs of SQL on Relational Database.

SKILLS:

- ✓ *Develop E-R model for real life applications.*
- ✓ *Design of relational databases for real world applications.*
- ✓ *Devise queries using relational algebra and SQL.*
- ✓ *Analyze transaction processing, concurrency control and recovery techniques*

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Develop an E-R model for real life applications.	Apply	1	1,10
2	Design and normalize databases for real time applications.	Create	1	1,3
3	Devise queries using Relational Algebra and SQL.	Analyze	2	2
4	Express queries using database tools like Oracle, DB2, MYSQL.	Apply	2	5,10

TEXT BOOKS:

1. Ramez, Elmasri and Shamkant B. Navathe, “Fundamentals of Database Systems”, 7th Edition, Pearson Education, 2016.
2. Raghu Rama Krishnan and Johannes Gehrke, “Database Management Systems”, 3rd Edition, Tata McGraw Hill, 2013.

REFERENCE BOOKS:

1. Abraham Silberschatz, Henry F.Korth and S.Sudarshan, “Database System Concepts”, 7th edition, Tata Mc Graw Hill,2019.
2. Allen G. Taylor “Database Development for Dummies” 1st Edition, 2011
3. C. J. Date “Introduction to Database Systems” 7th Edition, Addison Wesley, 2003.

24CS204 OBJECT-ORIENTED PROGRAMMING THROUGH JAVA



Source: https://www.datasciencecentral.com/wp-content/uploads/2021/10/866750 7462.jpeg

Hours Per Week :

L	T	P	C
3	0	2	4

PREREQUISITE KNOWLEDGE: Classes, Objects and Inheritance.

COURSE DESCRIPTION AND OBJECTIVES:

This course is about the fundamentals of Object-Oriented Programming (OOP) Concept and OOP-based software development methodology. Java as a class-based and pure OOP language is used to demonstrate and implement appropriate concepts and techniques. The students are exposed to the concepts, fundamental syntax, and the thought processes behind object- oriented programming. By end of the course, students will acquire the basic knowledge and skills necessary to implement object-oriented programming techniques in software development using Java.

MODULE-1

UNIT-1 **12L+0T+8P=20 Hours**

INTRODUCTION

History of Java, Byte code, JVM, Java buzzwords, OOP principles, Data types, Variables, Scope of variables, Operators, Control statements, Type conversion and casting, Arrays.

Concepts Of Classes and Objects: Introduction to methods, Method over loading, Constructors, Constructor overloading, Usage of static with data and method, Access control, this keyword, Garbage collection, String class, String Tokenizer.

UNIT-2 **12L+0T+8P=20 Hours**

INHERITANCE AND EXCEPTIONS

Inheritance: Types of inheritance, Member access rules, Usage of super keyword, Method overriding, Usage of final keyword, Abstract classes, Interfaces - differences between abstract classes and interfaces, defining an interface, implementing interface, applying interfaces, variables in interface and extending interfaces. Packages-defining, creating and accessing a package, importing packages, access control in packages.

Exception Handling: Concepts of exception handling, Types of exceptions, Usage of try, catch, throw, throws and finally keywords, Built-in exceptions, User defined exception.

PRACTICES:

- There is a telecommunication company called “Powered Air” who have approached you to build their Interactive Voice Response (IVR) system. write a Java program and be able to provide the following menu (given below):
  Note: User should provide an input for each menu display. Welcome to Powered Air service. What would you like to do?
  a. Know my balance. b. Know my validity date
  c. Know number of free calls available. d. More
  1. Prepaid Bill Request 2. Customer Preferences 3. GPRS activation
  4. Special Message Offers 5. Special GPRS Offers 6. 3G Activation
  7. Go back to Previous menu
  You are free to display your own messages in this IVR.
- Create a class Rectangle. The class has attributes length and width. It should have methods that calculate the perimeter and area of the rectangle. It should have read Attributes method to read length and width from user.

SKILLS:

- ✓ *To analyse and develop algorithm for real life problems using Java.*
- ✓ *Experience with developing and debugging programs in different IDEs.*
- ✓ *Develop multi-threaded applications.*
- ✓ *Creating web applications*

Hint: Area of rectangle = length * width, Perimeter of rectangle = 2*(length+width).

- Implement a Java Program that reads a line of integers, and then displays each integer, and the sum of all the integers (use StringTokenizer class).
- Implement a java program to print all tokens of a string on the bases of multiple separators (use StringTokenizer class).
- Using inheritance, one class can acquire the properties of others. Consider a class Animal that has only one method “walk”. Next, create a Bird class that also has a fly method. Finally, create a bird object that can both fly and walk.
- Using inheritance, Write the following code in your editor :

  1. A class named Arithmetic with a method named “add” that takes integers as parameters and returns an integer denoting their sum.

  2. A class named Adder that inherits from a superclass named Arithmetic.

  Note: Your classes should not be Public.
- When a subclass inherits from a superclass, it also inherits its methods; however, it can also override the superclass methods (as well as declare and implement new ones). Consider the Sports class having methods getName()[which returns name of sport] and getNumberOf TeamMembers()[which returns noof team members] create a Soccer class that inherits from the Sports class. We can override the get Name method and return a different subclass-specific string and override getNumberOfTeamMembers method and return noof team members
- Implement a java program to create an abstract class named Shape that contains an empty method named number Of Sides ( ).Provide three classes named Trapezoid, Triangle and Hexagon such that each one of the classes extends the class Shape. Each one of the classes contains only the method number Of Sides ( ) that shows the number of sides in the given geometrical figures.
- You are given an interface Advanced Arithmetic which contains a method signature int divisor_ sum(int n). You need to write a class called My Calculator which implements the interface. divisor_sum function just takes an integer as input and return the sum of all its divisors. For example divisors of 6 are 1, 2, 3 and 6, so divisor_sum should return 12. The value of n will be at most 1000.
- Implement a Java program for the following

  a) Creation of simple package.

  b) Accessing a package.
- Implement a Java program to read two numbers a,b from user and perform division a/b,if the user passes b value as zero, handle the exception using try and catch otherwise display the result.
- Create a class called Customer with data members account_number, balance (initialize with 10000), and member functions print(), deposit(), and withdraw(). Print method display account number and balance. If withdraw amount is less than current balance while withdrawing, throw an exception “In Sufficient Funds”. If the input is 1 do print. If the input is 2 withdraw (). If the input is 3 deposit. If the input is 4 terminate program.
- Implement a Java program which accepts age as input from the user and throws an exception “Not Eligible to Vote” when age is <=18 otherwise print “Eligible to Vote”.

MODULE-2

UNIT-1 **12L+0T+8P=20 Hours**

MULTI THREDING AND FRAMEWORK

Multithreading: Concepts of multi threading, Differences between process and thread, Thread life cycle, Creating multiple threads using Thread class and Runnable interface, Synchronization, Thread priorities, Inter thread communication.

Collection Framework: Collections Overview, Collection Interfaces - List, Set, Map, List – Array List, Linked List, Vector, Set - HashSet, TreeSet, Map - HashTable, HashMap, accessing a collection via an Iterator, comparator, comparable.

UNIT-2 **12L+0T+8P=20 Hours**

SWINGS

GUI Programming With Swing: Delegation event model-Events, Event sources, Event Listeners, Event classes, handling mouse and keyboard events.

Exploring Swing Controls: JLabel and Image Icon, JText Field, JButton, JCheckBox, JRadioButton, JTabbed Pane, JList, JCombo Box.

PRACTICES:

- Print in Order

  Suppose we have a class:

  public class Foo {

  public void first() { print("first"); }

  public void second() { print("second"); }

  public void third() { print("third"); }

  }

  The same instance of Foo will be passed to three different threads. Thread A will call first(), thread B will call second(), and thread C will call third(). Design a mechanism and modify the program to ensure that second() is executed after first(), and third() is executed after second().

  Note: We do not know how the threads will be scheduled in the operating system, even though the numbers in the input seem to imply the ordering. The input format you see is mainly to ensure our tests' comprehensiveness.

  Example 1:

  Input: nums = [1,2,3] Output: "firstsecondthird"

  Explanation: There are three threads being fired asynchronously. The input [1,2,3] means thread A calls first(), thread B calls second(), and thread C calls third(). "firstsecondthird" is the correct output.

  Example 2:

  Input: nums = [1,3,2] Output: "firstsecondthird"

  Explanation: The input [1,3,2] means thread A calls first(), thread B calls third(), and thread C calls second(). "firstsecondthird" is the correct output.

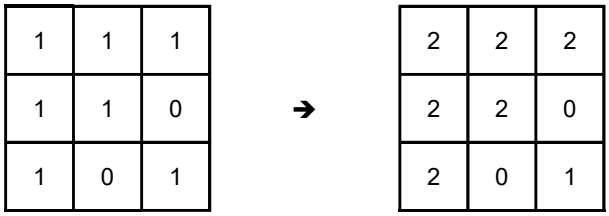
- **Flood Fill:**

  An image is represented by an m x n integer grid image where image[i][j] represents the pixel value of the image.

  You are also given three integers sr, sc, and color. You should perform a flood fill on the image starting from the pixel image[sr][sc].

  To perform a flood fill, consider the starting pixel, plus any pixels connected 4-directionally to the starting pixel of the same color as the starting pixel, plus any pixels connected 4-directionally to those pixels (also with the same color), and so on. Replace the color of all of the aforementioned pixels with color.

  Return the modified image after performing the flood fill.



  Example 1:

  Input: image = [[1,1,1],[1,1,0],[1,0,1]], sr = 1, sc = 1, color = 2

  Output: [[2,2,2],[2,2,0],[2,0,1]]

  Explanation: From the centre of the image with position (sr, sc) = (1, 1) (i.e., the red pixel), all pixels connected by a path of the same color as the starting pixel (i.e., the blue pixels) are colored with the new color.

Note the bottom corner is not coloured 2, because it is not 4-directionally connected to the starting pixel.

Example 2:

Input: image = [[0,0,0],[0,0,0]], sr = 0, sc = 0, color = 0

Output: [[0,0,0],[0,0,0]]

Explanation: The starting pixel is already colored 0, so no changes are made to the image.

- Count words in a given string

  The input parameter is a list of strings representing lines of text.

  Count how often the word occurs in the text.

  If the word "kitten" occurred in a text 23 times, then its entry would be "kitten - 23\n". Return statistics as a String containing all the entries.

  Omit all words which contain less than 4 letters and appear less than 10 (the words which are too small or to rare) The entries in the resulting String should be also sorted by their amount and then in alphabetical order if it is needed.
- Implement a Java program for handling mouse events when the mouse entered, exited, clicked, pressed, released, dragged and moved in the client area.
- Implement a Java program for handling key events when the key board is pressed, released, typed.
- Implement a Java swing program that reads two numbers from two separate text fields and display sum of two numbers in third text field when button "add" is pressed.
- Implement a Java program to design student registration form using Swing Controls. The form which having the following fields and button "save". Form Fields are: Name, RNO, Mail id, Gender, Branch, and Address.
- Implement a java program using swings to design a multiple choice question having three options (use radio button) ,display the message using dialog box "Your answer is wrong" if the user selects wrong option otherwise display ,"Your answer is correct."

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Apply object oriented concepts on real time scenarios.	Apply	1	1,2
2	Apply the concepts of Multithreading and Exception handling to develop efficient and error free codes	Apply	1, 2	1,2
3	Design and develop Java applications to solve real world problems by using modern tools and collection framework	Create	2	3,5
4	Design and develop GUI based applications using swings for internet and system based applications.	Create	2	3,5

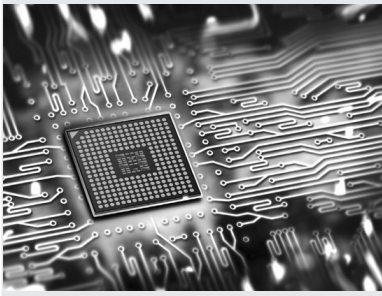
TEXT BOOKS:

1. Herbert Schildt, "Java the complete reference", 12th Edition, McGraw Hill, Education, 2021.
2. M.T. Somashekara, D. S. Guru, K.S. Manjunatha, "Object-Oriented Programming with Java", 1st Edition, PHI Learning, 2017.

REFERENCE BOOKS:

1. E. Balagurusamy, "Programming with Java", 6th Edition, McGraw Hill, 2019.
2. Mark Lassoff," Java Programming for Beginners: Learn the fundamentals of programming with Java", 1st Edition, Packt Publishing Limited,2017.
3. Philip Conrod, Lou Tylee," Learn Java GUI Applications : A JFC Swing Tutorial", 11th Edition, Kidware Software, 2019.

XXXXX DIGITAL LOGIC DESIGN



Source: https://collegeacademy.in/images/ex-blogs/deld.jpg

Hours Per Week :

L	T	P	C
3	0	2	4

PREREQUISITE KNOWLEDGE: Basics of Computers.

COURSE DESCRIPTION AND OBJECTIVES:

This course introduces the basic knowledge on number systems, analysis and design of combinational and sequential circuits. The course mainly focuses on designing digital circuits in optimized manner by using components like decoders, encodes, multiplexers. It also deals with design of sequential circuits and Programmable logic devices.

MODULE-1

UNIT-1 **12L+0T+8P=20 Hours**

INTRODUCTION

Number Systems: Binary Numbers, Number base Conversions, Complements, Binary codes.

Boolean Algebra: Fundamental concepts of Boolean algebra basic theorems and properties.

Gate-Level Minimization: Canonical and standard forms - SOP and POS forms, Logic gates, Algebraic simplification and realization with basic gates and universal gates, The map method – two, three, four variable K map; POS and SOP simplification; Don't care conditions; NAND and NOR implementation.

UNIT-2 **12L+0T+8P=20 Hours**

COMBINATIONAL LOGIC CIRCUITS

Combinational circuits analysis, design procedure; Half adder, Full adder, Half subtractor, Full subtractor, Binary adder/subtractor; BCD adder; Binary multiplier; Magnitude comparator; Decoders; Encoders; Multiplexers; De-Multiplexer.

PRACTICES:

- Design a combinational circuit with three inputs and one output. The output is 1 when the binary value of the inputs is less than 3. The output is 0 otherwise.
- Design a combinational circuit with three inputs x, y, z and three outputs A, B, C. When the binary inputs is 0, 1, 2 or 3, the binary output is one grater than the input. When the binary input is 4, 5, 6, or 7 then the binary output is one less than the input.
- Design a code converter that converts a decimal digit from the 8, 4, -2, -1 code to BCD.
- Implement a Full – Adder using 4 X 1 multiplexer.
- Design a 16 X 1 Multiplexer with five 4 X 1 multiplexer.
- Design a 5-to-32 line decoder with four 3-to-8-line decoders with enable and 2-to-4-line decoder.
- Design and realize 4bit parallel adder and subtractor.

MODULE-2

UNIT-1 **12L+0T+8P=20 Hours**

CIRCUITS, REGISTERS AND COUNTERS

Sequential Logic Circuits: Latches, Flip-Flops - SR, JK, D, T; Flip-flop conversion; Analysis of sequential circuits; Design procedure.

Registers and Counters: Shift registers; Ripple counters; Synchronous counters.

SKILLS:

- ✓ *Learn different data and number representations.*
- ✓ *Design of logical circuits using all types of gates.*
- ✓ *Minimizing of Boolean functions.*
- ✓ *Design of simple logical circuits*

UNIT-2 **12L+0T+8P=20 Hours**

MEMORY AND PROGRAMMABLE LOGIC DEVICES

Random access memory; Read only memory; Programmable logic array; Programmable array logic.

PRACTICES:

- Design a JK flip-flop using a D flip-flop.
- Design a sequential circuit with two D flip-flops A and B and, one input x. When x=0, the state of the circuit remains same. When x = 1, the circuit goes through the state transitions from 00 to 01 to 11 to 10 back to 00 and repeats.
- Design a 4-bit binary synchronous counter with D flip-flop.
- Design and realize a decimal counter.
- A sequential circuit has two J-K fiip-flops A and B. Two inputs x and y, and one output z. The flip-flop input equations and circuit output equation are:
  JA=Bx+B'y'
  KA=B'xy'
  JB=A'x
  KB=A+xy'
  Z=Ax'y'+Bx'y'
  a) Tabulate the state table. b) Derive the state equations.
- Realize the given two Boolean functions with a PLA:
  F1(A, B, C) = ∑(0, 1, 2, 4)
  F2(A, B, C) = ∑ (0, 5, 6, 7)
  F3(A, B, C) = ∑ (1, 3, 4, 5, 7)
- Tabulate the PAL programming table for the four Boolean functions listed below. Minimize the numbers of product terms.
  A(x, y, z) = ∑ (1, 3, 5, 6) B(x, y, z) = ∑ (0, 1, 6, 7)
  C(x, y, z) = ∑ (3, 5) D(x, y, z) = ∑ (1, 2, 4, 5, 7).

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Apply the knowledge of digital logic concepts to optimize digital circuits.	Apply	1	1
2	Apply Boolean algebra rules and karnaugh map method to reduce the Boolean functions.	Apply	1	1, 2
3	Design Combinational digital circuits for the given problem statement by applying the digital techniques.	Analyze	1	3
4	Design and analyze sequential digital circuits for the given problem statement and improve the performance by reducing the complexities.	Analyze	2	3
5	Categorize various types of Programmable Logic Devices.	Analyze	2	2

TEXT BOOK:

1. M Morris Mano and Michael D. Ciletti, "Digital Design", 5th Edition, Pearson Education, 2013.

REFERENCE BOOKS:

1. John F.Wakerly, "Digital Design Principles and Practices", 3rd Edition, Pearson/PHI, 2015
2. Charles H.Roth. "Fundamentals of Logic Design", 6th Edition, Thomson Learning, 2013.
3. John. M Yarbrough, "Digital Logic Applications and Design", Thomson Learning, 2006.

24CS205 FRONT-END APPLICATIOIN DEVELOPMENT



Source: https://www.geeksforgeeks.org/best-way-to-start-with-competitive-programming-geeksforgeeks-cp-live-course/

Hours Per Week :

L	T	P	C
0	2	2	2

PREREQUISITE KNOWLEDGE: OOPs through JAVA. No Previous Development Experience Required. Access to various Open -Source (Free)Software.

COURSE DESCRIPTION AND OBJECTIVES:

This course covers web development principles such as static webpage design, building a simple HTML text site, styling web pages with CSS, and programming websites using Javascript.

MODULE-1

UNIT-1 **0L+0T+8P=8 Hours**

WEB PAGE DESIGNING:

Web technologies, WWW, website, web browser, web server, global and local impact of web technologies on modern society.

HTML: Creating structured documents, Links and navigation, Tables, Forms, and Frames.

HTML 5: Introduction to HTML5, The HTML5 Canvas, HTML5 audio and Video.

UNIT-2 **0L+0T+8P=8 Hours**

CSS: INTRODUCTION

CSS Syntax, CSS selectors: (ID, Class, Tags, Attributes)

CSS Properties: Align-Content, Align-items, Align-self, Animation.

PRACTICES:

- Design a webpage having four frames named a) Top, b)Center, c)Bottom, and d) Left. The top frame should contain the company logo and title. The bottom frame should contain copyright information. The left frame should contain various links like Home, Products, Services, Branches, About, etc., When clicked on respective links, the content should display on the center frame.
- Design a catalog page that should contain the details of all the books available on the website in a table. The details should contain the following: a) Snapshot of Cover Page b) Author Name c) Publisher. d) Price. e) Add to cart button.
- Design a timetable schedule for your current semester using the Table tag.
- Design a HTML page for Student Registration Form using Form Elements that includes Form, input upload.
- Write a HTML program with at least two <h1>, two images, two buttons and appropriate CSS to display,
  - All <h1> with font-size 12pt, and italic in Verdana font using Inline CSS.
  - All <img> with border color red, thickness 15px using Document Level CSS
  - All <input type='button'> should change background color to red on mouse over them using External CSS.
- Design a web page to display the videos on-page, on user selection using frames and HTML5 tags.
- Design a web page to display different types of objects using HTML5 Canvas.

MODULE-2

UNIT-1 **0L+0T+8P=8 Hours**

JAVASCRIPT

JAVASCRIPT: Learning JavaScript-how to add scripts to your page, DOM, Variables, Operators, JavaScript Data Types, JavaScript Special Characters.

UNIT-2 **0L+0T+8P=8 Hours**

JavaScript Arrays, JavaScript Boolean, Loops, For-In, For, Do-While, While Loops, JavaScript Events. Built in Objects, Form and regular expression, Validation.

PRACTICES:

- Write Java script programs to demonstrate the following objects with atleast five methods: a) Math b) String c) Array c) Date
- Write a Java script program to display message on OnBlur and OnFocus events
- Design a HTML page having a textbox and four buttons viz Factorial Fibonacci Prime and Palindrome. When a button is pressed an appropriate java script function should be called to display the following
  - Factorial of that number.
  - Fibonacci series up to that number.
  - Prime numbers up to that number.
  - Is it Palindrome or not?
- Design a web page that contains a color pallet, when the user moves the mouse to the particular area, then it changes the background color of the web page.
- Write JavaScript Function to modify Style of the Paragraph Text.
- Design a registration page to validate the following fields using Java Script.Make sure the user name starts with an upper case letter.
  - The user name must have at least one digit.
  - Ensure that Email is valid.
  - Ensure that the password length is between 8 to 20 characters.
  - Make sure the password contains at least one upper case letter, one lower case, and one special character exclude [. dot, , comma, ; semi colon, :colon].

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Usage of HTML, HTML5, CSS, Java Script in web application development.	Apply	1,2	1
2	Create sand style webpages with html and css.	Understanding	1	2
3	Analyze JavaScript frameworks and how they can be utilized to save time when writing custom Javascript code.	Apply	2	4
4	Design and develop Script Forms,Validation.	create	2	1

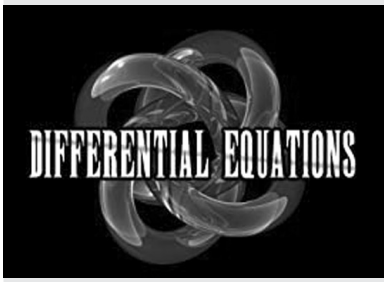
TEXT BOOKS:

1. Learning Web Design:A beginners guide to HTML,CSS,Javascript and Web Graphics,By Jennifer Niederst Robbins.
2. Eloquent JavaScript, by Marijn Haverbeke.
3. HTML And CSS: Design And Build Website by Jon Duckett.
4. JavaScript And JQuery: Interactive Front-End Web Development by Jon Duckett.
5. Front-End Development: The Big Nerd Ranch Guide, by Chris Aquino and Todd Gandee.

REFERENCE BOOKS:

1. JavaScript: The Good Parts, Book by Douglas Crockford
2. JavaScript and JQuery: Interactive Front-End Web Development, Book by Jon Duckett
3. JavaScript for dummies.
4. Head First JavaScript (English, Morrison Michael).

22ST202 PROBABILITY AND STATISTICS



Source: https://www.amazon.com/Differential-Equations/dp/B01H30X2JA

Hours Per Week :

L	T	P	C
3	2	0	4

PREREQUISITE KNOWLEDGE: Basic knowledge in statistics and mathematics.

COURSE DESCRIPTION AND OBJECTIVES:

To provide students with foundation in elementary topics of statistics and probability such as descriptive statistics, correlation, probability, random variables, correlation, regression, and testing of hypothesis. The course emphasizes statistics to solve engineering and management problems.

MODULE-1

UNIT-1 **12L+0T+8P=20 Hours**

DESCRIPTIVE STATISTICS

Basic Definitions, Frequencies, Graphical Representation, Histogram, Ogive curves; Measures of Central tendency, Arithmetic mean, Median, Mode, Mean deviation, Standard deviation; Symmetry and Skewness, Karl Pearson's Coefficient of skewness.

UNIT-2 **12L+0T+8P=20 Hours**

PROBABILITY AND RANDOM VARIABLES

Probability: Introduction, Definition (Classical and Axiomatic approach), Addition theorem, Conditional probability, Multiplication theorem and Bayes theorem.

Random Variables: Random variables, Discrete and Continuous variables and distribution function.

PRACTICES:

- Various graphical presentation techniques.
- Measures of central tendency.
- Skewness.
- Karl Pearson's coefficient of skewness.
- Applications of addition theorem.
- Applications of multiplication theorem.

MODULE-2

UNIT-1 **12L+0T+8P=20 Hours**

REGRESSION ANALYSIS AND DISTRIBUTIIONS

Correlation and Regression: Correlation, types, Pearson's coefficient of correlation, regression, regression lines.

Distributions: Introduction to distributions: Binomial, Poisson and Normal distributions with properties and applications.

UNIT-2 **12L+0T+8P = 20 Hours**

TESTING OF HYPOTHESIS

Testing large samples-one mean, two means, one proportion and two proportions. Testing small

SKILLS:

- ✓ *Collect the data from various data sources and evaluate mean, median, mode mean deviation and standard deviation.*
- ✓ *Identify the areas which we can apply the probability theory.*

samples- one mean, two means (independent and paired samples), Chi square tests-goodness of fit and independence of attributes.

PRACTICES:

- Correlation.
- Karl Pearson's coefficient of correlation.
- Regression and regression lines.
- Applications of statistical distributions.
- Testing the large sample tests-one mean and two sample means.
- One proportion and two proportion tests.
- Testing small samples-one, two samples and paired tests.
- Chi-square test for goodness of fit.
- Chi-square test for independence of attributes.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Apply measures of central tendency, skewness, and Karl Pearson's coefficient of skewness to study the statistical data sets.	Apply	1	1,2
2	Apply the probability theory and their applications to measure the uncertainty.	Apply	1	1,2
3	Study the relations between statistical variables and can fit the mathematical models for association.	Analyze	2	1,2,3
4	Test the statistical significances for various samples.	Evaluate	2	1,2,4
5	Identify the distribution type to measure the occurrences of chance.	Evaluate	2	1,4,5

TEXT BOOKS:

1. Sheldon M. Ross, An Introduction to Probability and Statistics for Engineers and Scientists, 3rd Edition, Academic Press, Elsevier.
2. S. C. Gupta and V. K. Kapoor, "Fundamentals of Mathematical Statistics", Sultan Chand & Sons, 2012.

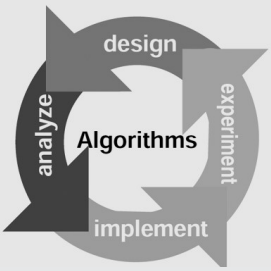
REFERENCE BOOKS:

1. P. R. Vittal, "Mathematical Statistics", Margham Publications, Chennai, 2018.
2. Kishore S. Trivedi, "Probability and Statistics with Realiability, Queueing and Computer Science Applications", 2nd edition, Wiley Student edition, 2008.
3. A. Singaravelu, "Probability and Statistics", 22nd edition, Meenakshi Agency, 2015.

XXXXXX DESIGN AND ANALYSIS OF ALGORITHMS

Hours Per Week :

L	T	P	C
3	0	2	4



Source: https://www.facebook.com/Design-and-Analysis-of-Algorithms-15539028781555564/

PREREQUISITE KNOWLEDGE: Programming for problem solving, Discrete Mathematical Structures, Data Structures.

COURSE DESCRIPTION AND OBJECTIVES:

This course offers the basic knowledge required to analyze the asymptotic performance of algorithms. In addition, this course provides the knowledge required to solve different problems using suitable design strategies such as the greedy method, divide and conquer, dynamic programming, backtracking and branch & bound. This course helps to understand the impact of the choice of data structures and algorithm design strategies on the performance. This course also provides the understanding of advanced graph applications and throws light on tractable and intractable problems.

MODULE-1

UNIT-1 **12L+0T+8P=20 Hours**

INTRODUCTION

Algorithm, Pseudo-code for expressing algorithms, Performance analysis – space and time complexity; Asymptotic notation - big oh notation, Omega notation, Theta notation and little oh notation; Analysis of recursive algorithms through recurrence relations- substitution method, Recursion tree method, Masters Theorem.

Disjoint sets: Disjoint set operations, Union and find algorithms.

UNIT-2 **12L+0T+8P=20 Hours**

DIVIDE & CONQUER AND GREEDY METHOD

Divide and Conquer: General method, Applications - Binary search, Quick sort, Merge sort and Strassen's matrix multiplication.

Greedy Method: Applications - job sequencing with deadlines, Knapsack problem, Minimum cost spanning trees.

PRACTICES:

- Sort a given set of elements using the following methods and determine the time required to sort the elements. Repeat the experiment for different values of n, the number of elements in the list to be sorted and plot a graph of the time taken versus n inputs. The elements can be read from a file or can be generated using the random number generator.
  a. Quick sort b. Merge sort
- Search for a given set of elements using the following methods and determine the time required to search the given element. Repeat the experiment for different values of n, the number of elements in the list to be sorted and plot a graph of the time taken versus no. of elements. The elements can be read from a file or can be generated using the random number generator.
  a. Linear Search b. Binary Search
- Implement the following using divide and conquer approach.
  To multiply two given square matrices.
  To multiply two given square matrices using Strassen's matrix multiplication.
- Design the Algorithm to solve Job sequencing with deadlines problem and Analyze its time complexity. Implement the above algorithm using Greedy method.

SKILLS:

- ✓ *Analyze the given algorithm concerning space and time complexities and compare it with other algorithms.*
- ✓ *Develop algorithms for solving problems using divide and conquer, greedy, dynamic programming, backtracking and branch & bound techniques.*
- ✓ *Application of existing design strategies to solve real-world problems.*

- Design the Algorithm to solve fractional Knapsack problem using Greedy method. Analyze the time complexity and implement the above algorithm.
- Design the Algorithm to find minimum spanning tree and its cost for an undirected graph. Analyze the time complexity and implement the above algorithm.

MODULE-2

UNIT-1 **12L+0T+8P=20 Hours**

DYNAMIC PROGRAMMING AND BACKTRACKING

Dynamic Programming: General method, Applications - optimal binary search trees, Matrix chain multiplication, 0/1knapsackproblem, All pairs shortest path problem, Travelling sales person problem.

Backtracking: General method, Applications - N-Queen problem, Sum of subsets problem, Graph colouring and Hamiltonian cycles.

UNIT-2 **12L+0T+8P=20 Hours**

BRANCH & BOUND AND P, NP, NP - HARD AND NP-COMPLETE

Branch and Bound: General method, Applications- Travelling sales person problem, 0/1 knapsack problem using LC branch and bound solution and FIFO branch and bound solution.

P, NP, NP - HARD and NP-Complete: Basic Concepts - Non-Deterministic Algorithms - The Classes NP-Hard and NP Complete- NP Hard Problems- Clique Decision Problem-Cook's Theorem.

PRACTICES:

- Design the Algorithm to find all pairs shortest path problem by using dynamic programming approach. Analyze its time complexity and implement the above algorithm.
- Design the Algorithm to find optimal binary search tree and its cost by using dynamic programming approach. Analyze its time complexity and implement the above algorithm.
- Design the Algorithm to find optimal order of matrix chain multiplication and its cost using dynamic programming approach. Analyze its time complexity and implement the above algorithm.
- Design the Algorithm to find optimal route for travelling sales person problem and its cost by using dynamic Programming approach. Analyze its time complexity and implement the above algorithm.
- Design the Algorithm to solve N-queens problem by using backtracking approach and Analyze its time complexity. Implement the above algorithm.
- Design the Algorithm to solve sum of subsets problem using backtracking approach and Analyze its time complexity. Implement the above algorithm.
- Design the Algorithm to solve 0/1 Knapsack problem using Branch and Bound method. Analyze the time complexity and Implement the above algorithm.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Analyze the efficiency of a given algorithm using time and space complexity theory. Understanding algorithmic design strategy like divide and conquer approach.	Analyze	1	1, 2, 12
2	Apply greedy algorithm Strategy for suit able problems and argue the correctness of such algorithms with respect to the global optimization.	Apply	1	1, 2,3, 5, 12
3	Apply the dynamic programming paradigm and identify the kind of problem best suited to solve using dynamic programming.	Apply	2	1, 2, 3, 5, 12
4	Compare and contrast the design principles of branch and bound with backtracking strategy.	Apply	2	1, 2,3,5, 12
5	Investigate computational complexity of different class of problems.	Analyze	2	1, 2,4,12

TEXT BOOKS:

1. Ellis Horowitz, SatrajSahni and Rajasekharan, “Fundamentals of Computer Algorithms”, 2nd Edition, Galgotia publications, 2006.
2. Thomas H. Coremen, Charles E. Leiserson and Ronald L. Rivest, “Introduction to Algorithm”, 2nd Edition, MIT press Ltd., 2014.

REFERENCE BOOKS:

1. Anony Levitin, “Introduction to Design and Analysis of Algorithms”, 3rd Edition, Pearson Education, 2016.
2. Donald E. Knuth, “The Art of Computer Programming”, 2nd Edition, Addison Wesley Publishing Company, 1998.
3. Ronald L. Graham, Donald E. Knuth and Oren Patashnik, “Concrete Mathematics”, 2nd Edition, Addison wesley Publishing Company,1998.
4. Dasgupta, Papadimitriou and Vazirani,“Algorithms”, 1st Edition, McGraw-Hill publishers, 2008.
5. Weiss, “Data Structures and Algorithm Analysis”, 1st Edition, Addison-Wesley Publishing Company, 2016.



Source: https://www.123rf.com/stock-photo/operating_system.html

22CS207 OPERATING SYSTEMS

Hours Per Week :

L	T	P	C
2	0	2	3

PREREQUISITE KNOWLEDGE: Knowledge of computers fundamentals, Computer organization & Digital logic and its design.

COURSE DESCRIPTION AND OBJECTIVES:

This course aims at concepts and principles of Operating Systems, its overall responsibility inacting as an interface between the system's hardware components and the user. Further, it also helps students to understand the different scheduling policies, process synchronization mechanisms, deadlock handling mechanisms and memory management techniques.

MODULE-1

UNIT-1 **10L+0T+10P=20 Hours**

LINUX FILE SYSTEM & PROCESS SCHEDULING

Introduction to LINUX File System: The LINUX file System, File System Hierarchy, File system Commands, File Attributes, File Permissions.

Filters: cmp, comm, diff, head, tail, find, cut, paste, sort, uniq.

Regular Expressions: grep, egrep, fgrep, Sed- line addressing, context addressing, text editing, substitution.

Introduction to Operating System: What Operating System do; Operating System Structure; Process concept-overview, Process Scheduling, Operations on Process; Inter Process Communication; Threads;

Process (CPU) Scheduling-Scheduling Criteria, Scheduling Algorithms; Multiple-Processor scheduling;

UNIT-2 **6L+0T+6P=12 Hours**

PROCESS SYNCHRONIZATION AND DEADLOCKS

Process Synchronization: The critical-section problem; Peterson's solution; Synchronization hardware; Semaphores; Classical problems of synchronization; Monitors.

Deadlocks: Deadlock characterization; Methods of handling deadlocks; Deadlock prevention; Deadlock avoidance; Deadlock detection and recovery.

PRACTICES:

- Use the cat command to create a file containing the following data. Call it mytable.txt usetabsto separate the fields.

  1425 ravi 15.65
  4320 ramu 26.27
  6830 sita 36.15
  1450 raju 21.86

  a. Use the cat command to display the file, mytable.txt.
  b. Use the vicomm and to correct any errors in the file, mytable.txt.
  c. Use the sort command to sort the file mytable.txt according to the first field.
  d. Call thesortedfilemytable.txt (same name)
  e. Printthefilemytable.txt.
  f. Use the cut &paste commands to swap fields 2and 3mytable.Call itmytable.txt (same name)

g. Print the new file, mytable.txt.

- Write a shell script that takes a command–line argument and reports on whether it is directory, a file, or something else.
- Write a shell script that accepts one or more file name as arguments and convertsall of them to uppercase, provided they exist in the current directory.
- Write a shell script that displays a list of all the files in the current directory to which the user has read, write and execute permissions.
- Write a shell script that computes the total and average marks of a student according to the following;
- Ifaveragemarks≥69thenresultis-Distinction‖.
- Ifaveragemarks≥59and≤70thenresultis-FirstClass‖.
- Ifaveragemarks≥49and≤60thenresultis-SecondClass‖If average marks ≤50 then result is -Pass‖.
- Note that any subject marks ≤ 40then result is-Fail‖.
- Accept student name and six subject marks through the keyboard.
- Write an interactive file-handling shell program. Let it offer the user the choice of copying, removing, renaming, or linking files. Once the user has made a choice, have the program ask the user for the necessary information, such as the file name, new name and so on.
- Write a shell script, which receives two file names as arguments. It should check whether the two file contents are same or not. If they are same then second file should be deleted.
- Write a shell script that accepts a file name starting and ending line numbers as arguments and displays all the lines between the given line numbers.
- Write a shell script that deletes all lines containing a specified word in one or more files supplied as arguments to it.
- Implementation of new process creation and its communications.
- Implement of thread creation and deletion.
- Implementation of FCFS scheduling.
- Implementation of SJF and RR Scheduling.
- Implementation of producer consumer problem.
- Implementation of Banker's algorithm for Dead lock avoidance.

SKILLS:

- ✓ *Manage open-source operating systems like Ubuntu, Fedora etc.*
- ✓ *Know the concepts of Processes scheduling and File Systems.*
- ✓ *Identification of different disk scheduling methodologies.*

MODULE-2

UNIT-1 **8L+0T+8P=16 Hours**

MEMORY MANAGEMENT

Memory Management: Basic concept tofmemory management, Swapping, Contiguous Memory Allocation, Paging, Structure of the Page Table, Segmentation.

Virtual Memory Management: Demand Paging, Page Replacement: Optimal, First in First Out (FIFO), Second Chance (SC), Not recently used (NRU), Least Recently used (LRU), Allocation of Frames.

UNIT-2 **8L+0T+8P=16 Hours**

SECONDARY STORAGE STRUCTURE

Secondary Storage Structure: Over view of mass-storage structure, disk structure, disk scheduling;

File System Interface - File concept, Access Methods, Directory & Disk Structure, File-System Mounting, File Sharing, Protection; File-system structure.

File System Implementation- Directory implementation, Allocation Methods, Free Space Management.

PRACTICES:

- Assume that you have a page-reference string for a process with m frames (initially all empty). The page-reference string has length p, and n distinct page numbers occur in it.
  a) What is a lower bound on the number of page faults?
  b) What is an upper bound on the number of page faults?
- Consider the following page-replacement algorithms. Rank these algorithms on a five-point scale from "bad" to "perfect" according to their page-fault rate. Separate those algorithms that

suffer from Belady's anomaly from those that do not.

a) LRU replacement. b) FIFO replacement.

c) Optimal replacement. d) Second-chance replacement.

- Consider the page reference string: 1, 2, 3, 4,2, 1, 5, 6, 2, 1, 2, 3,7, 6, 3, 2, 1, 2, 3, 6.
- How many page faults would occur for the following replacement algorithms, assuming one, two, three, four, five, six, and seven frames? Remember that all frames are initially empty, so your first unique pages will cost one fault each.
- LRU replacement.
- FIFO replacement.
- Optimal replacement.
- How many page fault soccur for your algorithm for the following reference string with four page frames? 1, 2, 3,4, 5, 3, 4, 1, 6,7, 8, 7, 8, 9, 7, 8,9, 5, 4, 5,4, 2.
- What is the minimum number of page faults for an optimal page replacement strategy for thereference string above with four page frames?
- Consider a demand-paged computer system where the degree of multiprogramming is currently fixed at four. The system was recently measured to determine utilization of the CPU and the paging disk. Three alternative results are shown below. For each case, what is happening?

  a) Can the degree of multiprogramming be increased to increase the CPU utilization? Is the paging helping?

  b) CPU utilization 13 percent; disk utilization 97 percent.

  c) CPU utilization 87percent; disk utilization 3 percent.

  d) CPU utilization 13 percent; disk utilization 3 percent.
- Implementation of Disk scheduling algorithm–FCFS.
- Implementation of Disk scheduling algorithm–SSTF and SCAN.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Classify the basic concepts of operating system and explore Linux ecosystem.	Analyze	1	1
2	Apply the concepts of process scheduling algorithms and process synchronization techniques to derive the efficiency of resource utilization.	Apply	1	1, 2, 3, 5, 12
3	Analyze the requirements for attempting Operating systems principles.	Analyze	1,2	1,2,12
4	Design the various memory management schemes For a given scenario.	Create	2	3,5
5	Apply the concepts of file system interface and implementation.	Apply	1,2	2,5

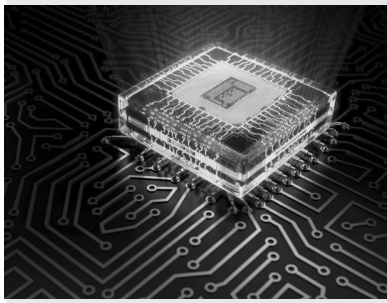
TEXT BOOKS:

1. Sumitabha Das, Unix concepts and applications‖, TMH Publications, 4th Edition, July 2017.
2. Abraham Silberschatz, Peter Baer Galvin and Greg Gagne, "Operating System Concepts", 9th Edition, John Wiley & SonsInc, 2013.

REFERENCE BOOKS:

1. Richard. Stevens and Stephen A Rago, "Advanced Programming in the Unix Environment", 3rd Edition, Addison-Wesley, 2013.
2. William Stallings, "Operating Systems-Internals and Design principles" PHI, 7th Edition, 2012.
3. Gary J. Nutt. Addison-Wesley, "Operating Systems: A Modern Perspective", 2nd Edition, 2001.
4. B.A. Forouzan & R.F.Giberg, - Unix and shell Programming‖, Thomson, 1st Edition, New Delhi, 2003.

XXXXX COMPUTER ORGANIZATION



Source: https://machinelearningmedium.com/assets/images/computer-architecture.jpg

Hours Per Week :

L	T	P	C
2	2	0	3

PREREQUISITE KNOWLEDGE: Digital logic design.

COURSE DESCRIPTION AND OBJECTIVES:

This course covers the basics of modern Computer Organization and Architecture. The emphasis is on understanding the design of computer and its components. The student will learn the concepts of data representation, micro-operations, memory organizations and input output organization.

MODULE-1

UNIT-1 **8L+8T+0P=16 Hours**

INTRODUCTION, RTL, DATA REPRESENTATION AND COMPUTER ARITHMETIC

Introduction, Register Transfer language & Data Representation: Organization and Architecture, Register Transfer, Bus and Memory Transfers, Data Representation-Fixed Point Representation, Floating Point Representation.

Computer Arithmetic: Fixed point arithmetic operations such as Addition and Subtraction, Multiplication Algorithms, Division Algorithms.

UNIT-2 **8L+8T+0P=16 Hours**

MICRO OPERATIONS AND BASIC COMPUTER ORGANIZATION AND DESIGN

Micro operations: Arithmetic Micro operations, Logic Micro Operations, Shift Micro Operations, Arithmetic Logic Shift Unit.

Basic Computer Organization and Design: Instruction Codes, Computer Register, Computer Instructions, Instruction Cycle, Memory – Reference Instructions. Register Reference Instructions, Input –Output and Interrupt.

PRACTICES:

- Design a Common bus system for eight registers with eight bits each using multiplexers.
- Design a Common bus system for four registers with four bits each using three state gate buffers.
- A digital computer has a common bus system for 16 registers of 32 bits each. The bus is constructed with multiplexers.

  How many selection inputs are there in each multiplexer?

  What size of the multiplexers are needed?

  How many multiplexers are there in the bus?
- Perform arithmetic operations (+42) + (-13) and (-42) – (-13) in binary using signed 2's complement representation for negative numbers.
- Find the product using Booth Multiplication Algorithm.
- a. (9) X (13) b. (9) X (-13) c. (-9) X (13) d. (-9) X (-13)
- Perform the division of 27 and 4 using Division algorithm.
- Design a 4- bit combinational circuit decrementer using 4 full adder circuits.
- Register A holds the 8-bit binary 11011001. Determine the B operand and the logic micro operation to be performed in order to change the value in A to :

  a) 01101101 b) 11111101

SKILLS:

- ✓ *Learn different data representations.*
- ✓ *Design digital circuitry for implementing different operations.*
- ✓ *Identify the types of memories and their uses.*
- ✓ *Study various data transfer mechanisms in digital computer and I/O.*

- An 8-bit register contains the binary value 10011100. What is the register value after an arithmetic shift right? Starting from the initial number 10011 100, determine the register value after an arithmetic shift left, and state whether there is an overflow.
- Starting from an initial value of R =11011101, determine the sequence of binary values in R after a logical shift-left, followed by a circular shift-right, followed by a logical shift-right and a circular shift-left.
- Design arithmetic logic shift unit that performs different operations on 4 bits.

MODULE-2

UNIT-1 **8L+8T+0P=16 Hours**

CPU AND MEMORY ORGANIZATION

Central Processing Unit: General Register Organization, STACK Organization. Instruction Formats, Addressing Modes, Data Transfer and Manipulation, Program Control.

Memory Organization: Memory Hierarchy, Main Memory, Auxiliary Memory, Associative Memory, Cache Memory.

UNIT-2 **8L+8T+0P=16 Hours**

I/O ORGANIZATION

Input-Output Organization: Peripheral Devices, Input-Output Interface, Asynchronous data transfer, Modes of Transfer, Priority Interrupts, Direct Memory Access.

PRACTICES:

- The content of PC in the basic computer is 3AF (all numbers are in hexadecimal). The content of AC is 7EC3. The content of memory at address 3AF is 932E. The content of memory at address 32E is 09AC. The content of memory at address 9AC is 8B9F.

  What is the instruction that will be fetched and executed next?

  Show the binary operation that will be performed in the AC when the instruction is executed.

  Give the contents of registers PC, AR, DR, AC, and IR in hexadecimal and the values of E, I, and the sequence counter SC in binary at the end of the instruction cycle.
- Implement the given expressions into different addressing architectures.
  Y=(A-B)/(C*D + E) b. Y=A-B+C*(D *E+F)
- How many 128 x 8RAM chips are needed to provide a memory capacity of 2048 byte?
- How many lines of the address bus must be used to address 2048 bytes of memory? How many of these lines will be common to all chips?
- How many lines must be decoded for chip select and design the size of the decoders.
- A computer uses RAM chips or 1024 x 1 capacity.
  How many chips are needed, and show the connection of memory capacity 1024 bytes?

  How many chips are needed to provide a memory capacity or 16K bytes? Explain in words how the chips are to be connected to the address bus.
- How many characters per second can be transmitted over a 1200-baud line in each of the following modes? (Assume a character code of eight bits.)
  Synchronous serial transmission.

  Asynchronous serial transmission with two stop bits.

  Asynchronous serial transmission with one stop bit.
- Information is inserted into a FIFO buffer at a rate of m bytes per second. The information is deleted at a rate of n byte per second. The maximum capacity of the buffer is k bytes.
  How long does it take for an empty buffer to fill up when m >n?

  How long does it take for a full buffer to empty when m <n?

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Analyze Computer Organization and Computer Architecture, different arithmetic operations.	Analyze	1	1, 2, 12
2	Design different digital circuits required to perform the micro operations.	Apply	1	1, 2, 3, 12
3	Design interface circuits for memory and peripheral, DMA and communication devices. Compare various modes of transfer.	Analyze	2	1, 2, 3, 4, 12
4	Evaluate the performance of a processor and memory in terms of speed, size and cost.	Evaluate	2	1, 2, 12

TEXT BOOKS:

1. M. Morris Mano, "Computer System Architecture", 3rd Edition update, Pearson, 2017.
2. William Stallings, "Computer Organization & Architecture: Designing for Performance", 11th Edition, Pearson, 2019.

REFERENCE BOOKS:

1. Carl Hamacher, ZvonkoVranesic, SafwatZaky, "Computer Organization", 5th Edition, McGraw Hill, 2002.
2. Vincent P. Heuring and Harry F Jordan, " Computer Systems Design and Architecture ", 2nd edition , Pearson/ Prentice Hall India 2004.
3. David A. Patterson and John L. Hennessy, "Computer Organization and Design-The Hardware/ Software Interface", ARM Edition, 5th Edition, Elsevier, 2009.c) L={ww^r | w is (0+1)*}

REFERENCE BOOKS:

1. Zed A Shaw, Learn C the Hard Way: Practical Exercises on the Computational Subjects You Keep Avoiding (Like C), Addison Wesley, 2015.
2. Christoph Dürr, Sorbonne University, Jill-Jênn Vie, Inria, Competitive programming in Python, Cambridge University Press, 2020.
3. Michael Sipser, "Introduction to Theory of Computation", 3rd Edition, Course Technology, 2012.

III

YEAR

COMPUTER SCIENCE AND ENGINEERING

B.Tech.

COURSE CONTENTS

I SEM & II SEM

I SEMESTER

22TP301	-	Soft Skills Laboratory
24CS301	-	Optimization Techniques
24CS302	-	Artificial Intelligence
24CS303	-	Computer Networks
24CS304	-	Mobile App Development
	-	Department Elective - 1
	-	Open Elective - 2
24EN301	-	BEC Certification
	-	Minor/Honours - 2

II SEMESTER

24CS305	-	Computing Ethics
24TP301	-	Quantitative Aptitude and Logical Reasoning
24CS306	-	Machine Learning
XXXXXXX	-	Cryptography and Network Security
	-	Department Elective - 2
	-	Department Elective - 3
	-	Open Elective - 3
24CS307	-	Inter Departmental Project
	-	Minor/Honours - 3

22TP301 SOFT SKILLS LABORATORY



Source: https://choosework.ssa.gov/blog/2019-07-23-soft-skills-an-intro-to-effective-communication

Hours Per Week :

L	T	P	C
0	0	2	1

PREREQUISITE KNOWLEDGE: Grasp on their own academic achievements.

COURSE DESCRIPTION AND OBJECTIVES:

To impart employability skills like resume preparation and facing interviews. To enable trainees to develop interpersonal and leadership skills and to train them on work place skills like making presentations, participating in group discussions etc.

MODULE-1

UNIT-1 **0L+0T+8P=8 Hours**

PERSONALITY DEVELOPMENT

Soft Skills: Need for soft skills, professionalism, employability skills; Communication: Need for effective communication - the process of communication, levels of communication, flow of communication, choice of diction and style with reference to setting (formal, semi-formal or informal); communication networks, barriers to communication, miscommunication, noise and ways to overcome the barriers; Career Planning: Job vs. career, SWOT analysis.

UNIT-2 **0L+0T+8P=8 Hours**

LANGUAGE AND VOCABULARY

Vocabulary Building: Word etymology, roots, prefixes & suffixes, synonyms & antonyms, collocations, one-word substitutes, analogies, idioms and phrases, contextual guessing of unfamiliar words, task-oriented learning; Reflection of language on Personality, Gender sensitive language in MNCs, Mind your language, Seven essential skills for a team player; attentive listening, intelligent questioning, gently persuading, respecting other's views, assisting others, sharing, participating actively.

PRACTICES:

- Self-Introduction.
- Personal and Academic SWOC.
- Johari Window.
- Giving and taking opinions of Self Vs others and assessing oneself.
- Goal setting.
- Short, Mid and Long Term goals planning the semester.
- Time management: four quadrant system.
- Stephen Covey Time Management Matrix planning a semester.
- Stress-management.
- Questionnaire to assess level of stress.
- 50 words towards resume preparation and interviews.
- Newly coined words.
- Gender sensitive words and Words acceptable in Indian context and objectionable international context.

MODULE-2

UNIT-1 **0L+0T+8P=8 Hours**

LANGUAGE IN ACTION

Functional English: Situational dialogues, Role plays (including small talk); Group Discussion: Articulation and flow of oral presentation, dynamics of group discussion, intervention, summarizing and conclusion, voice modulation, content generation, Key Word Approach (KWA), Social, Political, Economic,

SKILLS:

- ✓ *Balance social and emotional intelligence quotients though SWOC, JOHARI etc. activities.*
- ✓ *Prepare tailor made resume and face various job interviews with enriched personality traits.*
- ✓ *Career planning with clear personal and professional goals.*
- ✓ *Solve personal and professional life hiccups with confidence and maturity*

Legal and Technical Approach (SPELT), View Point of Affected Part (VAP), language relevance, fluency and coherence – 11th and 12th weeks; Resume preparation: Structure and presentation, defining career objective, projecting one's strengths and skill-sets, summarizing, formats and styles and covering letter-Statement of Purpose.

UNIT-2 **0L+0T+8P=8 Hours**

PREPARING FOR PRESENTATIONS AND INTERVIEWS

Facing Interviews: Interview process, understanding employer expectations, pre-interview planning, opening strategies, impressive self-introduction, answering strategies, other critical aspects such as body language, grooming, other types of interviews such as stress-based interviews, tele- interviews, video interviews, frequently asked questions (FAQs) including behavioral and HR questions and the aspect looked at by corporate during interviews; Presentation Skills: Selection of a topic, preparing an abstract, gathering information, organizing the information, drafting the paper, citing reference sources – writing striking introductions, discussing the methodology used, developing the argument, presentation style, language, presenting the paper and spontaneously answering audience questions.

PRACTICES:

- Opening and closing a telephonic conversation.
- Making an appointment.
- Making a query.
- Offering/Passing on information.
- Communicating with superiors.
- Expressing agreement/objection.
- Opening bank account (combination of prepared and impromptu situations given to each student).
- Group Discussions on various topics.
- Preparing SoP and Resume.
- Mock interviews on the FAQs including feedback.
- Oral presentation with the help of technology (Preparing PPT and presenting).

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Have the ability to introspect on individual strengths and weaknesses, and emerge as a balanced personality with improved self-awareness and self-worth.	Apply	1	12
2	Observe gender sensitive language and workplace etiquette in his professional life.	Analyze	1	9
3	Be able to prepare a resume and gain the confidence to face an interview.	Create	1, 2	10
4	Possess the interpersonal skills to conduct himself/herself effectively in everyday professional and social contexts.	Apply	2	8
5	Bring professionalism into his/her daily activities.	Create	2	8

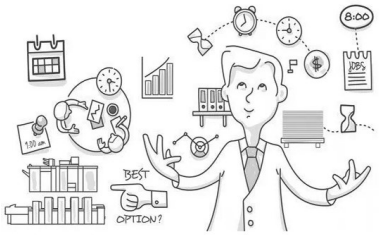
TEXT BOOKS:

1. Adrian Furnham, "Personality and intelligence at work", Psychology Press, 2008.
2. S. P. Dhanvel, "English and Soft skills", Orient Blackswan, 2011.

REFERENCE BOOKS:

1. Edward Holffman, "Ace the corporate personality", McGraw Hill, 2001.
2. John Adair Kegan Page, "Leadership for innovation", Kogan, 2007.
3. Krishna Mohan & NP Singh, "Speaking English effectively", Macmillan, 2008.
4. Rajiv K. Mishra, "Personality Development", Rupa & Co. 2004.

24CS301 OPTIMIZATION TECHNIQUES



Source: https://oworkers.com/8-process-optimization-techniques-how-to-get-started/

Hours Per Week :

L	T	P	C
2	2	0	3

PREREQUISITE KNOWLEDGE: Probability & Linear algebra.

COURSE DESCRIPTION AND OBJECTIVES:

This course introduces the fundamental concepts of Optimization Techniques and to make the learners aware of the importance of optimizations in real scenarios. The most important objective is to provide the concepts of various classical and modern methods for constrained and unconstrained problems in both single and multivariable. Finally, the basic idea behind the evolutionary algorithms like Ant Colony Optimization, Particle Swarm Optimization could be discussed for further study.

MODULE-1

UNIT-1 **12L+8T+0P=20 Hours**

INTRODUCTION TO CLASSICAL METHODS AND LINEAR PROGRAMMING PROBLEMS TERMINOLOGY

Design Variables – Constraints, Objective Function - Problem Formulation, Linear Programming Problem - Simplex method - Concept of Duality.

General Transportation Problem: The transportation table finding in initial basic feasible solution, North-West corner method, Least cost method, Row minima method, Column minima method.

UNIT-2 **12L+8T+0P=20 Hours**

SINGLE VARIABLE OPTIMIZATION

Problems Optimality Criterion - Bracketing Method - Region Elimination Methods -Interval Halving Method - Gradient Based Methods: Newton -Raphson Method - Bisection Method - Application to Root finding.

Multi Variable Optimization, Algorithms Optimality: Criteria - Unidirectional Search. Direct Search Methods: Hooke -Jeeves pattern search method. Gradient Based Methods: Cauchy's Steepest Descent Method - Newton's method, Quasi-Newton methods, L-BFGS.

PRACTICES:

A company makes two products (X and Y) using two machines (A and B). Each unit of X that is produced requires 50 minutes processing time on machine A and 30 minutes processing time on machine B. Each unit of Y that is produced requires 24 minutes processing time on machine A and 33 minutes processing time on machine B. At the start of the current week there are 30 units of X and 90 units of Y in stock. Available processing time on machine A is forecast to be 40 hours and on machine B is forecast to be 35 hours. The demand for X in the current week is forecast to be 75 units and for Y is forecast to be 95 units. Company policy is to maximise the combined sum of the units of X and the units of Y in stock at the end of the week. Formulate the problem of deciding how much of each product to make in the current week as a linear program. Solve this linear program graphically.

SKILLS:

- ✓ *Project scheduling process skills including defining project activities, and estimation of time and resources*
- ✓ *Resource optimization skills to adjust the project schedule as per the demand and supply issues of project resources*
- ✓ *Feasibility Analysis.*

Solve using the Simplex method the following problem:

Maximize Z = f(x,y) = 3x + 2y

subject to: $2x + y \leq 18$

$2x + 3y \leq 42$

$3x + y \leq 24$

$x \geq 0 , y \geq 0$

Luminous lamps have three factories - F1, F2, and F3 with production capacity 30, 50, and 20 units per week respectively. These units are to be shipped to four warehouses W1, W2, W3, and W4 with requirement of 20, 40, 30, and 10 units per week respectively. The transportation costs (in Rs.) per unit between factories and warehouses are given below. Find an initial basic feasible solution of the given transportation problem using northwest corner rule.

Factory	**Warehouse**				**Supply**
	W1	W2	W3	W4	
F1	1	2	1	4	30
F2	3	3	2	1	50
F3	4	2	5	9	20
Demand	20	40	30	10	

- A mobile phone manufacturing company has three branches located in three different regions, say Jaipur, Udaipur and Mumbai. The company has to transport mobile phones to three destinations, say Kanpur, Pune and Delhi. The availability from Jaipur, Udaipur and Mumbai is 40, 60 and 70 units respectively. The demand at Kanpur, Pune and Delhi are 70, 40 and 60 respectively. The transportation cost is shown in the matrix below (in Rs). Use the Least Cost method to find a basic feasible solution (BFS).

		DESTINATIONS			
		Kanpur	**Pune**	**Delhi**	**Supply**
Sources	Jaipur	4	5	1	40
	Udaipur	3	4	3	60
	Mumbai	6	2	8	70
	Demand	**70**	**40**	**60**	**170**

- Find Solution using Row minima method.

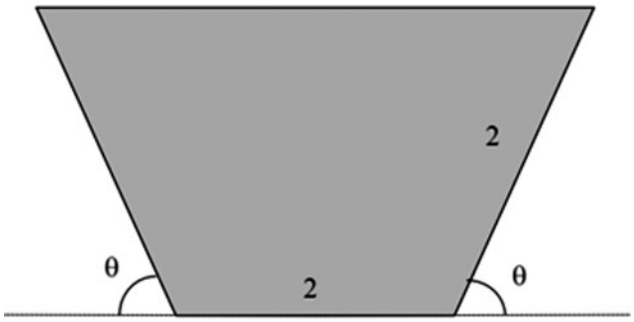
	D1	**D2**	**D3**	**D4**	**Supply**
S1	11	13	17	14	250
S2	16	18	14	10	300
S3	21	24	13	10	400
Demand	200	225	275	250	

- Find Solution using Column minima method.

	D1	D2	D3	D4	Supply
S1	11	13	17	14	250
S2	16	18	14	10	300
S3	21	24	13	10	400
Demand	200	225	275	250	

- Find the positive root of the following equation by (Bisection) method,
  $f(x)=\sin(x/2\pi)-\cos^2 x$
- Consider figure below. The cross-sectional area A of a gutter with equal base and edge length of 2 is given by:
  $A = 4\sin\theta\,(1+\cos\theta)$

Using an initial interval of $[0,\pi/2]$, find the interval after 3 iterations. Use an initial interval $\varepsilon = 0.2$.



MODULE-2

UNIT-1 **12L+8T+0P=20 Hours**

STOCHASTIC OPTIMIZATION PROBLEMS

Notion of regret, online to batch conversion, Methods offering vanishing regret - OGD, EG, OMD Convex Sets and Functions: Affine and convex sets, convexity preserving operations, separating and supporting hyper-planes, generalized inequalities, Operations preserving convexity, conjugate function, Quasi-convex functions, Log-concave and log-convex functions.

UNIT-2 **12L+8T+0P=20 Hours**

NON-CONVEX OPTIMIZATION PROBLEMS

Applications - sparse recovery, affine rank minimization, low-rank matrix completion, Convex approaches - relaxation-based methods, Non-convex approaches - projected gradient descent.

Algorithms: Unconstrained problems, equality constrained problems

PRACTICES:

- Show that each convex cone is indeed a convex set.
- Consider the linear system $0 \le x_i \le 1$ for $i = 1, \ldots, n$ and let P denote the solution set. Explain how to solve a linear programming problem $\max\{c^T x : x \in P\}$.
  What if the linear system was $a_i \le x_i \le b_i$ for $i = 1, \ldots, n$. Here we assume $a_i \le b_i$ for each i.
- Can projected gradient descent (PGD) be used to obtain a stationary solution?
- Show that strong smoothness does not imply convexity by constructing a nonconvex function f : R
- p → R that is 1-SS.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Understand the classical, linear programming and transportation problem along with game theory.	Understand	1	1
2	Analyze real world problems around you.	Analyze	1	1, 2
3	Apply optimization to real-world problems.	Apply	1	1, 2, 3
4	Implement dynamic programming in real world complex problems.	Implement	2	3, 4, 11, 12

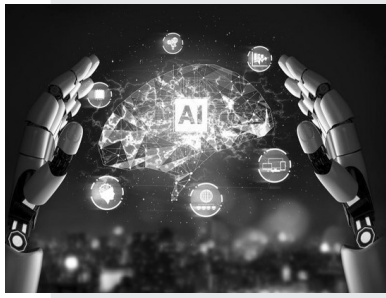
TEXT BOOKS:

1. Kanti Swarup, Man Mohan and P.K.Gupta, “Operations Research”, Sultan Chand &Sons, 2005.
2. S. S. Rao, “Engineering Optimization Theory and Practice”, 4th Edition, Wiley Publishers, 2009.

REFERENCE BOOKS:

1. M.C. Bhuvaneswari, “Application of Evolutionary Algorithms for Multi-Objective Optimization in VLSI and Embedded Systems”, Spinger, 2014.
2. Ashlock D, “Evolutionary Computation for Modeling and Optimization”, Springer, 2006.
3. Kalyanmoy Deb, “Optimization for Engineering Design”, Prentice Hall of India, 2nd Edition, 2012.
4. A. Ravindran and K.M. Rogsdell, G.V. Reklaites, “Engineering Optimization: Methods and Applications”,Wiley, 2nd Edition, 2006.
5. E. J. Haug and J.S. Arora, “Applied Optimal Design”, Wiley Publishers,1979.

24CS302 ARTIFICIAL INTELLIGENCE



Source: https://www.forbes.com/sites/bernardmarr/2020/08/03/3-important-ways-artificial-intelligence-will-transform-your-business-and-turbocharge-success/

Hours Per Week :

L	T	P	C
2	0	2	3

PREREQUISITE KNOWLEDGE: Probability and statistics.

COURSE DESCRIPTION AND OBJECTIVES:

The primary objective of this course is to introduce the basic principles, techniques, and applications of Artificial Intelligence. In addition to this, student will understand the building blocks of AI such as search, knowledge representation, inference, logic and learning. This course enables the students to develop a small AI system for real time problems.

MODULE-1

UNIT-1 **6L+6T+0P=12 Hours**

INTELLIGENT SYSTEMS

Introduction, what is AI, Examples of AI systems, Brief history of AI Agent, Agents and environments, Structure of agents, the concept of rationality, the nature of environments, Types of agents, problem solving approaches to typical AI problem.

UNIT-2 **10L+10T+0P=20 Hours**

PROBLEM SOLVING

State Space Problem; Searching: Uniform search, Informed Search: Solving problems by searching: Heuristic functions, Hill climbing, Best First Search, A* algorithm, AO* algorithm, Searching game trees: Min Max Search, Alpha Beta pruning.

PRACTICES:

- In the classical vacuum cleaner problem, we have two rooms and one vacuum cleaner. There is dirt in both the rooms and it is to be cleaned. The vacuum cleaner is present in any one of these rooms. Find the solution, how we can reach to reach a state in which both the rooms are clean and are dust free.
- In this problem, three missionaries and three cannibals must cross a river using a boat which can carry at most two people, under the constraint that, for both banks, that the missionaries present on the bank cannot be outnumbered by cannibals. The boat cannot cross the river by itself with no people on board. Find the solution, how to solve the given problem.
- You are given two jugs, a 4-gallon one and a 3-gallon one, a pump which has unlimited water which you can use to fill the jug, and the ground on which water may be poured. Neither jug has any measuring markings on it. Find the solution, how can you get exactly 2 gallons of water in the 4-gallon jug?
- There is a farmer who wishes to cross a river but he is not alone. He also has a goat, a wolf, and a cabbage along with him. There is only one boat available which can support the farmer and either of the goat, wolf or the cabbage. So at a time, the boat can have only two objects (farmer and one other). But the problem is, if the goat and wolf are left alone (either in the boat or onshore), the wolf will eat the goat. Similarly, if the Goat and cabbage are left alone, then goat will eat the cabbage. The farmer wants to cross the river with all three of his belongings: goat, wolf, and cabbage. What strategy he should use to do so?
- Either place a block that doesn't have other blocks stacked on top of it on another block with the same behaviour, or on the table. The initial and the goal state are described by the exact position of each block. Find the solution, how to solve the given problem.

SKILLS:

- ✓ *Analyze Intelligent systems.*
- ✓ *Apply problem solving techniques.*
- ✓ *Interface various knowledge representation.*
- ✓ *Create a dynamic planning.*

- Given a 3×3 board with 8 tiles (every tile has one number from 1 to 8) and one empty space. The objective is to place the numbers on tiles to match the final configuration using the empty space. We can slide four adjacent (left, right, above, and below) tiles into the empty space. Find the solution, how to solve the given problem by using using A* search algorithm.
- The rules of tic-tac-toe on the 3 × 3 field are as follows. Before the first turn all the field cells are empty. The two players take turns placing their signs into empty cells (the first player places Xs, the second player places Os). The player who places Xs goes first, the another one goes second. Find the solution, how to solve the given problem where the winner is the player who first gets three of his signs in a row next to each other (horizontal, vertical or diagonal).
- In crypt arithmetic problem, the digits (0-9) get substituted by some possible alphabets or symbols. The task in crypt arithmetic problem is to substitute each digit with an alphabet to get the result arithmetically correct. Find the solution, how to solve the given problem, where we can perform all the arithmetic operations on a given crypt arithmetic problem.

MODULE-2

UNIT-1 **10L+10T+0P=20 hours**

KNOWLEDGE REPRESENTATION AND PLANNING

Propositional logic: Inference in propositional logic, Resolution, Forward chaining, Backward chaining, First order logic: Reasoning patterns in First order logic, Resolution, Forward chaining, Backward chaining, The planning problem: Planning with state space search, Partial order planning, Planning graphs.

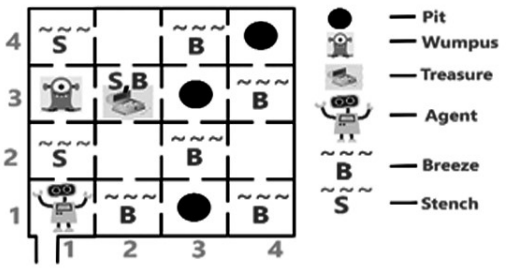
UNIT-2 **6L+6T+0P=12 hours**

LEARNING

Forms of learning: Supervised Learning, Unsupervised learning, Reinforcement learning, Learning Decision Trees, Ensemble Learning, Expert system.

PRACTICES:

- With logic programming, compare expressions and find out unknown values.
- The Wumpus world is a cave with 16 rooms (4×4). Each room is connected to others through walkways (no rooms are connected diagonally). The knowledge-based agent starts from Room [1, 1]. The cave has – some pits, a treasure and a beast named Wumpus. The Wumpus cannot move but eats the one who enters its room. If the agent enters the pit, it gets stuck there. The goal of the agent is to take the treasure and come out of the cave. The agent is rewarded, when the goal conditions are met. The agent is penalized, when it falls into a pit or being eaten by the Wumpus. Some elements support the agent to explore the cave, like -The Wumpus's adjacent rooms are stench. -The agent is given one arrow which it can use to kill the Wumpus when facing it (Wumpus screams when it is killed). – The adjacent rooms of the room with pits are filled with breeze. -The treasure room is always glittery. Find the Wumpus presented room.



- you are on one side of a river with a wolf, a goat, and a cabbage. You want to transport all three to the other side of the river, but you can only transport one object at a time. You cannot leave the wolf and the goat alone, or the cabbage and the goat alone; you are the only thing keeping them from eating each other. How can you transport everything from one side of the river to the other? Formulate it in terms of a Planning Domain Definition Language (PDDL).

- Impliment the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.
- Build an Artificial Neural Network by implementing the Back propagation algorithm and test the same using appropriate data sets.
- Implement k-nearest neighbors classification using python.
- Implement linear regression using python.
- Implement the naïve Bayesian classifier for a sample training dataset. Compute the accuracy of the classifier, considering few test data sets.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

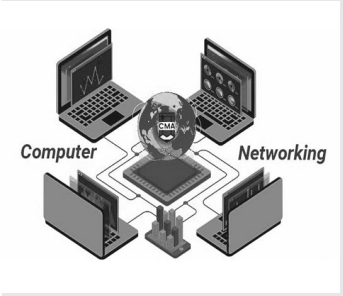
CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Apply AI search Models and Generic Search strategies for problem solving.	Apply	1	1,3
2	Inspect and analyze Logic for representing Knowledge and Reasoning of AI systems and Conduct investigation and implement project using AI learning techniques.	Analyze	1	2
3	Apply and evaluate the searching strategies to achieve the goal for a given situation..	Apply	2	6
4	Design different learning algorithms for improving the performance of AI systems.	Apply	2	4

TEXT BOOK:

1. S. Russel and P. Norvig, “Artificial Intelligence – A Modern Approach”, 4th Edition, Pearson Education, 2010.

REFERENCE BOOKS:

1. David Poole, Alan Mackworth, Randy Goebel, “Computational Intelligence : a logical approach”, Oxford University Press,2013.
2. G. Luger, “Artificial Intelligence: Structures and Strategies for complex problem solving”, 4th Edition, Pearson Education, 2008.
3. J. Nilsson, “Artificial Intelligence: A new Synthesis”, Elsevier Publishers. 2012.



Source: https://snabaynetworking.com/what-is-computer-network-and-its-types/

24CS303 COMPUTER NETWORKS

Hours Per Week :

L	T	P	C
3	0	2	4

PREREQUISITE KNOWLEDGE: JAVA programming and UNIX commands.

COURSE DESCRIPTION AND OBJECTIVES:

This course focuses on imparting knowledge about various protocols involved in LANs and WANs. In addition, it gives a good foundation on different protocols such as data link protocols, internet protocols, and transport protocols present in the respective layers of the data communication system.

MODULE-1

UNIT-1 **12L+0T+8P=20 Hours**

INTRODUCTION TO COMPUTER NETWORKS AND INTERNET

Understanding of network and Internet, the network edge, the network core, Understanding of Delay, Loss and Throughput in the packet switching network, protocols layers and their service model, History of the computer network.

UNIT-2 **12L+0T+8P=20 Hours**

DATA LINK LAYER

Introduction to link layer services, error detection, and correction techniques, Multiple access links and protocols, Switched Local Area Networks.

PRACTICES:

- Study different types of Network cables (Copper and Fiber) and prepare cables (Straight and Cross) to connect two or more systems.
  a) Use a crimping tool to connect jacks.
  b) Use a LAN tester to connect the cables.
- Install and configure Network Devices: HUB, Switch, and Routers (Consider both manageable and non-manageable switches. Perform logical configuration of the system and set the bandwidth of different ports).
- Install and Configure Wired and Wireless NIC and transfer files between systems in Wired LAN and Wireless LAN. Consider both ad-hoc and infrastructure modes of operation.
- Apply commands such as Ping, Tracert, Ipconfig, pathping, telnet, FTP, getmac, ARP, Hostname, Nbtstat, netdiag, and Nslookup to solve various problems.
- Packet Sniffer using Wireshark.
- Wireshark Lab: Ethernet and ARP
- Wireshark Lab: 802.11.

MODULE-2

UNIT-1 **12L+0T+8P = 20 Hours**

NETWORK LAYER

Introduction to forwarding and routing, Network Service models, Virtual and Datagram networks, study of router, IP protocol and addressing on the Internet, Routing algorithms, Broadcast and Multicast routing.

UNIT-2 **12L+0T+8P=20 Hours**

TRANSPORT LAYER

Introduction and transport layer services, Multiplexing and Demultiplexing, Connectionless transport (UDP), Principles of reliable data transfer, Connection-oriented transport (TCP), and Congestion control.

APPLICATION LAYER- Web and HTTP, FTP, E-mail, DNS, Socket programming with TCP and UDP.

PRACTICES:

- Find all the IP addresses on your network using Unicast, Multicast, and Broadcast on your network.
- Use Packet tracer software to build network topology and configure using Distance vector routing and Link State routing protocols.
- Implementation of one-way and two-way communication using TCP / UDP.
- Hello command is used to know whether the machine at the other end is working or not. Echo command is used to measure the round-trip time to the neighbor. Implement Hello and Echo commands using JAVA.
- Using Packet Tracer to experience all the simulations.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Build the basic concepts of Network hardware, software and reference models.	Apply	1	1, 2, 12
2	Evaluate different physical layer media and switching methods.	Evaluation	1	1, 2, 5, 12
3	Implement various protocols with modern tools.	Apply	1	1, 2, 3, 5, 12
4	Apply different protocols to perform end-to-end delivery and interaction with users.	Analyze	2	1, 2, 12
5	Analyze various design issues, protocols and functionalities of network layer.	Analyze	2	1, 2, 12
6	Demonstrate various protocols involved in data link layer operations.	Apply	2	1,2, 5

TEXT BOOKS:

1. Kurose and Ross, “Computer Networking- A Top-Down approach”, 6th Edition, Pearson, 2017.
2. Behrouz Forouzan, “Computer Networks- A Top-Down approach”, McGraw Hill, 2014.

REFERENCE BOOKS:

1. Andrew S. Tanenbaum, “Computer Networks”, 5th edition. Pearson Education, 2014.
2. Behrouz A. Forouzan, “Data communications and Networking”, 5th edition, TMH, 2017.
3. William Stallings, “Data and Computer Communications”, 10th edition, Pearson Education, 2017.
4. Fred Halsall, “Computer Networking and the Internet”, 5th edition, Addison Wesley, 2005.

SKILLS:

- ✓ *Establish local area networks with different topologies.*
- ✓ *Design of new routing protocols.*
- ✓ *Network troubleshooting such as installing network interface card drivers, setting IP addresses, subnet masking, etc.*



Source: https://www.tatvasoft.com.au/blog/mobile-application-development-methodology/

24CS304 MOBILE APPLICATION DEVELOPMENT

Hours Per Week :

L	T	P	C
0	2	2	2

PREREQUISITE KNOWLEDGE: OOPs through Java, DDL & DML Commands – DBMS.

COURSE DESCRIPTION AND OBJECTIVES:

This course guides the student in designing and building a mobile application using Android™. The main objective of this course is to let the student learn basic Android programming concepts while building a variety of apps, starting with basic to making use of advanced concepts.

MODULE–1

UNIT-1 **0L+8T+16P=24 Hours**

INTRODUCTION

Introduction to Mobile Application Development- Mobile Applications and Device Platforms, Alternatives for Building Mobile Apps; Introduction to Android, Android versions, Android Architecture.

Application Development Process- Developers Workflow basics, Installing the Android SDK Tools; Anatomy of an Android Application.

Basic Building blocks - Activities, Services, Broadcast Receivers & Content providers; Intents & Fragments.

View Group- Layout: Linear Layout, Relative Layout, Frame Layout, Grid Layout, constraint Layout, Table Layout, and Absolute Layout.

UNIT-2 **0L+8T+16P=24 Hours**

VIEWS

Views: Basic Views; Picker Views- Time Picker View, Data Picker View; List Views – List View, Spinner View; Scroll View.

Activities: Creating an activity, Understanding the activity life cycle using Log and Toast, applying styles and themes to an activity, and hiding the activity title.

Linking Activities using Intents: Introduction to Intents and its types with examples, passing data between activities with intents, Activity Navigation- Implement up navigation with parent activities.

Fragments: Introduction to Fragment, the life cycle of a fragment, Adding fragments dynamically, Interaction between fragments.

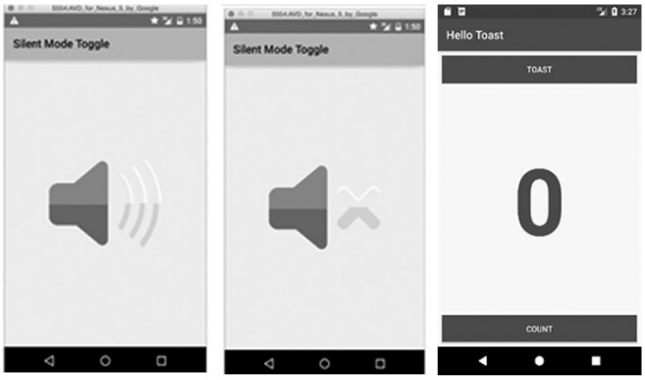
PRACTICES:

- Setting up Android Studio:
  a) Installing Android Studio
  b) Select an empty activity to simulate the “Welcome App” Using Android Studio.
  c) Exploring the interface of the Android Studio to understand the Project Structure.
- Develop an Android application using controls like Button, TextView, and EditText for designing a calculator having basic functionality like Addition, Subtraction, Multiplication, and Division.
- **Design the HelloToast app:** The HelloToast app consists of two Button elements and one Text View. When the user taps the first Button, it displays a short message (a Toast) on the screen. Tapping the second Button increases a “click” counter displayed in the TextView, which starts at zero.

SKILLS:

- ✓ *Design mobile applications for user requirements.*
- ✓ *Use of suitable advanced components to design mobile apps.*
- ✓ *Utilization of activities, intents, layouts, and views for content.*

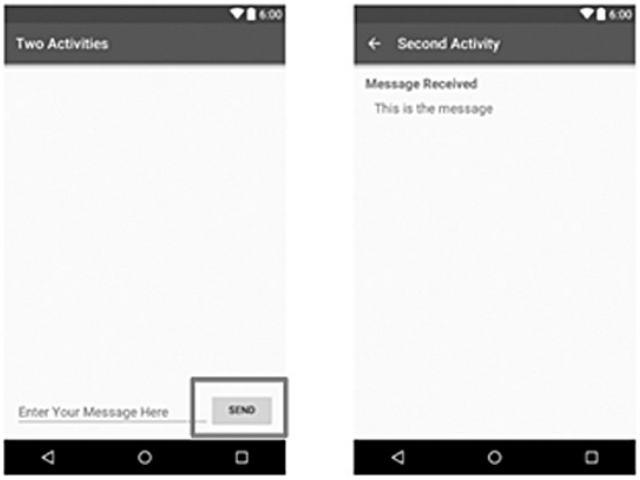
- **Design Silent Model Toggle application:** This app allows the user to toggle the ringer mode on the phone by simply pressing a button.



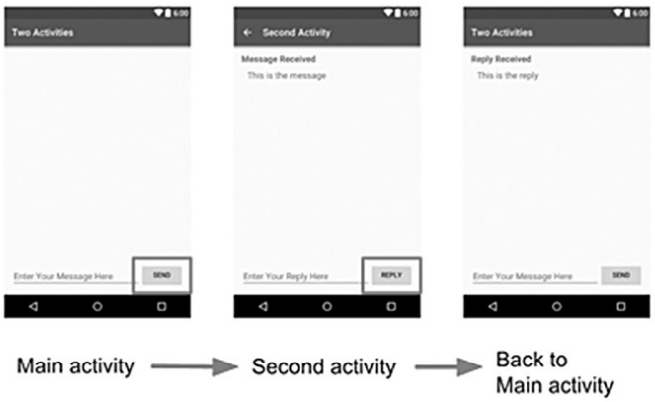
- In this assignment, students will create and build an app called Two Activities. Students will build the app in four stages.

a) In the first stage, you create an app whose main activity contains one button, Send. When the user clicks this button, your main activity uses an intent to start the second activity.

b) In the second stage, you add an EditText view to the main activity. The user enters a message and clicks Send. The main activity uses an intent to start the second activity and sends the user's message to the second activity. The second activity displays the message received.



c) In the final stage of creating the Two Activities app, you add an EditText and a Reply button to the second activity. The user can now type a reply message and tap Reply, and the reply is displayed on the main activity. At this point, you use an intent to pass the reply from the second activity to the main activity.



d) Implement all the Activity lifecycle call back methods to print messages to logcat when those methods are invoked. These log messages will allow you to see when the Activity lifecycle changes state, and how those lifecycle state changes affect your app as it runs.

- **Design an application with implicit intents:** Create a new app with one Activity and three options for actions: open a website, open a location on a map, and share a snippet of text. All the text fields are editable (EditText) but contain default values.



- **Design Droid Café:** In this practical, the student will create and build a new app starting with the Basic Activity template that imitates a dessert-ordering app. The user can tap an image to perform an action-in this case, display a Toast message-as shown in the figure below. The user can also tap a shopping cart button to proceed to the next Activity.
- **Experiment with the android:** inputType attribute for EditText elements. You add EditText elements for a person's name and address and use attributes to define single-line and multiple-line elements that make suggestions as you enter text. You also add an EditText that shows a numeric keypad for entering a phone number.
- Other types of input controls include interactive elements that provide user choices. You add radio buttons to Droid Cafe for choosing only one delivery option from several options. You also offer a spinner input control for selecting the label (Home, Work, Other, Custom) for the phone number.



MODULE–2

UNIT-1 **0L+8T+16P=24 Hours**

CREATING A FEATURE-RICH APPLICATION

Creating a Feature-Rich Application: Display Orientation – Anchor Views, resizing and repositioning Views, Managing changes to Screen Orientation; Notifications; Action bar; Dialog box; Adapters- Array Adapters and Base Adapters; Recycler View.

UNIT-2 **0L+8T+16P=24 Hours**

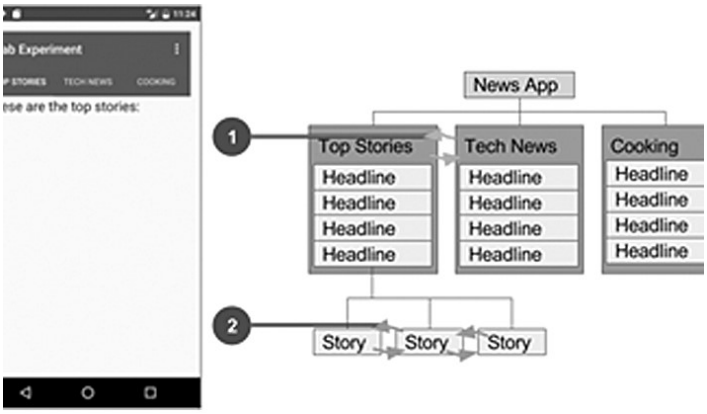
SQLITE DATABASE

SQLite Database – Creating the database, Dealing with CRUD;

Firebase-Getting Started with Firebase, Add Firebase to your Android project, Firebase database-Introduction to Firebase database, set up Firebase Real-time Database for Android, Read and Write Data on Android; Publish the App in Play store.

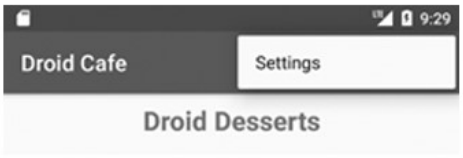
PRACTICES:

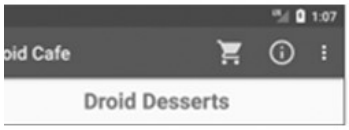
- Design an application to keep data when the user rotates the device, and when the screen is rotated: When the user rotates the device, Android will normally destroy and re-create the current Activity. You want to keep some data across this cycle, but all the fields in your Activity are lost during it.
- Create a Splash Screen for the existing project- Droid Café from Module-1.
- Design a News App- Consider the following screen as reference:



NOTE: Use Recycle View to display the news under each category.

- Adding more features to Droid Café: In the previous assignments, you created an app called Droid Café, using the Basic Activity template. This template also provides a skeletal options menu in the app bar at the top of the screen.
  a) Update that menu option as shown in the following images:
  b) Add notification option: The app must notify the user when the user places the order.
- Provide user authentication for the Droid Café using Firebase – Authentication or SQLite.







- Save all the user preferences in the Firebase Real time Database to fetch whenever required.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Apply views, intents, and fragments to an existing application.	Apply	1	2
2	Evaluate an existing app to enrich it with new features.	Evaluate	2	2,3
3	Analyse methods for storing, sharing, and retrieving data in an Android app.	Analyse	2	5
4	Design and publish a mobile app in the play store with a database forgiven real-time scenarios using modern tools- Android Studio, and Firebase.	Create	2	5,10

TEXT BOOKS:

1. John Horton “Android Programming for Beginners: Build in-depth, full-featured Android apps starting from zero programming experience”, 3rd Edition, 2021.
2. Wei-Meng Lee, “Beginning Android Application Development”, 1st Edition, John Wiley & Sons, 2012.

REFERENCE BOOKS:

1. Michael Burton,” Android App Development for Dummies “, 3rd Edition, A Wiley Brand, 2020.
2. Dawn Griffiths & David Griffiths, “Headfirst Android Development A Brain-Friendly Guide” 2nd Edition, O’Reilly, 2015.
3. https://aws.amazon.com/mobile/mobile-application-development/
4. https://google-developer-training.github.io/android-developer-fundamentals-course-concepts/.

24CS305 COMPUTING ETHICS



Source: Figure 2 from Computer and Information Security Ethics -- Models | Semantic Scholar

Hours Per Week :

L	T	P	C
1	2	0	2

PREREQUISITE KNOWLEDGE: General awareness on ethical guidelines.

COURSE DESCRIPTION AND OBJECTIVES:

The course will provide students with an understanding on Engineering Ethics and the nature of moral issues and dilemmas faced by Computer Science and engineers in their professional lives. It will give them an awareness on professional rights and responsibilities of an engineer and acquaint them on the Code of Conduct and Ethics prescribed by professional bodies like IEEE, ACM etc. for its members..

MODULE-1

UNIT-1 **0L+8T+8P=16 Hours**

INTRODUCTION

Introduction to Sociotechnical Computer Ethics.

Introduction: Why Computer Ethics? The Sociotechnical Systems Perspective, Sociotechnical Computer Ethics, Micro- and Macro-Level Analysis.

Ethics and Information Technology

Introduction: Doing Ethics, Ethical Theories and Concepts.

UNIT-2 **12L+0T+8P=20 Hours**

ETHICS IN IT CONFIGURED SOCIETIES

Technology as the Instrumentation of Human Action, Three Features of IT-Configured Activities, IT Configured Domains of Life, Democracy and the Internet.

Information Flow, Privacy and Surveillance

Introduction: Information Flow with and without Information Technology, why care about Privacy, Is Privacy over? Strategies for shaping Personal Information Flow.

PRACTICES:

- Surprises about Social Networking
- RFID and caring for the Elderly People
- Turing does not need to know
- Turnitin Dot Com
- Email Privacy and Advertising
- Workplace Spying
- Data Mining and e-Business

MODULE-2

UNIT-1 **0L+8T+8P=16 Hours**

DIGITAL INTELLECTUAL PROPERTY

The complexities of Digital Property, Protecting Property rights in Software, The Philosophical Basis of Property, Free and Open Source Software, PS Versus FOSS.

Introduction: Law and Order in the Internet, Sociotechnical Order, Online Crime, Hackers and Hacker

SKILLS:

✓ .

Ethic, Sociotechnical Security.

High Throughput Computing- Task Programming: Task computing, Task-based application models, Aneka task-based programming.

UNIT-2 **0L+8T+8P=16 Hours**

PROFESSIONAL ETHICS

Introduction: Why Professional Ethics, The Paradigm of Professions, Characteristics of Professions, Sorting out Computing and its Status as a Profession, Codes of Ethics, Professional Relationships, A Legal perspective on professionalism in Computing.

PRACTICES:

Laboratory session of this course is designed in such a way that the student should complete three projects of the given type by performing the below experiments.

- Obtaining pirated Software Abroad
- Free Software that follows Proprietary Software
- Using Public Domain Software in Proprietary Software
- Wiki Warfare
- Yahoo and Nazi Memorabilia
- Bot Roast
- Software Safety
- Conflicts of Interest
- Security in a Custom Database.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Engage in an informed critical reflection on the nature of professionalism and ethical challenges inherent in engineering profession	Under-standing	1	8
2	Understand the role of professional bodies, and the code of ethics and industrial standards pre-scribed for engineers	Under-standing	2	8
3	Apply awareness of professional rights and re-sponsibilities of an engineer to conduct themselves ethically within an organization	Applying	1	8
4	Apply understanding of safety norms to highlight ethical issues in risky situation	Applying	2	8

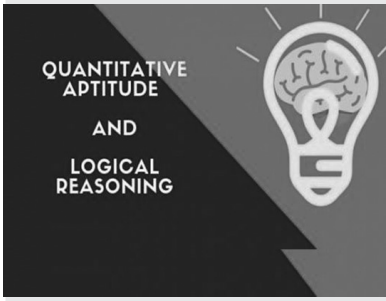
TEXT BOOKS:

1. Deborah G. Johnson, Keith W. Miller and Prentice Hall, “Computer Ethics”, 4th Edition, Pearson, 2009.

REFERENCE BOOKS:

1. Charles E. Harris, Michael S. Pritchard and Michael J. Rabins, “Engineering Ethics: Concepts and Cases”, 4th edition, Wadsworth Thompson Learning, 2009.
2. M. Govindarajan, S.Natarajan and V. S. Senthil Kumar, “Engineering Ethics”, Prentice Hall of India, Reprint 2013.

24TP301 QUANTITATIVE APTITUDE AND LOGICAL REASONING



Source: https://images.app.goo.gl/kvtVgA8TkvDCqLhj7

Hours Per Week :

L	T	P	C
1	2	0	2

PREREQUISITE KNOWLEDGE: Basic logical thinking and Problem solving ability.

COURSE DESCRIPTION AND OBJECTIVES:

The Students will be introduced to various Arithmetic and Reasoning Problems. The students will have acquaintance with various problems like Time & Work, Time & distance, Percentages, Profit & Loss etc. besides solving puzzles and Critical Reasoning.

MODULE-1

UNIT-1 **0L+4T+12P=16 Hours**

Number system, LCM & HCF of numbers, Percentage, Ratio and proportion, Profit, loss and discount, Average & Mixtures, Simple Interest & Compound interest.

UNIT-2 **0L+4T+12P=16 Hours**

Time and work, Time & distance, Problems on trains, Problems on ages, Permutation & Combinations, Probability.

PRACTICES:

- Each concept would be taught in detail in the class followed by 10 problems solved in the class.
- Students would have to solve 10 additional problems as a homework assignment in each concept.

MODULE-2

UNIT-1 **0L+4T+12P=16 Hours**

Number series, Letter series, Analogy, Odd man out, Coding and decoding, Syllogisms- Statement & Conclusions, Puzzle test.

UNIT-2 **0L+4T+12P=16 Hours**

Blood relations, Direction sense test, Order & Ranking, Seating Arrangements, Calendar & Clocks.

PRACTICES:

- Each concept would be taught in detail in the class followed by 10 problems solved in the class. Students would have to solve 10 additional problems as homework assignments in each concept.

SKILLS:

- ✓ *Helps in developing and improving problem-solving skills.*
- ✓ *Flexing and honing logical abilities.*
- ✓ *Allow students to develop critical thinking skills.*

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Meet the demands of current job market besides equipping them higher studies like CAT, GMAT etc.	Apply	1	2, 5
2	Solve Arithmetic and Reasoning Problems within shortest possible time without paper work.	Apply	1	2, 5
3	Exhibit better analytical skills and aptitude skills.	Analyse	2	2, 4
4	Develop interpretational skills.	Evaluation	2	2, 4

TEXT BOOKS:

1. R. S. Aggarwal, "Quantitative Aptitude for Competitive Examinations", S. CHAND Publications- Revised Edition, 2017.
2. ARIHANT, "A New Approach to Verbal & Non-Verbal Reasoning", Arihant Publication- Revised Edition, 2021.

REFERENCE BOOKS:

1. Trishna Knowledge Systems, "Quantitative Aptitude for Competitive Examinations", Pearson Publication, 2013.
2. R. S. Aggarwal, "A Modern Approach to Verbal & Non-Verbal Reasoning", Revised Edition, S. CHAND Publications, 2018.COURSE OUTCOMES:

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3	Exhibit better analytical skills and aptitude skills.	Analyse	2	2, 4
4	Develop interpretational skills.	Evaluation	2	2, 4

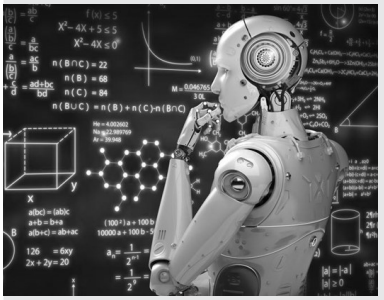
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2. R. S. Aggarwal, "A Modern Approach to Verbal & Non-Verbal Reasoning", Revised Edition, S. CHAND Publications, 2018.

24CS306 MACHINE LEARNING



Source: https://www.forbes.com/sites/kalevleetaru/2019/01/15/why-machine-learning-needs-semantics-not-just-statistics/

Hours Per Week :

L	T	P	C
2	2	2	4

PREREQUISITE KNOWLEDGE: Probability and Linear algebra, Python programming.

COURSE DESCRIPTION AND OBJECTIVES:

This course provides a broad introduction to various machine learning concepts including Supervised learning (parametric/non-parametric algorithms, support vector machines, kernels, neural networks) and Unsupervised learning (clustering, dimensionality reduction) methods. Students will get an understanding of various challenges of Machine Learning and will be able to decide on model complexity. Numerous case studies introduced in this course allow the students to apply machine-learning algorithms in computer vision, medical imaging, audio, and text domains. Laboratory experiments of this course will introduce students to advanced Machine Learning Python libraries such as Scikit-Learn, Matplotlib, and many other recent ML-related APIs. The course is designed such that the students get enough hands-on experience with a major focus on the practical implementation of theoretical concepts.

MODULE-1

UNIT-1 **8L+8T+8P=24 Hours**

INTRODUCTION

What is machine learning? Machine learning applications; Types of Learning: Supervised learning; Un-supervised learning; Reinforcement learning, Applications of Probability and statistics.

Model Training Essentials: Re-sampling methods: Bias–Variance Trade-off. Hypothesis Testing and Variable Selection, Sub sampling and Upsampling, SMOTE; Cross Validation (validation set, Leave-One-Cut (LOO), k-fold strategies) and bootstrap; Evaluation measures-Error functions, Confusion Matrix, Accuracy, Precision and Recall, F1 Score.

Regression Analysis: Linear Regression, Simple and Multiple Linear Regression, Polynomial Regression, Logistic Regression, Multi nominal Regression. Ordinary Least Squares Method, Model Shrinkage-Ridge, and LASSO regreassion.

UNIT-2 **8L+8T+8P=24 Hours**

FEATURE SELECTION

Feature Selection Strategies: Problem statement and Uses, Filter methods, Wrapper methods, Embedded methods. Branch and bound algorithm, Sequential forward/backward selection algorithms.

Dimensionality Reduction: Singular value decomposition, matrix factorization, Linear discriminant analysis, Principal components analysis.

PRACTICES:

- Apply the following tasks to any given dataset:
  a) Load and visualize data.
  b) Check out and replace missing values.
  c) Encode the Categorical data.
  d) Splitting the dataset into Training and Test set.
  e) Splitting the dataset into k-folds.
  f) Feature scaling.

SKILLS:

- ✓ *Statistical data analysis.*
- ✓ *Classify / Cluster data.*
- ✓ *Tool usage for developing ML applications.*

- House price prediction:
  a) Create a model that predicts a continuous value (price) from input features (square footage, number of bedrooms and bathrooms).
  b) Implement a univariate Model using Least Squares and plot best-fit line.
  c) Implement a multivariate Model using Least Squares and plot best-fit line.
  d) Retrieve model error and model coefficients.
  e) Observe Variance Inflation Factor (VIF).
  f) Implement Ridge regression model.
  g) Implement LASSO regression model.
  h) Report your observations on the above models for house prediction.
- Heart disease prediction:
  a) Implement a logistic regression model to predict whether an individual is suffering from heart disease or not.
  b) Evaluate and compare model performance using the following validation approaches:
  i. Validation set approach.
  ii. K-fold cross validation.
  iii. Stratified K-fold cross validation.
  iv. LOO strategy.
  c) Plot Confusion matrix.
  d) Report performance of the model in terms of the following metrics:
  i. Accuracy.
  ii. Precision-Recall.
  iii. F1 Score.
  e) Report your observations and explain when to use what type of measures.
- Implement the Polynomial Regression algorithm to fit data points. Select the appropriate data set for your experiment and draw graphs.
- Working with imbalanced datasets:
  a) Load an imbalanced dataset and visualize imbalance in the data as a bar plot.
  b) Implement KNN model for classification.
  c) Balance the dataset using:
  i. Random Over sampling.
  ii. Random Under sampling.
  iii. SMOTE.
  d) Implement KNN model for classifying data balanced in the above steps.
  e) Report your observations on the performance of models trained using balanced and imbalanced data.
- Perform effective feature selection in a given dataset using any one of the feature selection techniques.
- Dimension Reduction:
  a) Load a dataset and Implement Bayes classification model.
  b) Apply dimension reduction using:
  i. Principal Component Analysis
  ii. Linear Discriminant Analysis
  c) Apply the model on data with reduced dimension.
  d) Compare and contrast model performance in each case.

MODULE-2

UNIT-1 **8L+8T+8P=24 Hours**

CLASSIFICATION

Classification: Binary, Multi-class and Multi-label Classification; K-Nearest Neighbours, Support Vector Machines, Decision Trees, The Naïve Bayes' Classifier, Class Imbalance, Perceptron ANN model.

Ensemble Methods: Ensemble Learning Model Combination Schemes, Voting, Error-Correcting Output Codes, Bagging: Random Forest Trees, Boosting: Adaboost, Stacking.

UNIT-2 **8L+8T+8P=24 Hours**

CLUSTERING

Clustering: Different distance functions and similarity measures, K-means clustering, Medoids, Hierarchical Clustering-Single linkage and Complete linkage clustering, Graph based Clustering -MST, DBSCAN, Spectral clustering.

PRACTICES:

- Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples. Read the training data from a .CSV file.
- Implement the naïve Bayesian classifier for a sample training data set stored as a.csv file. Compute the accuracy of the classifier, considering few test data sets.
- Assuming a set of spam or not-spam mails that need to be classified, use the naïve Bayesian classifier model to perform this task. Calculate the accuracy, precision, and recall for your data set.
- Implement k-Nearest Neighbor algorithm to classify the iris data set. Print both correct and wrong predictions. Python ML library classes can be used for this problem.
- Demonstrate the working of the decision tree-based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample?
- Build a model using SVM with different kernels.
- Implement and build models using the following Ensemble techniques.
  a) Bagging.
  b) Boosting: Adaboost, Stacking.
- Build a model to perform Clustering using K-means after applying PCA and determining the value of K using the Elbow method.
- Unsupervised Modeling:
  a) Cluster the data using the following models:
  i. Spectral Clustering.
  ii. K-medoids.
  iii. DBSCAN.
  iv. Hierarchical Clustering.
  b) Compare and contrast model performance in each case.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Apply a wide variety of learning algorithms such as Probabilistic, Discriminative and Generative algorithms for a given application.	Apply	1, 2	1
2	Design an end-to-end Machine-learning model to realize solutions for real-world problems.	Apply	1	3
3	Implement various machine learning models using advanced ML tools.	Create	1, 2	5
4	Analyze and evaluate the performance of various machine learning models approaches on different kinds of data.	Analyze	2	2

TEXT BOOKS:

1. Ethem Alpaydin, “Introduction to Machine Learning”, 3rd Edition, The MIT Press, 2014.
2. Flach, Peter. “Machine learning: the art and science of algorithms that make sense of data”. Cambridge University Press, 2012.

REFERENCE BOOKS:

1. Murphy, Kevin P. Machine learning: a probabilistic perspective. MIT press, 2012.
2. Aurélien Géron, “Hands-on Machine Learning with Scikit Learn and Tensor Flow”, O’reilly, 2017.
3. Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani, “An Introduction to Statistical Learning with Applications in R”, Springer, 2013. (ISLR).

XXXXXX CRYPTOGRAPHY AND NETWORK SECURITY



Source: https://www.brainkart.com/subject/CRYPTOGRAPHY-AND-NETWORK-SECURITY-PRINCIPLES-AND-PRACTICE_136/

Hours Per Week :

L	T	P	C
3	0	2	4

PREREQUISITE KNOWLEDGE: Computer networks.

COURSE DESCRIPTION AND OBJECTIVES:

This course focuses on the modern concepts of network security using various cryptographic algorithms and underlying network security applications. It enables to understand various symmetric and asymmetric cryptographic techniques. It focuses on providing security services in applications such as e-mail functioning, web security and secure electronic transactions protocol and system security.

MODULE-1

UNIT-1 **12L+0T+8P=20 Hours**

INTRODUCTION

Introduction To Computer and Network Security Concepts: Computer Security Concepts, Security attacks, Security services, Security mechanisms, Fundamental Security Design Principles, Attack Surfaces and Attack trees, A model for network security.

Classical Encryption Techniques: Symmetric cipher model, Substitution techniques, Transposition techniques

UNIT-2 **12L+0T+8P=20 Hours**

SYMMETRIC AND ASYMMETRIC CRYPTOGRAPHY

Symmetric Ciphers: Block cipher principles, Data encryption standard, Strength of DES, Blockcipher design principles, AES cipher, Multiple encryption and triple DES, Block cipher modes ofoperation, RC4.

Asymmetric Ciphers and Cryptographic Hash Functions: Principles of public keycryptosystems, RSA algorithm, Diffie-Hellman Key Exchange, Message Authentication requirements,Authentication functions, Message authentication Codes, Hash functions, Security of hash functionsand MACs, Digital signature standard.

PRACTICES:

- Implement Substitution and Transposition Ciphers
  - Ceaser cipher
  - Playfair cipher
  - Hill cipher
  - Rail fence cipher
- Implement Symmetric Cipher
  - S-DES
  - RC4
- Implement Asymmetric Cipher
  - RSA
  - Diffie-Hellman
  - Hash Function

SKILLS:

- ✓ *Design various security services for appropriate applications.*
- ✓ *Identifying the appropriate firewall, password management and anti-virus models for specific applications.*
- ✓ *Test and resolve threats and malfunctions in network.*
- ✓ *Apply different security mechanisms for web applications.*
- ✓ *Build authentication system for security protocols.*

MODULE-2

UNIT-1 **12L+0T+8P=20 Hours**

SECURITY APPLICATIONS

Network Security Applications: Kerberos, X.509 authentication service, Public key infrastructure,

E-Mail Security: Pretty good privacy, S/MIME.

IP Security Overview: IP security architecture, Authentication header, Encapsulating security payload, Combining security associations, key management.

UNIT-2 **12L+0T+8P=20 Hours**

WEB AND SYSTEM SECURITY

Web Security: Secure socket layer and transport layer security, HTTPS, Secure Shell (SSH)

System Security: Intruders, Intrusion detection, Malicious software, Firewalls

PRACTICES:

- Configure IP Address in a system in LAN (TCP/IP Configuration)
- Configure DNS to establish interconnection between systems
- Configuring Windows Firewall
- Adding users, setting permissions
- Configure Mail server
- Demonstrate the usage of Wireshark to identify abnormal activity in network communication.
- Demonstrate usage of NMAP (Zenmap) Tool in Network Scanning.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Apply cryptographic techniques in various security service solutions effectively in everyday professional and social contexts.	Apply	1,2	1,2
2	Analyze the usage of secure protocols to safeguard sensitive data using internet.	Analyze	1,2	1,2
3	Usage of tools to Identify abnormal activity in network communication to take appropriate action.	Apply	2	5
4	Apply various security protocols to safe guard the data internet using SSL/TCL.	Apply	2	1,2

TEXT BOOK:

1. William Stallings, “Cryptography and Network security”, 7th Edition, Pearson Education, 2017.

REFERENCE BOOKS:

1. William Stallings “Network Security Essentials Applications and Standards”, 2nd Edition, Pearson Education, 2009.
2. Eric Malwald, “Fundamentals of Network Security”, 4th Edition, Pearson Education, 2010.
3. Buchmann, “Introduction to Cryptography”, 2nd Edition, Pearson Education, 2009.
4. Charlie Kaufman, “Radis Perlman and Mike Speciner, Network Security - Private Communication in a Public World”, 1st Edition, Pearson Education, 2009.

IV

YEAR

B.Tech.

COMPUTER SCIENCE AND ENGINEERING

I SEMESTER

24CS401	-	Software Project Management
24CS402	-	Parallel & Distributed Computing
22CS406	-	Privacy Preserving and Intrusion Detection
	-	Department Elective - 4
	-	Department Elective - 5
	-	Department Elective - 6
	-	Minors/Honours - 4

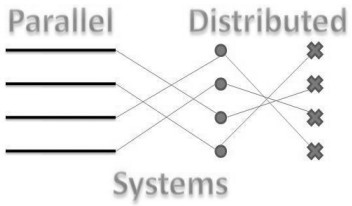
II SEMESTER

22CS404 / 22CS405	-	Internship / Project Work
	-	Minors/Honours - 4

COURSE CONTENTS

I SEM & II SEM

24CS401 SOFTWARE PROJECT MANAGEMENT



Source: http://uceou.edu/PDS/About%20PDS.html

Hours Per Week :

L	T	P	C
2	2	0	3

PREREQUISITE KNOWLEDGE: Fundamental of programming and operating system, computer networking, and computer Organization..

COURSE DESCRIPTION AND OBJECTIVES:

This course covers the challenges faced in constructing parallel and distributed applications, Various implementation techniques, paradigms, architectures and parallel algorithms. And current trends in parallel and distributed computing like Open MP, POSIX Threads, Apache Hadoop (DFS).

MODULE-1

UNIT-1 **12L+8T+0P=20 Hours**

INTRODUCTION TO PARALLEL COMPUTING

The idea of Parallelism, Power and potential of parallelism, examining sequential and parallel programs, Scope and issues of parallel and distributed computing, Goals of parallelism, Parallelism and concurrency using multiple instructions streams.

Parallel Architecture: Pipeline architecture, Array processor, Multi-processor architecture, Systolic architecture, Dataflow architecture, Architectural classification schemes, Memory access classification, Memory Issues: Shared vs. distributed, Symmetric multiprocessing (SMP), SIMD, Vector processing, GPU co-processing, Flynn's Taxonomy, Instruction Level support for parallel programming, Multiprocessor caches and Cache Coherence, Non-Uniform Memory Access (NUMA).

UNIT-2 **12L+8T+0P=20 Hours**

PARALLEL ALGORITHM DESIGN PRINCIPLES AND PROGRAMMING

Need for communication and coordination/synchronization, Scheduling and contention, Independence and partitioning, Task- Based Decomposition, Data Parallel Decomposition, Characteristics of task and interaction, Load balancing, Data Management, parallel algorithm models, Sources of overhead in parallel programs, Performance metrics for parallel algorithm implementations, Parallel algorithmic patterns like divide and conquer, Map and Reduce, Specific algorithms like parallel Merge Sort, Parallel graph Algorithms.

PRACTICES:

- Identify Multiple Instruction Single Data, or MISD. How would an MISD system work? Give an example.
- Suppose a shared-memory system uses snooping cache coherence and write-back caches. Also suppose that core 0 has the variable x in its cache, and it executes the assignment x = 5. Finally suppose that core 1 doesn't have x in its cache, and after core 0's update to x, core 1 tries to execute y = x. What value will be assigned to y? Why?
- Consider a simplified version of bucket-sort. You are given an array A of n random integers in the range [1...r] as input. The output data consist of r buckets, such that at the end of the algorithm, Bucket i contains indices of all the elements in A that are equal to i.

  Describe a decomposition based on partitioning the input data (i.e., the array A) and an appropriate mapping onto p processes. Describe briefly how the resulting parallel algorithm would work.

  Describe a decomposition based on partitioning the output data (i.e., the set of r buckets) and an appropriate mapping onto p processes. Describe briefly how the resulting parallel algorithm would work.

SKILLS:

- ✓ *Recognize parallelism in computational problems.*
- ✓ *Know different parallel systems and their classification.*
- ✓ *Design parallel algorithms for different applications.*
- ✓ *Compare replication schemes with respect to performance, availability and consistency concerns.*
- ✓ *Design, implement, and debug distributed systems.*
- ✓ *Implement parallel algorithms using MPI and OpenMP environments.*
- ✓ *Element parallel algorithms using MPI and OpenMP environments 3.0 3.5*
- ✓ *AVERAGE.*

- Consider seven tasks with running times of 1, 2, 3, 4, 5, 5, and 10 units, respectively. Assuming that it does not take any time to assign work to a process, compute the best- and worst-case speedup for a centralized scheme for dynamic mapping with two processes.

MODULE-2

UNIT-1 **12L+8T+0P=20 Hours**

INTRODUCTION TO DISTRIBUTED SYSTEMS

Goals of the Distributed Systems, Relation to parallel systems, synchronous versus asynchronous execution, design issues and challenges, Types of Distributed Systems, Distributed System Models, Hardware and software concepts related to distributed systems, middleware models.

Distributed Computing and Communication design principles: A Model of distributed executions, Models of communication networks, Global state of distributed system, Models of process communication. Communication and Coordination: Shared Memory, Consistency, Atomicity, Message- 08 Passing, Consensus, Conditional Actions, Critical Paths, Scalability, and cache coherence in multiprocessor systems, synchronization mechanism.

UNIT-2 **12L+8T+0P=20 Hours**

PARALLEL AND DISTRIBUTED PROGRAMMING FRAMEWORKS

Overview of CUDA, Open MP, POSIX Threads, Apache Hadoop (DFS), and current trends in parallel and distributed computing.

PRACTICES:

- Give five types of hardware resource and five types of data or software resource that can usefully be shared. Give examples of their sharing as it occurs in practice in distributed systems.
- The host computers used in peer-to-peer systems are often simply desktop computers in users' offices or homes. What are the implications of this for the availability and security of any shared data objects that they hold and to what extent can any weaknesses be overcome through the use of replication?
- Consider two communication services for use in asynchronous distributed systems. In service A, messages may be lost, duplicated or delayed and checksums apply only to headers. In service B, messages may be lost, delayed or delivered too fast for the recipient to handle them, but those that are delivered arrive with the correct contents. Describe the classes of failure exhibited by each service. Classify their failures according to their effects on the properties of validity and integrity. Can service B be described as a reliable communication service?
- Illustrate distributed design through a substantial case study, examining in detail the design of the Google infrastructure, a platform and associated middleware that supports both Google search and a set of associated web services and applications including Google Apps.
- Implementation of the parallel algorithms (on a PC-cluster under Linux platform). The programs will be based on POSIX Thread, MPI programming, Hadoop, Apache Spark etc.

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

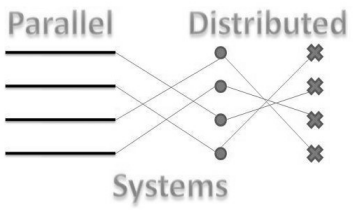
CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Apply the fundamentals of parallel and parallel computing including architectures and paradigms.	Apply	1	1
2	Analyse the various design principles of parallel algorithms.	Analyse	1	2
3	Learn the intricacies of distributed programming.	Under-stand	2	1
4	Develop and execute basic parallel and distributed applications using basic programming models and tools.	create	2	5

TEXT BOOKS:

1. DISTRIBUTED SYSTEMS Concepts and Design Fifth Edition. George Coulouris. Cambridge University. Jean Dollimore formerly of Queen Mary, University of London.
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3. Ian Foster: Designing and Building Parallel Programs – Concepts and tools for Parallel Software Engineering, Pearson Publisher, 1st Edition, 2019.
4. Parallel Programming in C with MPI and Open MP Michael J.Quinn, McGrawHill Higher Education.
5. https://hpc.llnl.gov/training/tutorials/introduction-parallel-computing-tutorial.
6. https://www.geeksforgeeks.org/introduction-to-parallel-computing/.
7. https://nptel.ac.in/.
8. https://www.coursera.org/.



Source: http://uceou.edu/PDS/About%20PDS.html

24CS402 PARALLEL & DISTRIBUTED COMPUTING

Hours Per Week :

L	T	P	C
2	2	0	3

PREREQUISITE KNOWLEDGE: Fundamental of programming and operating system, computer networking, and computer Organization..

COURSE DESCRIPTION AND OBJECTIVES:

This course covers the challenges faced in constructing parallel and distributed applications, Various implementation techniques, paradigms, architectures and parallel algorithms. And current trends in parallel and distributed computing like Open MP, POSIX Threads, Apache Hadoop (DFS).

MODULE-1

UNIT-1 **12L+8T+0P=20 Hours**

INTRODUCTION TO PARALLEL COMPUTING

The idea of Parallelism, Power and potential of parallelism, examining sequential and parallel programs, Scope and issues of parallel and distributed computing, Goals of parallelism, Parallelism and concurrency using multiple instructions streams.

Parallel Architecture: Pipeline architecture, Array processor, Multi-processor architecture, Systolic architecture, Dataflow architecture, Architectural classification schemes, Memory access classification, Memory Issues: Shared vs. distributed, Symmetric multiprocessing (SMP), SIMD, Vector processing, GPU co-processing, Flynn's Taxonomy, Instruction Level support for parallel programming, Multiprocessor caches and Cache Coherence, Non-Uniform Memory Access (NUMA).

UNIT-2 **12L+8T+0P=20 Hours**

PARALLEL ALGORITHM DESIGN PRINCIPLES AND PROGRAMMING

Need for communication and coordination/synchronization, Scheduling and contention, Independence and partitioning, Task- Based Decomposition, Data Parallel Decomposition, Characteristics of task and interaction, Load balancing, Data Management, parallel algorithm models, Sources of overhead in parallel programs, Performance metrics for parallel algorithm implementations, Parallel algorithmic patterns like divide and conquer, Map and Reduce, Specific algorithms like parallel Merge Sort, Parallel graph Algorithms.

PRACTICES:

- Identify Multiple Instruction Single Data, or MISD. How would an MISD system work? Give an example.
- Suppose a shared-memory system uses snooping cache coherence and write-back caches. Also suppose that core 0 has the variable x in its cache, and it executes the assignment x = 5. Finally suppose that core 1 doesn't have x in its cache, and after core 0's update to x, core 1 tries to execute y = x. What value will be assigned to y? Why?
- Consider a simplified version of bucket-sort. You are given an array A of n random integers in the range [1...r] as input. The output data consist of r buckets, such that at the end of the algorithm, Bucket i contains indices of all the elements in A that are equal to i.

  Describe a decomposition based on partitioning the input data (i.e., the array A) and an appropriate mapping onto p processes. Describe briefly how the resulting parallel algorithm would work.

  Describe a decomposition based on partitioning the output data (i.e., the set of r buckets) and an appropriate mapping onto p processes. Describe briefly how the resulting parallel algorithm would work.

- Consider seven tasks with running times of 1, 2, 3, 4, 5, 5, and 10 units, respectively. Assuming that it does not take any time to assign work to a process, compute the best- and worst-case speedup for a centralized scheme for dynamic mapping with two processes.

MODULE-2

UNIT-1 **12L+8T+0P=20 Hours**

INTRODUCTION TO DISTRIBUTED SYSTEMS

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UNIT-2 **12L+8T+0P=20 Hours**

PARALLEL AND DISTRIBUTED PROGRAMMING FRAMEWORKS

Overview of CUDA, Open MP, POSIX Threads, Apache Hadoop (DFS), and current trends in parallel and distributed computing.

PRACTICES:

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- Implementation of the parallel algorithms (on a PC-cluster under Linux platform). The programs will be based on POSIX Thread, MPI programming, Hadoop, Apache Spark etc.

SKILLS:

- ✓ *Recognize parallelism in computational problems.*
- ✓ *Know different parallel systems and their classification.*
- ✓ *Design parallel algorithms for different applications.*
- ✓ *Compare replication schemes with respect to performance, availability and consistency concerns.*
- ✓ *Design, implement, and debug distributed systems.*
- ✓ *Implement parallel algorithms using MPI and OpenMP environments.*
- ✓ *Element parallel algorithms using MPI and OpenMP environments 3.0 3.5*
- ✓ *AVERAGE.*

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Apply the fundamentals of parallel and parallel computing including architectures and paradigms.	Apply	1	1
2	Analyse the various design principles of parallel algorithms.	Analyse	1	2
3	Learn the intricacies of distributed programming.	Under-stand	2	1
4	Develop and execute basic parallel and distributed applications using basic programming models and tools.	create	2	5

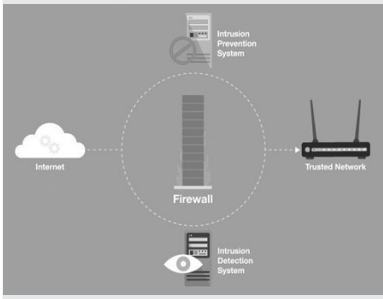
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4. Parallel Programming in C with MPI and Open MP Michael J.Quinn, McGrawHill Higher Education.
5. https://hpc.llnl.gov/training/tutorials/introduction-parallel-computing-tutorial.
6. https://www.geeksforgeeks.org/introduction-to-parallel-computing/.
7. https://nptel.ac.in/.
8. https://www.coursera.org/.

22CS406 PRIVACY PRESERVING AND INTRUSION DETECTION



Source: https://www.kirkpatrickprice.com/blog/idps-techniques

Hours Per Week :

L	T	P	C
2	2	0	3

PREREQUISITE KNOWLEDGE: Cryptography and Network Security.

COURSE DESCRIPTION AND OBJECTIVES:

This course aims to understand modern concepts related to Intrusion Detection System. The course compares alternative tools and approaches for Intrusion Detection through quantitative analysis to determine the best tool or approach to reduce risk from intrusion Students undergoing this course can identify and describe the parts of all intrusion detection systems and characterize new and emerging IDS technologies according to the basic capabilities all intrusion detection systems share.

MODULE-1

UNIT-1 **8L+8T+0P=16 Hours**

INTRODUCTION

Basic Concepts of Security, Introduction to Intrusions, Need of Intrusion Detection, Classification of Intrusion Detection Systems, Sources of Vulnerabilities, Attacks against various security objectives, countermeasures of attacks.

UNIT-2 **8L+8T+0P=16 Hours**

INTRUSION DETECTION AND PREVENTION TECHNOLOGIES

Host-based intrusion detection system (HIDS), Network-based IDS, Information Sources for IDS, Host and Network Vulnerabilities and Countermeasures. Intrusion detection techniques, misuse detection: pattern matching, rule-based and state-based anomaly detection: statistical based, machine learning based, data mining-based hybrid detection.

IDS and IPS Architecture: Tiered architectures, Single-tiered, Multi-tiered, Peer-to-Peer. Sensor: sensor functions, sensor deployment and security.

PRACTICES:

- Installing Snort into the Operating System
- Configuring and Starting the Snort IDS
- Writing and Adding a Snort Rule
- Triggering an Alert for the New Rule

MODULE- 2

UNIT-1 **8L+8T+0P=16 Hours**

AGENTS

Agent functions, agent deployment and security. Manager component: manager functions, manager deployment and security. Information flow in IDS and IPS, defending IDS/IPS, Case study on commercial and open-source IDS.

UNIT-2 **8L+8T+0P=16 Hours**

ALERT MANAGEMENT AND CORRELATION DATA FUSION

Alert correlation, Pre-process, Correlation Techniques, Post-process, Alert Correlation architectures. Cooperative Intrusion Detection, Cooperative Discovery of Intrusion chain, Abstraction-based Intrusion

SKILLS:

- ✓ *Installation and configuration of IDS.*
- ✓ *Compare various anomaly detection techniques.*
- ✓ *Evaluate security of intrusion detection tool.*

Detection, Interest-based communication and cooperation, agent-based cooperation.

PRACTICES:

- Demo of Eavesdropping attack and its Prevention using SSH.
- Demonstrate how to provide secure data storage, secure data transmission and for creating digital signatures (GnuPG)
- Setup a honey pot and monitor the honeypot on network (KF Sensor)
- Installation of rootkits and study about the variety of option.
- Perform wireless audit on an access point or a router and decrypt WEP and WPA. (NetStumbler)
- Install any open-source IDS and study the logs

COURSE OUTCOMES:

Upon successful completion of this course, students will have the ability to:

CO No.	Course Outcomes	Blooms Level	Module No.	Mapping with POs
1	Apply Intrusion Detection tools and techniques in order to improve their security posture.	Apply	2	1,2
2	Apply the knowledge to the architecture, configuration, and analysis of specific intrusion detection systems.	Apply	2	1, 2
3	Analyse appropriate situations and scenarios where intrusion detection may be applied to achieve an increased level of situational awareness and information assurance.	Analyse	1	1,2
4	Evaluate the security of an organization for better performance.	Evaluate	1	1,2

TEXT BOOKS:

1. C. Endorf, E. Schultz and J. Mellander, Intrusion Detection & Prevention, McGraw-Hill/Osborne , 2006.

REFERENCE BOOKS:

1. Ali A. Ghorbani, Network intrusion detection and prevention concepts and techniques, Springer, 2010.
2. J. M. Kizza, Computer Network Security, Springer, 2005.
3. Chris Sanders and Jason Smith, Applied Network Security Monitoring: Collection, Detection, and Analysis, Syngress, 2013.